

1

Partial Differentiation

1.1 INTRODUCTION

Area of a rectangle depends upon its length and breadth, hence we can say that area is the function of two variables *i.e.* its length and breadth.

z is called a function of two variables x and y if z has one definite value for every pair of values of x and y . Symbolically, it is written as

$$z = f(x, y)$$

The variable x and y are called independent variables while z is called the dependent variable.

Similarly, we can define z as a function of more than two variables.

Geometrically: Let $z = f(x, y)$

where x, y belong to an area A of the xy -plane. For each point (x, y) corresponds a value of z . These values of (x, y, z) form a surface in space.

Hence, the function $z = f(x, y)$ represents a surface.

1.2 LIMIT

The function $f(x, y)$ is said to tend to the limit l as $x \rightarrow a$ and $y \rightarrow b$ if and only if the limit l is independent of the path followed by the point (x, y) as $x \rightarrow a$ and $y \rightarrow b$. Then

$$\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = l$$

The function $f(x, y)$ in region R is said to tend to the limit l as $x \rightarrow a$ and $y \rightarrow b$ if and only if corresponding to a positive number $\epsilon \in (a, b)$, there exists another positive number δ such that

$$|f(x, y) - l| < \epsilon \quad \text{for} \quad 0 < (x - a)^2 + (y - b)^2 < \delta^2$$

for every point (x, y) in R .

1.3 WORKING RULE TO FIND THE LIMIT

Step 1. Find the value of $f(x, y)$ along $x \rightarrow a$ and $y \rightarrow b$.

Step 2. Find the value of $f(x, y)$ along $y \rightarrow b$ and $x \rightarrow a$.

If the values of $f(x, y)$ in step 1 and step 2 remain the same, the limit exists otherwise not.

Step 3. If $a \rightarrow 0$ and $b \rightarrow 0$, find the limit along $y = mx$ or $y = mx^n$. If the value of the limit does not contain m then limit exists. If it contains m , the limit does not exist.

Note. (i) Put $x = 0$ and then $y = 0$ in f . Find its value f_1 .

(ii) Put $y = 0$ and then $x = 0$ in f . Find the value f_2 .

If $f_1 \neq f_2$, limit does not exist.

If $f_1 = f_2$, then

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|--|---|
| <p>(iii) Put $y = mx$ and find the limit f_3.
 If $f_1 = f_2 \neq f_3$, then limit does not exist.
 If $f_1 = f_2 = f_3$, then</p> | <p>(iv) Put $y = mx^2$ and find the limit f_4.
 If $f_1 = f_2 = f_3 \neq f_4$, then limit does not exist.
 If $f_1 = f_2 = f_3 = f_4$, then limit exists.</p> |
|--|---|

Example 1. Evaluate $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2}$

Solution. (i) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2} = \lim_{y \rightarrow 0} \frac{0}{0 + y^2} = 0 = f_1$ (say)

(ii) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y}{x^4 + y^2} = \lim_{x \rightarrow 0} \frac{0}{x^4 + 0} = 0 = f_2$ (say)

Here, $f_1 = f_2$, therefore

(iii) Put $y = mx$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 mx}{x^4 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{mx}{x^2 + m^2} = 0 = f_3 \quad (\text{say})$$

Here, $f_1 = f_2 = f_3$, therefore

(vi) Put $y = mx^2$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 mx^2}{x^4 + m^2 x^4} = \frac{m}{1 + m^2} = f_4$$

Here, $f_1 = f_2 = f_3 \neq f_4$

Thus, limit does not exist.

Ans.

Example 2. Evaluate $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^3 + y^3)$.

Solution. (i) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^3 + y^3) = \lim_{y \rightarrow 0} (0 + y^3) = 0 = f_1$ (say)

(ii) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^3 + y^3) = \lim_{x \rightarrow 0} (x^3 + 0) = 0 = f_2$ (say)

Here, $f_1 = f_2$, therefore

(iii) Put $y = mx$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^3 + y^3) = \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow mx} (x^3 + y^3) \right] = \lim_{x \rightarrow 0} (x^3 + m^3 x^3) = 0 = f_3 \quad (\text{say})$$

Here, $f_1 = f_2 = f_3$, therefore

(iv) Put $y = mx^2$

$$\begin{aligned} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^3 + y^3) &= \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow mx^2} (x^3 + y^3) \right] = \lim_{x \rightarrow 0} (x^3 + m^3 x^6) \\ &= \lim_{x \rightarrow 0} x^3 (1 + m^3 x^3) = 0 = f_4 \end{aligned} \quad (\text{say})$$

Here, $f_1 = f_2 = f_3 = f_4$

Thus, limit exists with value 0.

Ans.

Example 3. Evaluate $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{3x^2 y}{x^2 + y^2 + 5}$.

Solution. $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{3x^2y}{x^2 + y^2 + 5} = \lim_{x \rightarrow 1} \left[\lim_{y \rightarrow 2} \frac{3x^2y}{x^2 + y^2 + 5} \right] = \lim_{x \rightarrow 1} \frac{3x^2(2)}{x^2 + (2)^2 + 5}$

$$= \lim_{x \rightarrow 1} \frac{6x^2}{x^2 + 9} = \frac{6}{1 + 9} = \frac{3}{5}$$

Ans.

Example 4. Evaluate $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \frac{2x - 3}{x^3 + 4y^3}$.

Solution. (i) $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \frac{2x - 3}{x^3 + 4y^3} = \lim_{y \rightarrow 3} \left[\lim_{x \rightarrow \infty} \frac{2x - 3}{x^3 + 4y^3} \right]$

$$= \lim_{y \rightarrow 3} \left[\lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} - \frac{3}{x^3}}{1 + 4\left(\frac{y}{x}\right)^3} \right] = \lim_{y \rightarrow 3} \left(\frac{0 - 0}{1 + 4(0)} \right) = 0 = f_1 \quad (\text{say})$$

(ii) $\lim_{\substack{y \rightarrow 3 \\ x \rightarrow \infty}} \frac{2x - 3}{x^3 + 4y^3} = \lim_{x \rightarrow \infty} \left[\lim_{y \rightarrow 3} \frac{2x - 3}{x^3 + 4y^3} \right]$

$$= \lim_{x \rightarrow \infty} \frac{2x - 3}{x^3 + 108} = \lim_{x \rightarrow \infty} \frac{\frac{2}{x^2} - \frac{3}{x^3}}{1 + \frac{108}{x^3}} = \frac{0 - 0}{1 + 0} = 0 = f_2 \quad (\text{say})$$

Here, $f_1 = f_2$.

Hence, the limit exists with value 0.

Ans.

EXERCISE 1.1

Evaluate the following limits:

- | | | | |
|--|----------------------------------|---|----------------------------------|
| 1. $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{2x^2 + y^2}{2xy}$ | Ans. $\frac{3}{4}$ | 2. $\lim_{\substack{x \rightarrow 2 \\ y \rightarrow 3}} \frac{x^3 + y^2}{x^2 - y}$ | Ans. 17 |
| 3. $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \frac{2xy - 3}{x^3 + 4y^3}$ | Ans. 0 | 4. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{y - x^2}; x \neq 0, y \neq 0$ | Ans. Limit does not exist |
| 5. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x - y}{x^2 + y^2}; x \neq 0, y \neq 0$ | Ans. Limit does not exist | | |
| 6. $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 1}} \frac{xy - 2x}{xy - 2y}$ | Ans. 1 | 7. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^3 + 2y^3}{x^2 + 4y^2}; x \neq 0, y \neq 0$ | Ans. 0 |
| 8. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2y^3}{x^2 + y^2}; x \neq 0, y \neq 0$ | Ans. 0 | 9. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy + 2}{x^2 + y^2}; x \neq 0, y \neq 0$ | Ans. 0 |
| 10. $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 1}} \frac{3x(y - 2)}{2y(x - 2)}$ | Ans. $\frac{1}{2}$ | | |

1.4 CONTINUITY

A function $f(x, y)$ is said to be continuous at a point (a, b) if

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = f(a, b)$$

A function is said to be continuous in a domain if it is continuous at every point of the domain.

1.5 WORKING RULE FOR CONTINUITY AT A POINT (a, b)

Step 1. $f(a, b)$ should be well defined

Step 2. $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ should exist.

Step 3. $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$

Example 5. Test the function $f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & \text{when } x \neq 0, \quad y \neq 0 \\ 0 & \text{when } x = 0, \quad y = 0 \end{cases}$

for continuity.

Solution. **Step 1.** The function is well defined at $(0, 0)$.

$$\begin{aligned} \text{Step 2. } \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + y^2} = \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow mx} \frac{x^3 - y^3}{x^2 + y^2} \right] \\ &= \lim_{x \rightarrow 0} \frac{x^3 - m^3 x^3}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{x(1 - m^3)}{1 + m^2} = 0 \end{aligned}$$

Thus, limit exists at $(0, 0)$.

Step 3. limit of $f(x, y)$ at origin = value of the function at origin.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + y^2} = f(0, 0) = 0$$

Hence, the function f is continuous at the origin.

Ans.

Example 6. Discuss the continuity of $f(x, y) = \begin{cases} \frac{x}{\sqrt{x^2 + y^2}}, & x \neq 0, \quad y \neq 0 \\ 2, & x = 0, \quad y = 0 \end{cases}$

at the origin.

Solution. Here, we $f(x, y) = \begin{cases} \frac{x}{\sqrt{x^2 + y^2}}, & x \neq 0, \quad y \neq 0 \\ 2, & x = 0, \quad y = 0 \end{cases}$

Step 1. The function $f(x, y)$ at $(0, 0)$ is well defined.

$$\text{Step 2. } \lim_{(x,y) \rightarrow (0,0)} \frac{x}{\sqrt{x^2 + y^2}} = \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow mx} \frac{x}{\sqrt{x^2 + y^2}} \right] = \lim_{x \rightarrow 0} \frac{x}{\sqrt{x^2 + m^2 x^2}} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1 + m^2}}$$

For different values of m the limit f is not unique.

so the $\lim_{(x,y) \rightarrow (0,0)} \frac{x}{\sqrt{x^2 + y^2}}$ does not exist.

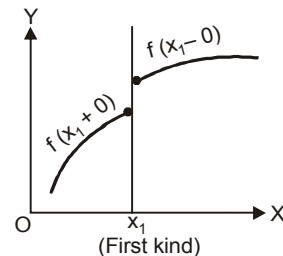
Hence $f(x, y)$ is not continuous at origin.

Ans.

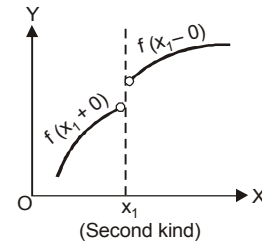
1.6 TYPES OF DISCONTINUITY

(Gujarat Univ. I sem. Jan. 2009)

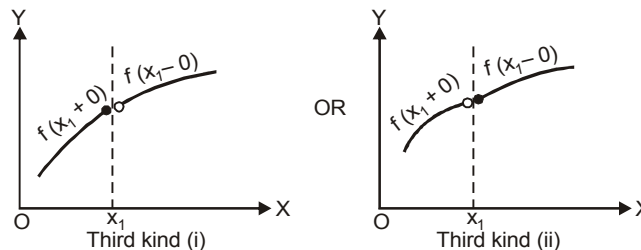
1. First Kind. $f(x)$ is said to have discontinuity of first kind at the point $x = x_1$ if Right limit $f(x_1 + 0)$ and left limit $f(x_1 - 0)$ exist but are not equal.



2. Second Kind. $f(x)$ is said to have discontinuity of the second kind at $x = x_1$ if neither right limit $f(x_1 + 0)$ exists nor left limit $f(x_1 - 0)$ exists.

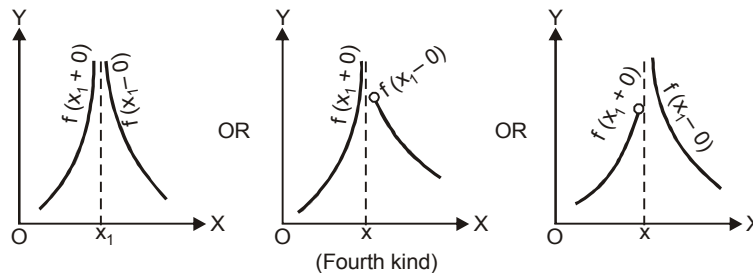


3. Third Kind (Mixed discontinuity). $f(x)$ is said to have mixed discontinuity at the point $x = x_1$ if only one of the two limits right limit $f(x_1 + 0)$ and left limit $f(x_1 - 0)$ exists and not the other.

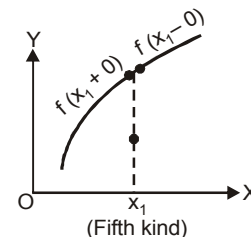


4. Fourth Kind (Infinite discontinuity). $f(x)$ is said to have infinite discontinuity at the point $x = x_1$ if either one or both limits right limit and left limit $f(x_1 - 0)$ is infinite.

If both limits do not exist and if $f(x_1 \pm h)$ oscillates between limits one of which is infinite as $\pm h \rightarrow 0$. It is also a point of infinite discontinuity.



5. Fifth Kind (Removable discontinuity). If right limit $f(x_1 + 0)$ is equal to left limit $f(x_1 - 0)$ is not equal to $f(x_1)$, then $f(x)$ is said to have removable discontinuity.



EXERCISE 1.2

Test for continuity:

$$1. f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & \text{when } x \neq 0, y \neq 0 \\ 0, & \text{when } x = 0, y = 0 \end{cases}$$

at origin.

Ans. Continuous at origin.

$$2. f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & \text{when } x \neq 0, y \neq 0 \\ 0, & \text{when } x = 0, y = 0 \end{cases}$$

at origin.

Ans. Not continuous at origin.

$$3. f(x, y) = \begin{cases} \frac{x^3 y^3}{x^3 + y^3}, & \text{when } x \neq 0, y \neq 0 \\ 0, & \text{when } x = 0, y = 0 \end{cases}$$

at origin.

Ans. Not continuous at origin.

$$4. f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & \text{when } x \neq 0, y \neq 0 \\ 0, & \text{when } x = 0, y = 0 \end{cases}$$

at origin.

Ans. Not continuous at origin.

$$5. f(x, y) = \begin{cases} x^3 + y^3, & \text{when } x \neq 0, y \neq 0 \\ 0, & \text{when } x = 0, y = 0 \end{cases}$$

at origin.

Ans. Continuous at origin.

$$6. f(x, y) = \begin{cases} \frac{x^2 + 2y}{x + y^2}, & \\ 1 & \text{when } x = 1, y = 2 \end{cases}$$

at the point (1, 2).

Ans. Continuous at (1, 2).

$$7. \text{ Show that the function } f(x, y) = \begin{cases} 2x^2 + y, & (x, y) \neq (1, 2) \\ 0, & (x, y) = (1, 2) \end{cases}$$

is discontinuous at (1, 2).

$$8. \text{ Show that the function } f(x, y) = \begin{cases} (x + y) \sin\left(\frac{1}{x + y}\right), & x + y \neq 0 \\ 0, & x + y = 0 \end{cases}$$

is continuous at (0, 0) but its partial derivatives of first order do not exist at (0, 0).

(A.M.I.E.T.E., Dec. 2007)

1.7 PARTIAL DERIVATIVES

Let $z = f(x, y)$ be function of two independent variables x and y . If we keep y constant and x varies then z becomes a function of x only. The derivative of z with respect to x , keeping y as constant is called partial derivative of 'z', w.r.t. 'x' and is denoted by symbols.

$$\frac{\partial z}{\partial x}, \frac{\partial f}{\partial x}, f_x(x, y) \text{ etc.}$$

$$\text{Then } \frac{\partial z}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}$$

The process of finding the partial differential coefficient of z w.r.t. 'x' is that of ordinary differentiation, but with the only difference that we treat y as constant.

Similarly, the partial derivative of 'z' w.r.t. 'y' keeping x as constant is denoted by

$$\frac{\partial z}{\partial y}, \frac{\partial f}{\partial y}, f_y(x, y) \text{ etc.}$$

$$\frac{\partial z}{\partial y} = \lim_{\delta y \rightarrow 0} \frac{f(x, y + \delta y) - f(x, y)}{\delta y}$$

$$\text{Notation. } \frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q, \quad \frac{\partial^2 z}{\partial x^2} = r, \quad \frac{\partial^2 z}{\partial x \partial y} = s, \quad \frac{\partial^2 z}{\partial y^2} = t$$

Example 7. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, then find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

$$\text{Solution. } u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{1 - \left(\frac{x}{y}\right)^2}} \cdot \frac{1}{y} + \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{1}{\sqrt{y^2 - x^2}} - \frac{y}{x^2 + y^2}$$

$$x \frac{\partial u}{\partial x} = \frac{x}{\sqrt{y^2 - x^2}} - \frac{xy}{x^2 + y^2} \quad \dots(1)$$

$$\frac{\partial u}{\partial y} = \frac{1}{\sqrt{1 - \left(\frac{x}{y}\right)^2}} \left(-\frac{x}{y^2}\right) + \frac{1}{1 + \left(\frac{y}{x}\right)^2} \frac{1}{x} = -\frac{x}{y\sqrt{y^2 - x^2}} + \frac{x}{x^2 + y^2}$$

$$y \frac{\partial u}{\partial y} = -\frac{x}{\sqrt{y^2 - x^2}} + \frac{xy}{x^2 + y^2} \quad \dots(2)$$

On adding (1) and (2), we have $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$ **Ans.**

Example 8. Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial \theta}$ if $u = e^{r \cos \theta} \cdot \cos (r \sin \theta)$

Solution. $u = e^{r \cos \theta} \cdot \cos (r \sin \theta)$

$$\begin{aligned} \frac{\partial u}{\partial r} &= e^{r \cos \theta} \cdot [-\sin (r \sin \theta) \cdot \sin \theta] + [\cos \theta \cdot e^{r \cos \theta}] \cdot \cos (r \sin \theta) \\ &\quad \text{(keeping } \theta \text{ as constant)} \\ &= e^{r \cos \theta} \cdot [-\sin (r \sin \theta) \cdot \sin \theta + \cos (r \sin \theta) \cdot \cos \theta] \\ &= e^{r \cos \theta} \cdot \cos (r \sin \theta + \theta) \end{aligned}$$

Ans.

$$\begin{aligned} \frac{\partial u}{\partial \theta} &= e^{r \cos \theta} \cdot [-\sin (r \sin \theta) \cdot r \cos \theta] + [-r \sin \theta \cdot e^{r \cos \theta}] \cdot \cos (r \sin \theta) \\ &\quad \text{(keeping } r \text{ as constant)} \\ &= -r e^{r \cos \theta} \cdot [\sin (r \sin \theta) \cdot \cos \theta + \sin \theta \cos (r \sin \theta)] \\ &= -r e^{r \cos \theta} \cdot \sin (r \sin \theta + \theta) \end{aligned}$$

Ans.

Example 9. If $u = (1 - 2xy + y^2)^{-1/2}$ prove that, $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$.

Solution. $u = (1 - 2xy + y^2)^{-1/2}$... (1)

Differentiating (1) partially w.r.t. 'x', we get

$$\begin{aligned} \frac{\partial u}{\partial x} &= -\frac{1}{2} (1 - 2xy + y^2)^{-3/2} (-2y) \\ x \frac{\partial u}{\partial x} &= xy (1 - 2xy + y^2)^{-3/2} \end{aligned} \quad \dots(2)$$

Differentiating (1) partially w.r.t. 'y', we get

$$\begin{aligned} \frac{\partial u}{\partial y} &= -\frac{1}{2} (1 - 2xy + y^2)^{-3/2} (-2x + 2y) \\ y \frac{\partial u}{\partial y} &= (xy - y^2) (1 - 2xy + y^2)^{-3/2} \end{aligned} \quad \dots(3)$$

Subtracting (3) from (2), we get

$$\begin{aligned} x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} &= xy (1 - 2xy + y^2)^{-3/2} - (xy - y^2) (1 - 2xy + y^2)^{-3/2} \\ &= y^2 (1 - 2xy + y^2)^{-3/2} = y^2 u^3. \end{aligned}$$

Proved.

Example 10. If $z = e^{ax + by} \cdot f(ax - by)$, prove that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$.

(A.M.I.E.T.E., Summer 2004)

Solution. $z = e^{ax+by} f(ax-by)$... (1)

Differentiating (1) w.r.t. 'x', we get

$$\frac{\partial z}{\partial x} = a e^{ax+by} f(ax-by) + e^{ax+by} .a f'(ax-by)$$

$$b \frac{\partial z}{\partial x} = a b e^{ax+by} f(ax-by) + a b e^{ax+by} f'(ax-by) \quad \dots (2)$$

Differentiating (1) w.r.t. 'y', we get

$$\frac{\partial z}{\partial y} = b e^{ax+by} f(ax-by) + e^{ax+by} .(-b) f'(ax-by)$$

$$a \frac{\partial z}{\partial y} = a b e^{ax+by} f(ax-by) - a b e^{ax+by} f'(ax-by) \quad \dots (3)$$

On adding (2) and (3), we get

$$b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2ab e^{ax+by} f(ax-by)$$

$$\Rightarrow b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2a b z \quad \text{Proved.}$$

1.8 PARTIAL DERIVATIVES OF HIGHER ORDERS

Let $z = f(x, y)$ then $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ being the functions of x and y can further be differentiated partially with respect to x and y .

Symbolically

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x^2} \quad \text{or} \quad \frac{\partial^2 f}{\partial x^2} \quad \text{or} \quad f_{xx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial y \partial x} \quad \text{or} \quad \frac{\partial^2 f}{\partial y \partial x} \quad \text{or} \quad f_{yx}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial^2 z}{\partial x \partial y} \quad \text{or} \quad \frac{\partial^2 f}{\partial x \partial y} \quad \text{or} \quad f_{xy}$$

Note. $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Example 11. Prove that $y = f(x+at) + g(x-at)$ satisfies

$$\frac{\partial^2 y}{\partial t^2} = a^2 \left(\frac{\partial^2 y}{\partial x^2} \right)$$

where f and g are assumed to be at least twice differentiable and a is any constant.

(U.P., I Sem; Jan 2011, A.M.I.E., Summer 2000)

Solution. $y = f(x+at) + g(x-at)$... (1)

Differentiating (1) w.r.t. 'x' partially, we get

$$\frac{\partial y}{\partial x} = f'(x+at) + g'(x-at)$$

$$\frac{\partial^2 y}{\partial x^2} = f''(x+at) + g''(x-at)$$

Differentiating (1) w.r.t. 't' partially, we get

$$\frac{\partial y}{\partial t} = f'(x+at).a + g'(x-at) (-a)$$

$$\begin{aligned}\frac{\partial^2 y}{\partial t^2} &= a^2 f''(x + at) + g''(x - at) a^2 \\ &= a^2 [f''(x + at) + g''(x - at)] = a^2 \frac{\partial^2 y}{\partial x^2}\end{aligned}$$

Hence $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$ **Proved.**

Example 12. If $z = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$, prove that $\frac{\partial^2 z}{\partial y \partial x} = \frac{x^2 - y^2}{x^2 + y^2}$.

Solution. $z = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$ (U.P., I Semester Comp. 2002)

$$\begin{aligned}\frac{\partial z}{\partial x} &= 2x \tan^{-1}\left(\frac{y}{x}\right) + x^2 \frac{1}{1 + \frac{y^2}{x^2}} \left(-\frac{y}{x^2}\right) - y^2 \frac{1}{1 + \frac{x^2}{y^2}} \left(\frac{1}{y}\right) \\ &= 2x \tan^{-1}\left(\frac{y}{x}\right) - \frac{x^2 y}{x^2 + y^2} - \frac{y^3}{x^2 + y^2} \\ &= 2x \tan^{-1}\left(\frac{y}{x}\right) - y \frac{(x^2 + y^2)}{x^2 + y^2} = 2x \tan^{-1}\left(\frac{y}{x}\right) - y \\ \frac{\partial^2 z}{\partial y \partial x} &= 2x \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(\frac{1}{x}\right) - 1 = 2 \frac{x^2}{x^2 + y^2} - 1 = \frac{x^2 - y^2}{x^2 + y^2}\end{aligned}$$

Proved.

Example 13. If $u = e^{xyz}$, find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$. (A.M.I.E. Winter 2000)

Solution.

$$u = e^{xyz}$$

$$\frac{\partial u}{\partial z} = e^{xyz} (x y)$$

$$\Rightarrow \frac{\partial^2 u}{\partial y \partial z} = e^{xyz} (x) + e^{xyz} (x z) (x y) = e^{xyz} (x + x^2 y z)$$

$$\begin{aligned}\frac{\partial^3 u}{\partial x \partial y \partial z} &= e^{xyz} (1 + 2x y z) + e^{xyz} (y z) (x + x^2 y z) \\ &= e^{xyz} [1 + 2x y z + x y z + x^2 y^2 z^2] \\ &= e^{xyz} [1 + 3x y z + x^2 y^2 z^2]\end{aligned}$$

Ans.

Example 14. If $v = (x^2 + y^2 + z^2)^{\frac{m}{2}}$, then find the value of m ($m \neq 0$) which will make

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0.$$

Solution. We have, $v = (x^2 + y^2 + z^2)^{\frac{m}{2}}$

$$\frac{\partial v}{\partial x} = \frac{m}{2} (x^2 + y^2 + z^2)^{\frac{m}{2}-1} (2x) = mx (x^2 + y^2 + z^2)^{\frac{m}{2}-1}$$

$$\frac{\partial^2 v}{\partial x^2} = m \left(\frac{m}{2} - 1 \right) x (x^2 + y^2 + z^2)^{\frac{m}{2}-2} (2x) + m (x^2 + y^2 + z^2)^{\frac{m}{2}-1}$$

$$\begin{aligned}
&= m(m-2)x^2(x^2+y^2+z^2)^{\frac{m}{2}-2} + m(x^2+y^2+z^2)^{\frac{m}{2}-1} \\
&= m(x^2+y^2+z^2)^{\frac{m}{2}-2} [(m-2)x^2 + x^2 + y^2 + z^2] \quad \dots(1)
\end{aligned}$$

$$\text{Similarly, } \frac{\partial^2 v}{\partial y^2} = m(x^2+y^2+z^2)^{\frac{m}{2}-2} [(m-2)y^2 + x^2 + y^2 + z^2] \quad \dots(2)$$

$$\frac{\partial^2 v}{\partial z^2} = m(x^2+y^2+z^2)^{\frac{m}{2}-2} [(m-2)z^2 + x^2 + y^2 + z^2] \quad \dots(3)$$

On adding (1), (2) and (3), we get

$$\begin{aligned}
\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} &= m(x^2+y^2+z^2)^{\frac{m}{2}-2} [(m-2)(x^2+y^2+z^2) + 3(x^2+y^2+z^2)] \\
0 &= m(x^2+y^2+z^2)^{\frac{m}{2}-1} [m-2+3] \quad \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0 \right]
\end{aligned}$$

$$\begin{aligned}
0 &= m(m+1)(x^2+y^2+z^2)^{\frac{m}{2}-1} \\
0 &= m(m+1) \Rightarrow m = 0, -1 \quad (m \neq 0)
\end{aligned}$$

Hence,

$$m = -1$$

Ans.

Example 15. If $u = (1-2xy+y^2)^{-\frac{1}{2}}$, prove that $\frac{\partial}{\partial x} \left((1-x^2) \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(y^2 \frac{\partial u}{\partial y} \right) = 0$.

Solution. We have, $u = (1-2xy+y^2)^{-\frac{1}{2}}$... (1)

$$\begin{aligned}
\frac{\partial u}{\partial x} &= -\frac{1}{2}(1-2xy+y^2)^{-\frac{3}{2}}(-2y) = \frac{y}{(1-2xy+y^2)^{\frac{3}{2}}} \\
(1-x^2) \frac{\partial u}{\partial x} &= \frac{(1-x^2)y}{(1-2xy+y^2)^{\frac{3}{2}}} \\
\frac{\partial}{\partial x} \left((1-x^2) \frac{\partial u}{\partial x} \right) &= \frac{(1-2xy+y^2)^{\frac{3}{2}}(-2xy) - (1-x^2)y \left(\frac{3}{2}(1-2xy+y^2)^{\frac{1}{2}}(-2y) \right)}{(1-2xy+y^2)^3}
\end{aligned}$$

Cancelling $(1-2xy+y^2)^{\frac{1}{2}}$ from numerator and denominator, we have

$$\begin{aligned}
&= \frac{(1-2xy+y^2)(-2xy) + 3(1-x^2)y^2}{(1-2xy+y^2)^{\frac{5}{2}}} = \frac{-2xy + 4x^2y^2 - 2xy^3 + 3y^2 - 3x^2y^2}{(1-2xy+y^2)^{\frac{5}{2}}} \\
&= \frac{x^2y^2 - 2xy^3 - 2xy + 3y^2}{(1-2xy+y^2)^{\frac{5}{2}}} \quad \dots(2)
\end{aligned}$$

Differentiating (1) partially w.r.t. y , we get

$$\frac{\partial u}{\partial y} = -\frac{1}{2}(1-2xy+y^2)^{-\frac{3}{2}}(-2x+2y) = \frac{x-y}{(1-2xy+y^2)^{\frac{3}{2}}}$$

$$y^2 \frac{\partial u}{\partial y} = \frac{xy^2 - y^3}{(1 - 2xy + y^2)^{\frac{3}{2}}}$$

$$\frac{\partial}{\partial y} \left(y^2 \frac{\partial u}{\partial y} \right) = \frac{(1 - 2xy + y^2)^{\frac{3}{2}} (2xy - 3y^2) - (xy^2 - y^3) \left(\frac{3}{2} (1 - 2xy + y^2)^{\frac{1}{2}} (-2x + 2y) \right)}{(1 - 2xy + y^2)^3}$$

Dividing numerator and denominator by $(1 - 2xy + y^2)^{\frac{1}{2}}$, we get

$$\begin{aligned} \frac{\partial}{\partial y} \left(y^2 \frac{\partial u}{\partial y} \right) &= \frac{(1 - 2xy + y^2)(2xy - 3y^2) + (xy^2 - y^3) 3(x - y)}{(1 - 2xy + y^2)^{\frac{5}{2}}} \\ &= \frac{(2xy - 4x^2y^2 + 2xy^3 - 3y^2 + 6xy^3 - 3y^4) + 3x^2y^2 - 3xy^3 - 3xy^3 + 3y^4}{(1 - 2xy + y^2)^{\frac{5}{2}}} \\ &= \frac{-x^2y^2 + 2xy^3 + 2xy - 3y^2}{(1 - 2xy + y^2)^{\frac{5}{2}}} \quad \dots(3) \end{aligned}$$

On adding (2) and (3), we get

$$\frac{\partial}{\partial x} (1 - x^2) \frac{\partial u}{\partial x} + \frac{\partial}{\partial y} \left(y^2 \frac{\partial u}{\partial y} \right) = \frac{x^2y^2 - 2xy^3 - 2xy + 3y^2}{(1 - 2xy + y^2)^{\frac{5}{2}}} + \frac{-x^2y^2 + 2xy^3 + 2xy - 3y^2}{(1 - 2xy + y^2)^{\frac{5}{2}}} = 0$$

Proved.

Example 16. Prove that if $f(x, y) = \frac{1}{\sqrt{y}} e^{-\frac{(x-a)^2}{4y}}$ then

$$f_{xy}(x, y) = f_{yx}(x, y).$$

Solution. $f(x, y) = \frac{1}{\sqrt{y}} e^{-\frac{(x-a)^2}{4y}} \quad \dots(1)$

Differentiating $f(x, y)$ partially w.r.t. x , we get

$$f_x(x, y) = \frac{1}{\sqrt{y}} \cdot \frac{[-2(x-a)]}{4y} e^{-\frac{(x-a)^2}{4y}} = \frac{-(x-a)}{2y^{3/2}} e^{-\frac{(x-a)^2}{4y}}$$

Differentiating again partially w.r.t. 'y' by product rule, we have

$$\begin{aligned} f_{yx}(x, y) &= \frac{3(x-a)}{4y^{5/2}} e^{-\frac{(x-a)^2}{4y}} - \frac{(x-a)^3}{8y^{7/2}} e^{-\frac{(x-a)^2}{4y}} \\ &= \frac{(x-a)}{8y^{7/2}} e^{-\frac{(x-a)^2}{4y}} [6y - (x-a)^2] \quad \dots(2) \end{aligned}$$

Differentiating (1) partially w.r.t. 'y', we have

$$f_y(x, y) = -\frac{1}{2y^{3/2}} e^{-\frac{(x-a)^2}{4y}} + \frac{(x-a)^2}{4y^{5/2}} e^{-\frac{(x-a)^2}{4y}}$$

Differentiating again partially w.r.t. 'x', we have

$$\begin{aligned}
 f_{xy}(x, y) &= \frac{(x-a)}{4y^{5/2}} e^{-\frac{(x-a)^2}{4y}} + \frac{(x-a)}{2y^{5/2}} e^{-\frac{(x-a)^2}{4y}} - \frac{(x-a)^3}{8y^{7/2}} e^{-\frac{(x-a)^2}{4y}} \\
 &= \frac{(x-a)}{8y^{7/2}} e^{-\frac{(x-a)^2}{4y}} [2y + 4y - (x-a)^2] \\
 &= \frac{(x-a)}{8y^{7/2}} e^{-\frac{(x-a)^2}{4y}} [6y - (x-a)^2] \quad \dots(3)
 \end{aligned}$$

From (2) and (3), we have $f_{xy}(x, y) = f_{yx}(x, y)$

Proved.

Example 17. If $u = x^y$, show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$.

Solution.

$$u = x^y$$

$$\Rightarrow \log u = \log x^y = y \log x$$

Differentiating partially, we have

$$\begin{aligned}
 \frac{1}{u} \cdot \frac{\partial u}{\partial x} &= \frac{y}{x}, \quad \text{and} \quad \frac{1}{u} \cdot \frac{\partial u}{\partial y} = \log x \\
 \Rightarrow \frac{\partial u}{\partial x} &= u \frac{y}{x}, \quad \Rightarrow \frac{\partial u}{\partial y} = u \log x \\
 \frac{\partial^2 u}{\partial y \partial x} &= \frac{1}{x} \left[u + y \cdot \frac{\partial u}{\partial y} \right] = \frac{u}{x} + \frac{y}{x} \cdot \frac{\partial u}{\partial y} = \frac{u}{x} + \frac{uy \log x}{x} \quad \left(\text{As } \frac{\partial u}{\partial y} = u \log x \right) \\
 \frac{\partial^3 u}{\partial x \partial y \partial x} &= -\frac{u}{x^2} + \frac{1}{x} \cdot \frac{\partial u}{\partial x} + y \cdot \left\{ \frac{x \cdot \left(\frac{\partial u}{\partial x} \cdot \log x + \frac{u}{x} \right) - u \log x}{x^2} \right\} \\
 &= -\frac{u}{x^2} + \frac{1}{x} \frac{\partial u}{\partial x} + \frac{y \log x}{x} \frac{\partial u}{\partial x} + \frac{uy}{x^2} - \frac{uy \log x}{x^2} \\
 &= -\frac{u}{x^2} + \frac{uy}{x^2} + \frac{uy^2 \log x}{x^2} + \frac{uy}{x^2} - \frac{uy \log x}{x^2} \\
 &= -\frac{u}{x^2} + \frac{2uy}{x^2} + \frac{uy^2 \log x}{x^2} - \frac{uy \log x}{x^2} \quad \dots(1) \\
 \frac{\partial u}{\partial y} &= u \log x \\
 \frac{\partial^2 u}{\partial x \partial y} &= \frac{u}{x} + \log x \cdot \frac{\partial u}{\partial x} = \frac{u}{x} + \log x \cdot \frac{uy}{x} \quad \left[\frac{\partial u}{\partial x} = \frac{uy}{x} \right] \\
 \frac{\partial^3 u}{\partial x^2 \partial y} &= -\frac{u}{x^2} + \frac{1}{x} \cdot \frac{\partial u}{\partial x} + y \cdot \left\{ \frac{x \cdot \left(\frac{u}{x} + \log x \cdot \frac{\partial u}{\partial x} \right) - u \log x}{x^2} \right\} \quad (1) \\
 &= -\frac{u}{x^2} + \frac{uy}{x^2} + \frac{uy}{x^2} + \frac{y \log x}{x} \frac{\partial u}{\partial x} - \frac{uy \log x}{x^2} \\
 &= -\frac{u}{x^2} + \frac{2uy}{x^2} + \frac{y \log x}{x} \frac{uy}{x} - \frac{uy \log x}{x^2}
 \end{aligned}$$

$$= -\frac{u}{x^2} + \frac{2uy}{x^2} + \frac{uy^2 \log x}{x^2} - \frac{uy \log x}{x^2} \quad \dots(2)$$

From (1) and (2), we get $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$ **Proved.**

Example 18. If $u = \log (x^3 + y^3 + z^3 - 3xyz)$, show that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = -\frac{9}{(x + y + z)^2} \quad (U.P. I Semester, winter 2003)$$

Solution. $u = \log (x^3 + y^3 + z^3 - 3xyz)$...(1)

Differentiating (1) partially w.r.t. 'x', we get

$$\frac{\partial u}{\partial x} = \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(2)$$

Similarly, $\frac{\partial u}{\partial y} = \frac{3y^2 - 3zx}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(3)$

$$\frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(4)$$

On adding (2), (3) and (4), we get

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz} \\ &= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)} = \frac{3}{x + y + z} \end{aligned}$$

$$\begin{aligned} \Rightarrow \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u &= \frac{3}{x + y + z} \\ \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \frac{3}{x + y + z} \\ &= \frac{\partial}{\partial x} \frac{3}{x + y + z} + \frac{\partial}{\partial y} \frac{3}{x + y + z} + \frac{\partial}{\partial z} \frac{3}{x + y + z} \\ &= -3(x + y + z)^{-2} - 3(x + y + z)^{-2} - 3(x + y + z)^{-2} \\ &= -\frac{9}{(x + y + z)^2} \quad \text{Proved.} \end{aligned}$$

1.9 WHICH VARIABLE IS TO BE TREATED AS CONSTANT

Let $x = r \cos \theta$, $y = r \sin \theta$

To find $\frac{\partial r}{\partial x}$, we need a relation between r and x .

$$r = x \sec \theta \quad \dots(1)$$

Differentiating (1) w.r.t. 'x' keeping θ as constant

$$\frac{\partial r}{\partial x} = \sec \theta \quad \dots(2)$$

Here, we have $r^2 = x^2 + y^2$...(3)

Differentiating (3) w.r.t. 'x' keeping y as constant.

$$2r \frac{\partial r}{\partial x} = 2x \quad \text{or} \quad \frac{\partial r}{\partial x} = \frac{x}{r} = \cos \theta \quad \dots(4)$$

From (2), $\frac{\partial r}{\partial x} = \sec \theta$ and from (4), $\frac{\partial r}{\partial x} = \cos \theta$. These two values of $\frac{\partial r}{\partial x}$ make confusion. To avoid the confusion we use the following notations.

Notation. (i) $\left(\frac{\partial r}{\partial x}\right)_\theta$ means the partial derivative of r with respect to x , keeping θ as constant.

From (3), $\left(\frac{\partial r}{\partial x}\right)_\theta = \sec \theta$

(ii) $\left(\frac{\partial r}{\partial x}\right)_y$ means the partial derivative of r with respect to x keeping y as constant.

From (4), $\left(\frac{\partial r}{\partial x}\right)_y = \cos \theta$

(iii) When no indication is given regarding the variables to be treated as constant

$$\frac{\partial}{\partial x} \text{ means } \left(\frac{\partial}{\partial x}\right)_y, \quad \frac{\partial}{\partial y} \text{ means } \left(\frac{\partial}{\partial y}\right)_x.$$

$$\frac{\partial}{\partial r} \text{ means } \left(\frac{\partial}{\partial r}\right)_\theta, \quad \frac{\partial}{\partial \theta} \text{ means } \left(\frac{\partial}{\partial \theta}\right)_r.$$

Example 19. If $x = r \cos \theta$, $y = r \sin \theta$, find

$$(i) \left(\frac{\partial x}{\partial r}\right)_\theta \quad (ii) \left(\frac{\partial y}{\partial \theta}\right)_r \quad (iii) \left(\frac{\partial r}{\partial x}\right)_y \quad (iv) \left(\frac{\partial \theta}{\partial y}\right)_x$$

Solution. (i) $\left(\frac{\partial x}{\partial r}\right)_\theta$ means the partial derivative of x with respect to r , keeping θ as constant.

$$x = r \cos \theta \quad \left(\frac{\partial x}{\partial r}\right)_\theta = \cos \theta$$

(ii) $\left(\frac{\partial y}{\partial \theta}\right)_r$ means the partial derivative of y with respect to θ , treating r as constant.

$$y = r \sin \theta \quad \left(\frac{\partial y}{\partial \theta}\right)_r = r \cos \theta$$

(iii) $\left(\frac{\partial r}{\partial x}\right)_y$ means the partial derivative of r with respect to x , treating y as constant. We have to express r as a function of x and y .

$$r = \sqrt{x^2 + y^2} \quad (\text{From the given equations})$$

$$\left(\frac{\partial r}{\partial x}\right)_y = \frac{1}{2} \frac{1}{\sqrt{x^2 + y^2}} \cdot 2x = \frac{x}{\sqrt{x^2 + y^2}}$$

(iv) Before finding $\left(\frac{\partial \theta}{\partial y}\right)_x$ we have to express θ in terms of x and y .

$$\theta = \tan^{-1} \frac{y}{x} \quad (\text{From the given equations})$$

$$\left(\frac{\partial \theta}{\partial y}\right)_x = \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

Ans.

EXERCISE 1.3

1. If $z^3 - 3yz - 3x = 0$, show that $z \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$; $z \left[\frac{\partial^2 z}{\partial x \partial y} + \left(\frac{\partial z}{\partial x} \right)^2 \right] = \frac{\partial^2 z}{\partial y^2}$
2. If $z(z^2 + 3x) + 3y = 0$, prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{2z(x-1)}{(z^2+x)^3}$.
3. If $z = \log(e^x + e^y)$, show that $rt - s^2 = 0$.
4. If $f(x, y) = x^3y - xy^3$, find $\left[\frac{1}{\frac{\partial f}{\partial x}} + \frac{1}{\frac{\partial f}{\partial y}} \right]_{x=1, y=2}$ Ans. $-\frac{13}{22}$
5. If $\theta = t^n e^{\frac{r^2}{4t}}$, find what value of n will make $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$. Ans. $n = -\frac{3}{2}$
6. Show that the function $u = \arctan(y/x)$ satisfies the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.
7. If $z = y f(x^2 - y^2)$ show that $y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} = \frac{xz}{y}$.
8. Show that $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$, where $z = x \cdot f(x+y) + y \cdot g(x+y)$.
9. If $u = \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$. Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.
10. If $u(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$, find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$. Ans. $\frac{2}{(x^2 + y^2 + z^2)^2}$
11. If $x = e^r \cos \theta$ and $y = e^r \sin \theta$
 Prove that $\frac{\partial x}{\partial r} = \frac{1}{r} \frac{\partial y}{\partial \theta}$, $\frac{\partial y}{\partial r} = -\frac{1}{r} \frac{\partial x}{\partial \theta}$
 Hence deduce that $\frac{\partial^2 x}{\partial r^2} + \frac{1}{r} \frac{\partial x}{\partial r} + \frac{1}{r} \frac{\partial^2 x}{\partial \theta^2} = 0$
12. If $x = r \cos \theta$, $y = r \sin \theta$, prove that
 (i) $\frac{\partial r}{\partial x} = \frac{\partial x}{\partial r}$, $r \cdot \frac{\partial \theta}{\partial x} = \frac{1}{r} \cdot \frac{\partial x}{\partial \theta}$ (ii) $\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$ (c) $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r} \left[\left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 \right]$
13. If $v = (x^2 - y^2) \cdot f(xy)$, prove that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = (x^4 - y^4) f''(xy)$
14. If $ux + vy = 0$ and $\frac{u}{x} + \frac{v}{y} = 1$, show that $\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = \frac{x^2 + y^2}{y^2 - x^2}$
15. If $z = x^y + y^x$, verify that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$
16. If $u = f(ax^2 + 2hxy + by^2)$ and $v = \phi(ax^2 + 2hxy + by^2)$ show that $\frac{\partial}{\partial y} \left(u \frac{\partial v}{\partial x} \right) = \frac{\partial}{\partial x} \left(u \frac{\partial v}{\partial y} \right)$.
17. If $u = r^m$, where $r^2 = x^2 + y^2 + z^2$, find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$. Ans. $m(m+1)r^{m-2}$
18. If $x = \frac{r}{2}(e^\theta + e^{-\theta})$, $y = \frac{r}{2}(e^\theta - e^{-\theta})$ prove that $\frac{\partial x}{\partial r} = \frac{\partial r}{\partial x}$

19. If $u(x, t) = a e^{-gx} \sin(nt - gx)$, satisfies the equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$, show that $ag = \sqrt{\frac{n}{2}}$.

20. If $u = \log(\tan x + \tan y)$, prove that, $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} = 2$.

21. If $u = \frac{1}{x}[(x - y) + \psi(x + y)]$, then show that $\frac{\partial}{\partial x} \left(\frac{x^2 \partial u}{\partial x} \right) = x^2 \frac{\partial^2 u}{\partial y^2}$

22. If $u = e^{xyz} f\left(\frac{xz}{y}\right)$, prove that

$$(i) \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2xyz u, \quad (ii) \quad y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2xyz u$$

$$\text{Also deduce that } x \frac{\partial^2 u}{\partial z \partial x} = y \frac{\partial^2 u}{\partial z \partial y}$$

(A.M.I.E., Summer 2001)

23. If $u = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, then show that

$$\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2$$

1.10 HOMOGENEOUS FUNCTION

A function $f(x, y)$ is said to be homogeneous function in which the power of each term is the same.

A function $f(x, y)$ is a homogeneous function of order n , if the degree of each of its terms in x and y is equal to n . Thus

$$a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \dots + a_{n-1} x y^{n-1} + a_n y^n \quad \dots(1)$$

is a homogeneous function of order n .

The polynomial function (1) which can be written as

$$x^n \left[a_0 + a_1 \left(\frac{y}{x} \right) + a_2 \left(\frac{y}{x} \right)^2 + \dots + a_{n-1} \left(\frac{y}{x} \right)^{n-1} + a_n \left(\frac{y}{x} \right)^n \right] = x^n \phi \left(\frac{y}{x} \right) \quad \dots(2)$$

(i) The function $x^3 \left[1 + \frac{y}{x} + 3 \left(\frac{y}{x} \right)^2 + 5 \left(\frac{y}{x} \right)^3 \right]$ is a homogeneous function of order 3.

(ii) $\frac{\sqrt{x} + \sqrt{y}}{x^2 + y^2} = \frac{\sqrt{x} \left[1 + \sqrt{\frac{y}{x}} \right]}{x^2 \left[1 + \left(\frac{y}{x} \right)^2 \right]} = x^{-3/2} \cdot \frac{1 + \sqrt{\frac{y}{x}}}{1 + \left(\frac{y}{x} \right)^2}$ is a homogeneous function of order $-3/2$.

(iii) $\sin^{-1} \frac{\sqrt{x} + \sqrt{y}}{x^2 + y^2}$ is not a homogeneous function as it cannot be written in the form of

$x^n f\left(\frac{y}{x}\right)$ so that its degree may be pronounced. It is a function of homogeneous expression.

1.11 EULER'S THEOREM ON HOMOGENEOUS FUNCTION

(U.P. I Semester, Dec. 2006)

Statement. If z is a homogeneous function of x, y of order n , then

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z$$

Proof. Since z is a homogeneous function of x, y of order n .

$\therefore z$ can be written in the form

$$z = x^n \cdot f\left(\frac{y}{x}\right) \quad \dots(1)$$

Differentiating (1) partially w.r.t. 'x', we have

$$\begin{aligned} \frac{\partial z}{\partial x} &= nx^{n-1} \cdot f\left(\frac{y}{x}\right) + x^n \cdot f'\left(\frac{y}{x}\right) \left(-\frac{y}{x^2}\right) \\ \Rightarrow \frac{\partial z}{\partial x} &= nx^{n-1} \cdot f\left(\frac{y}{x}\right) - x^{n-2} y \cdot f'\left(\frac{y}{x}\right) \end{aligned}$$

Multiplying both sides by x , we have

$$x \frac{\partial z}{\partial x} = n x^n \cdot f\left(\frac{y}{x}\right) - x^{n-1} y \cdot f'\left(\frac{y}{x}\right) \quad \dots(2)$$

Differentiating (1) partially w.r.t. 'y', we have

$$\frac{\partial z}{\partial y} = x^n f'\left(\frac{y}{x}\right) \cdot \frac{1}{x}$$

Multiplying both sides by y , we get

$$y \cdot \frac{\partial z}{\partial y} = x^{n-1} y \cdot f'\left(\frac{y}{x}\right) \quad \dots(3)$$

Adding (2) and (3), we have

$$\begin{aligned} x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y} &= n x^n f\left(\frac{y}{x}\right) \\ \Rightarrow x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y} &= n z \end{aligned} \quad \text{Proved.}$$

Note. If u is a homogeneous function of x, y, z of degree n , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

I. Deduction from Euler's theorem

If z is a homogeneous function of x, y of degree n and $z = f(u)$, then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)} \quad (\text{Nagpur University, Winter 2003})$$

Proof. Since z is a homogeneous function of x, y of degree n , we have, by Euler's theorem,

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz \quad \dots(1)$$

Now $z = f(u)$, given

$$\therefore \frac{\partial z}{\partial x} = f'(u) \frac{\partial u}{\partial x} \quad \text{and} \quad \frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y}$$

Substituting in (1), we get

$$x f'(u) \frac{\partial u}{\partial x} + y f'(u) \frac{\partial u}{\partial y} = n f(u)$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)}$$

Note. If $v = f(u)$ where v is a homogeneous function in x, y, z of degree n , then

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{n f(u)}{f'(u)}$$

Example 20. Verify Euler's theorem for $z = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$. (U.P. Ist Semester, Dec. 2009)

Solution. Here, we have

$$z = \frac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}} = \frac{x^{1/3} \left[1 + \left(\frac{y}{x} \right)^{1/3} \right]}{x^{1/2} \left[1 + \left(\frac{y}{x} \right)^{1/2} \right]} = x^{-1/6} \phi \left(\frac{y}{x} \right) \quad \dots(1)$$

Thus z is homogeneous function of degree $-\frac{1}{6}$.

$$\text{By Euler's theorem } x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = -\frac{1}{6} z. \quad \dots(2)$$

Differentiating (1) w.r.t. 'x', we get

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right) \left(\frac{1}{3} x^{-\frac{2}{3}} \right) - \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right) \left(\frac{1}{2} x^{-\frac{1}{2}} \right)}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} = \frac{\frac{1}{3} x^{-\frac{1}{6}} + \frac{1}{3} x^{-\frac{2}{3}} y^{\frac{1}{2}} - \frac{1}{2} x^{-\frac{1}{6}} - \frac{1}{2} x^{-\frac{1}{2}} y^{\frac{1}{3}}}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \\ x \frac{\partial z}{\partial x} &= \frac{\frac{1}{3} x^{\frac{5}{6}} + \frac{1}{3} x^{\frac{1}{3}} y^{\frac{1}{2}} - \frac{1}{2} x^{\frac{5}{6}} - \frac{1}{2} x^{\frac{1}{2}} y^{\frac{1}{3}}}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \quad \dots(3) \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= \frac{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right) \left(\frac{1}{3} y^{-\frac{2}{3}} \right) - \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right) \left(\frac{1}{2} y^{-\frac{1}{2}} \right)}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} = \frac{\frac{1}{3} x^{\frac{1}{2}} y^{-\frac{2}{3}} + \frac{1}{3} y^{-\frac{1}{6}} - \frac{1}{2} x^{\frac{1}{3}} y^{-\frac{1}{2}} - \frac{1}{2} y^{-\frac{1}{6}}}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \\ \Rightarrow y \frac{\partial z}{\partial y} &= \frac{\frac{1}{3} x^{\frac{1}{2}} y^{\frac{1}{3}} + \frac{1}{3} y^{\frac{5}{6}} - \frac{1}{2} x^{\frac{1}{3}} y^{\frac{1}{2}} - \frac{1}{2} y^{\frac{5}{6}}}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \quad \dots(4) \end{aligned}$$

Adding (3) and (4), we get

$$\begin{aligned} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= \frac{\frac{1}{3} x^{\frac{5}{6}} + \frac{1}{3} x^{\frac{1}{3}} y^{\frac{1}{2}} - \frac{1}{2} x^{\frac{5}{6}} - \frac{1}{2} x^{\frac{1}{2}} y^{\frac{1}{3}} + \frac{1}{3} x^{\frac{1}{2}} y^{\frac{1}{3}} + \frac{1}{3} y^{\frac{5}{6}} - \frac{1}{2} x^{\frac{1}{3}} y^{\frac{1}{2}} - \frac{1}{2} y^{\frac{5}{6}}}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \\ &= \frac{-\frac{1}{6} \left[x^{\frac{5}{6}} + y^{\frac{5}{6}} + x^{\frac{1}{3}} y^{\frac{1}{2}} + x^{\frac{1}{2}} y^{\frac{1}{3}} \right]}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} = \frac{-\frac{1}{6} \left[x^{\frac{1}{2}} \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right) + y^{\frac{1}{2}} \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right) \right]}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} \end{aligned}$$

$$= \frac{-\frac{1}{6} \left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right) \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right)}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)^2} = -\frac{1}{6} \frac{x^{\frac{1}{3}} + y^{\frac{1}{3}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}}$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = -\frac{1}{6} z \quad \dots(5)$$

From (2) and (5), Euler's theorem is verified.

Verified.

Example 21. If $u = \cos^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$, show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0. \quad (U.P. Ist Semester, Dec. 2009)$$

Solution. Here, we have, $u = \cos^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$

u is not a homogeneous function but if $z = \cos u$, then

$$u = \cos^{-1} z = \frac{x+y}{\sqrt{x} + \sqrt{y}} = \frac{x \left(1 + \frac{y}{x} \right)}{\sqrt{x} \left(1 + \sqrt{\frac{y}{x}} \right)} = x^{\frac{1}{2}} \frac{\left(1 + \frac{y}{x} \right)}{\left(1 + \sqrt{\frac{y}{x}} \right)} = x^{\frac{1}{2}} \phi \left(\frac{y}{x} \right).$$

z is a homogeneous function in x, y of degree $\frac{1}{2}$.

By Euler's theorem, we have $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{2} z$

$$x \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + y \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} = \frac{1}{2} z$$

$$x \frac{\partial u}{\partial x} (-\sin u) + y \frac{\partial u}{\partial y} (-\sin u) = \frac{1}{2} \cos u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \cot u. \quad \Rightarrow \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0 \quad \text{Proved.}$$

Example 22. If $u = \sin^{-1} \left[\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right]$, show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0. \quad (U.P. I Sem., Winter 2003)$$

Solution. We have, $u = \sin^{-1} \left[\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}} \right]$

Here, u is not a homogeneous function but if $v = \sin u = \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$ then v is a homogeneous function in x, y, z of degree -3 .

By Euler's Theorem $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = n v$

$$x \frac{\partial v}{\partial u} \frac{\partial u}{\partial x} + y \frac{\partial v}{\partial u} \frac{\partial u}{\partial y} + z \frac{\partial v}{\partial u} \frac{\partial u}{\partial z} = -3 v \quad \dots(1)$$

Putting the value of $\frac{\partial v}{\partial u}$ in (1), we get

$$x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} + z \cos u \frac{\partial u}{\partial z} = -3 \sin u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = -3 \frac{\sin u}{\cos u} = -3 \tan u$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0 \quad \text{Proved.}$$

Example 23. If $u = \log_e \left(\frac{x^4 + y^4}{x + y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$.

(Nagpur University, Summer 2008, Uttarakhand, I Semester 2008)

Solution. We have, $u = \log_e \left(\frac{x^4 + y^4}{x + y} \right)$

Here, u is not a homogeneous function but if

$$z = e^u = \frac{x^4 + y^4}{x + y} = \frac{x^4 \left[1 + \left(\frac{y}{x} \right)^4 \right]}{x \left[1 + \left(\frac{y}{x} \right) \right]} = x^3 \phi \left(\frac{y}{x} \right)$$

Then z is a homogeneous function of degree 3.

By Euler's Deduction formula I

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \frac{f(u)}{f'(u)} = 3 \frac{e^u}{e^u} = 3 \quad \text{Proved.}$$

Example 24. If $f(x, y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2}$, prove that

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2f = 0. \quad (A.M.I.E. Summer 2004)$$

Solution.

$$\begin{aligned} f(x, y) &= \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2 + y^2} \\ &= \frac{1}{x^2} \left(\frac{y}{x} \right)^0 + \frac{1}{x^2} \left(\frac{y}{x} \right) - \frac{1}{x^2} \frac{\log \frac{y}{x}}{\left[1 + \left(\frac{y}{x} \right)^2 \right]} \end{aligned}$$

$f(x, y)$ is a homogeneous function of degree -2 .

By Euler's Theorem

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = -2f \Rightarrow x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + 2f = 0 \quad \text{Proved.}$$

Example 25. If z be a homogeneous function of degree n , show that

$$(i) \quad x \cdot \frac{\partial^2 z}{\partial x^2} + y \cdot \frac{\partial^2 z}{\partial x \partial y} = (n-1) \frac{\partial z}{\partial x} \quad (ii) \quad x \cdot \frac{\partial^2 z}{\partial x \partial y} + y \cdot \frac{\partial^2 z}{\partial y^2} = (n-1) \frac{\partial z}{\partial y}$$

$$(iii) \quad x^2 \cdot \frac{\partial^2 z}{\partial x^2} + 2xy \cdot \frac{\partial^2 z}{\partial x \partial y} + y^2 \cdot \frac{\partial^2 z}{\partial y^2} = n(n-1)z. \quad (\text{Uttarakhand Ist Semester, Dec. 2006})$$

Solution. By Euler's Theorem $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z$... (1)

Differentiating (1), partially w.r.t. 'x', we get

$$\frac{\partial z}{\partial x} + x \frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 z}{\partial x \partial y} = n \frac{\partial z}{\partial x}$$

$$\Rightarrow x \frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 z}{\partial x \partial y} = (n-1) \frac{\partial z}{\partial x} \quad \text{Proved (i) ... (2)}$$

Differentiating (1), partially w.r.t. 'y', we have

$$x \frac{\partial^2 z}{\partial y \partial x} + \frac{\partial z}{\partial y} + y \frac{\partial^2 z}{\partial y^2} = n \frac{\partial z}{\partial y}$$

$$\Rightarrow x \frac{\partial^2 z}{\partial x \partial y} + y \frac{\partial^2 z}{\partial y^2} = (n-1) \frac{\partial z}{\partial y} \quad \text{Proved (ii) ... (3)}$$

Multiplying (2) by x, we have

$$x^2 \frac{\partial^2 z}{\partial x^2} + xy \frac{\partial^2 z}{\partial x \partial y} = (n-1) x \frac{\partial z}{\partial x} \quad \text{... (4)}$$

Multiplying (3) by y, we have

$$xy \frac{\partial^2 z}{\partial y \partial x} + y^2 \frac{\partial^2 z}{\partial y^2} = (n-1) y \frac{\partial z}{\partial y} \quad \text{... (5)}$$

Adding (4) and (5), we get

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = (n-1) \left(x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} \right)$$

$$= (n-1) n z \quad \text{[From (1)]}$$

$$= n(n-1) z \quad \text{Proved (iii)}$$

Example 26. If $f(x, y)$ and $\phi(x, y)$ are homogeneous functions of x, y of degree p and q respectively and $u = f(x, y) + \phi(x, y)$, show that

$$f(x, y) = \frac{1}{P(P-q)} \left[x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \right] - \frac{q-1}{P(P-q)} \left[x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right]$$

(A.M.I.E.T.E. Winter 2000)

Solution. Since f and ϕ are homogeneous functions of degree p and q respectively, we have

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = P f \quad \text{... (1)}$$

$$x \frac{\partial \phi}{\partial x} + y \frac{\partial \phi}{\partial y} = q \phi \quad \text{... (2)}$$

On adding (1) and (2), we get

$$x \left[\frac{\partial f}{\partial x} + \frac{\partial \phi}{\partial x} \right] + y \left[\frac{\partial f}{\partial y} + \frac{\partial \phi}{\partial y} \right] = P f + q \phi$$

i.e., $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = P f + q \phi \quad \text{... (3)}$

Also $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = P(P-1) f \quad \text{... (4)}$

$$\begin{aligned} u &= f(x, y) + \phi(x, y) \\ \frac{\partial u}{\partial x} &= \frac{\partial f}{\partial x} + \frac{\partial \phi}{\partial x} \\ \frac{\partial u}{\partial y} &= \frac{\partial f}{\partial y} + \frac{\partial \phi}{\partial y} \end{aligned}$$

And
$$x^2 \frac{\partial^2 \phi}{\partial x^2} + 2xy \frac{\partial^2 \phi}{\partial x \partial y} + y^2 \frac{\partial^2 \phi}{\partial y^2} = q(q-1)\phi \quad \dots(5)$$

On adding (4) and (5), we obtain

$$\begin{aligned} x^2 \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 \phi}{\partial x^2} \right) + 2xy \left(\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 \phi}{\partial x \partial y} \right) + y^2 \left(\frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 \phi}{\partial y^2} \right) &= P(P-1)f + q(q-1)\phi \\ x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= P(P-1)f + q(q-1)\phi \end{aligned} \quad \left| \begin{array}{l} \therefore u = f + \phi \\ \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 \phi}{\partial x^2} \\ \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 \phi}{\partial y^2} \end{array} \right.$$

Dividing by $P(P-q)$, we get

$$\Rightarrow \frac{1}{P(P-q)} \left[x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \right] = \frac{1}{P(P-q)} [P(P-1)f + q(q-1)\phi]$$

Subtracting $\frac{q-1}{P(P-q)} \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right)$ from both sides, we get

$$\begin{aligned} \Rightarrow \frac{1}{P(P-q)} \left[x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \right] - \frac{(q-1)}{P(P-q)} \left[x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right] \\ = \frac{1}{P(P-q)} [P(P-1)f + q(q-1)\phi] - \frac{(q-1)}{P(P-q)} \left[x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right] \\ = \frac{1}{P(P-q)} [P(P-1)f + q(q-1)\phi - (q-1)[Pf + q\phi]] \quad [\text{From (3)}] \\ = \frac{1}{P(P-q)} [P^2 - P - Pq + P]f + (q^2 - q - q^2 + q)\phi \\ = \frac{1}{P(P-q)} [(P^2 - Pq)f] = \frac{P(P-q)}{P(P-q)} f = f(x, y) \quad \text{Proved.} \end{aligned}$$

Example 27. If $z = x^n f\left(\frac{y}{x}\right) + y^{-n} \phi\left(\frac{x}{y}\right)$ then prove that

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n^2 z. \quad (\text{Nagpur University, Summer 2003})$$

Solution.
$$z = x^n f\left(\frac{y}{x}\right) + y^{-n} \phi\left(\frac{x}{y}\right)$$

$$z = u + v \quad \dots(1)$$

where, $u = x^n f\left(\frac{y}{x}\right)$ and $v = y^{-n} \phi\left(\frac{x}{y}\right)$

Since u is a homogeneous function of x, y of degree n .

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad \dots(2)$$

$$\text{and } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u \quad \dots(3)$$

As v is a homogeneous function of x, y of degree $-n$.

$$x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = -nv \quad \dots(4)$$

and $x^2 \frac{\partial^2 v}{\partial x^2} + 2xy \frac{\partial^2 v}{\partial x \partial y} + y^2 \frac{\partial^2 v}{\partial y^2} = -n(-n-1)v = n(n+1)v$... (5)

On adding (2) and (4), we get

$$\begin{aligned} x \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} \right) + y \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right) &= nu - nv \\ x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= nu - nv \end{aligned} \quad \begin{aligned} z &= u + v \\ \frac{\partial z}{\partial x} &= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} \\ \frac{\partial z}{\partial y} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \end{aligned} \quad \dots (6) \quad [\text{From (1)}]$$

On adding (3) and (5), we get

$$\begin{aligned} x^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \right) + 2xy \left(\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 v}{\partial x \partial y} \right) + y^2 \left(\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 v}{\partial y^2} \right) &= n(n-1)u + n(n+1)v \\ x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} &= n(n-1)u + n(n+1)v \end{aligned} \quad \begin{aligned} \frac{\partial^2 z}{\partial x^2} &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \end{aligned} \quad \dots (7) \quad [\text{From (1)}]$$

On adding (6) and (7), we have

$$\begin{aligned} x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= n(n-1)u + n(n+1)v + nu - nv \\ &= nu(n-1+1) + nv(n+1-1) \\ &= n^2u + n^2v = n^2(u+v) = n^2z \end{aligned}$$

Proved.

II. Deduction: Prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) [g'(u) - 1] \quad (\text{Nagpur University, Winter 2003})$$

where, $g(u) = n \frac{f(u)}{f'(u)}$

Proof. By Euler's deduction formula I

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= n \frac{f(u)}{f'(u)} \quad \left(\text{Given } n \frac{f(u)}{f'(u)} = g(u) \right) \\ x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= g(u) \end{aligned} \quad \dots (1)$$

Differentiating (1) partially w.r.t. 'x', we have

$$\begin{aligned} \left(x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \cdot 1 \right) + y \frac{\partial^2 u}{\partial x \partial y} &= g'(u) \frac{\partial u}{\partial x} \\ \Rightarrow x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} &= [g'(u) - 1] \frac{\partial u}{\partial x} \end{aligned} \quad \dots (2)$$

Similarly, on differentiating (1) partially w.r.t. 'y', we have

$$y \frac{\partial^2 u}{\partial y^2} + x \frac{\partial^2 u}{\partial y \partial x} = [g'(u) - 1] \frac{\partial u}{\partial y} \quad \dots (3)$$

Multiplying (2) by x, (3) by y and adding, we get

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= [g'(u) - 1] \left[x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right] \\ &= [g'(u) - 1] g(u) \\ &= g(u) [g'(u) - 1] \end{aligned} \quad \begin{aligned} &[\text{From (1)}] \\ &\mathbf{Proved.} \end{aligned}$$

Example 28. If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, prove that

$$(i) \quad x \cdot \frac{\partial u}{\partial x} + y \cdot \frac{\partial u}{\partial y} = \sin 2u \quad (A.M.I.E., \text{ Winter } 2001)$$

$$(ii) \quad x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \cos 3u \sin u. \quad (M.U. 2009; \text{ Nagpur University, } 2002)$$

Solution. Here u is not a homogeneous function. We however write

$$z = \tan u = \frac{x^3 + y^3}{x - y} = \frac{x^3 \left[1 + \left(\frac{y}{x}\right)^3 \right]}{x \left[1 - \left(\frac{y}{x}\right) \right]} = x^2 \cdot \frac{1 + \left(\frac{y}{x}\right)^3}{1 - \left(\frac{y}{x}\right)} = x^2 \phi\left(\frac{y}{x}\right)$$

so that z is a homogeneous function of x, y of order 2.

(i) By Euler's Theorem

[Here $f(u) = \tan u$]

$$\begin{aligned} \therefore \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{n f(u)}{f'(u)} \quad \dots(1) \\ &= \frac{2 \tan u}{\sec^2 u} = \frac{2 \sin u \cos^2 u}{\cos u} = 2 \sin u \cos u = \sin 2u \end{aligned}$$

(ii) By deduction II

$$x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u)[g'(u) - 1]$$

Here

$$\sin 2u = g(u)$$

$$\begin{aligned} \therefore \quad x^2 \cdot \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= \sin 2u (2 \cos 2u - 1) = 2 \sin 2u \cos 2u - \sin 2u \\ &= \sin 4u - \sin 2u = 2 \cos 3u \sin u \quad \text{Proved.} \end{aligned}$$

Example 29. If $u = \sin^{-1}\left[\frac{x + y}{\sqrt{x} + \sqrt{y}}\right]$

Prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{-\sin u \cos 2u}{4 \cos^3 u}$.

Solution. We have, $u = \sin^{-1} \frac{x + y}{\sqrt{x} + \sqrt{y}}$

$$\text{Let } z = \sin u = \frac{x + y}{\sqrt{x} + \sqrt{y}} = \frac{x \left[1 + \frac{y}{x} \right]}{\sqrt{x} \left[1 + \sqrt{\frac{y}{x}} \right]} = x^{1/2} \phi\left(\frac{y}{x}\right)$$

$$z = f(u) = \sin u$$

z is a homogeneous function of degree $\frac{1}{2}$.

By Euler's deduction I

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= n \frac{f(u)}{f'(u)} \quad \Rightarrow \quad x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \frac{\sin u}{\cos u} \\ x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{1}{2} \tan u \end{aligned}$$

Let $g(u) = \frac{1}{2} \tan u$

By Euler's deduction II

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= g(u) [g'(u) - 1] = \frac{1}{2} \tan u \left(\frac{1}{2} \sec^2 u - 1 \right) \\ &= \frac{1}{4} \frac{\sin u}{\cos u} \left(\frac{1}{\cos^2 u} - 2 \right) = \frac{1}{4} \frac{\sin u}{\cos^3 u} (1 - 2 \cos^2 u) = \frac{-\sin u \cos 2u}{4 \cos^3 u} \end{aligned} \quad \text{Proved.}$$

EXERCISE 1.4

1. Verify Euler's theorem in case

(i) $f(x, y) = ax^2 + 2hxy + by^2$ (ii) $u = (\sqrt{x} + \sqrt{y})(x^n + y^n)$

2. If $v = \frac{x^3 y^3}{x^3 + y^3}$, show that $x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} = 3v$.

3. If $u = \log \frac{x^3 + y^3}{x^2 + y^2}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$.

4. If $z = (x^2 + y^2)/\sqrt{(x+y)}$, prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{3}{2}z$.

5. If $f(x, y) = x^4 y^2 \sin^{-1} \frac{y}{x}$, then find the value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$.
(A.M.I.E.T.E., Winter 2001) Ans. $6f(x, y)$

6. If $u = \sin^{-1} \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$, show that $\frac{\partial u}{\partial x} = -\frac{y}{x} \frac{\partial u}{\partial y}$.

7. If $u = \sec^{-1} \left(\frac{x^3 - y^3}{x + y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$, then evaluate

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \quad (\text{A.M.I.E.T.E., Winter 2001}) \quad \text{Ans. } -2 \cot u (2 \operatorname{cosec}^2 u + 1).$$

8. If $x = e^u \tan v$, $y = e^u \sec v$, find the value of

$$\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \left(x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} \right) \quad (\text{A.M.I.E., Summer 2001}) \quad \text{Ans. } 0$$

[Hint: Eliminate u and apply formula I. Again eliminate v and apply the formula]

9. If $u = \sin^{-1} \left[\frac{x^{1/4} + y^{1/4}}{x^{1/6} + y^{1/6}} \right]$, prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{1}{144} \tan u [\tan^2 u - 1].$$

10. Find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$ if $u = \sin^{-1} (x^3 + y^3)^{2/5}$.

Ans. $\frac{5}{6} \tan u \left(\frac{6}{5} \sec^2 u - 1 \right)$

11. If $u = \tan^{-1} \left(\frac{y^2}{x} \right)$, find

(i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$, and (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$ Ans. (i) $\frac{xy^2}{x^2 + y^4}$ (ii) $\frac{-2xy^6}{(x^2 + y^4)^2}$

12. If $u = \tan^{-1} \frac{\sqrt{x^3 + y^3}}{\sqrt{x} + \sqrt{y}}$, find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$ Ans. $-2 \sin^3 u \cos u$

13. If $u = f\left(\frac{y}{x}\right) + \sqrt{x^2 + y^2}$, find the value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$. Ans. 0

14. If $z = xy/(x + y)$, find the value of $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2}$. Ans. 0

15. Verify Euler's theorem on homogeneous function when $f(x, y, z) = 3x^2yz + 5xy^2z + 4z^4$

16. If $u = x\phi\left(\frac{Y}{X}\right) + \psi\left(\frac{Y}{X}\right)$, prove by Euler's theorem on homogeneous function that

$$X^2 \frac{\partial^2 u}{\partial x^2} + 2XY \frac{\partial^2 u}{\partial x \partial y} + Y^2 \frac{\partial^2 u}{\partial y^2} = 0.$$

17. Given $F(u) = V(x, y, z)$ where V is a homogeneous function of x, y, z of degree n , prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n \frac{F(u)}{F'(u)}$$

18. State and prove Euler's theorem, and verify for $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ (A.M.I.E., Summer 2000)

19. If $u = \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2} + \cos \frac{xy + yz}{x^2 + y^2 + z^2}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{4x^2 y^2 z^2}{x^2 + y^2 + z^2}$

1.12 TOTAL DIFFERENTIAL

In partial differentiation of a function of two or more variables, only one variable varies. But in total differentiation, increments are given in all the variables.

1.13 TOTAL DIFFERENTIAL CO-EFFICIENT

Let $z = f(x, y)$...(1)

If $\delta x, \delta y$ be the increments in x and y respectively, let δz be the corresponding increment in z .

Then $z + \delta z = f(x + \delta x, y + \delta y)$...(2)

Subtracting (1) from (2), we have

$$\delta z = f(x + \delta x, y + \delta y) - f(x, y) \quad \text{...(3)}$$

Adding and subtracting $f(x, y + \delta y)$ on R.H.S. of (3), we have

$$\delta z = f(x + \delta x, y + \delta y) - f(x, y + \delta y) + f(x, y + \delta y) - f(x, y)$$

$$\delta z = \frac{f(x + \delta x, y + \delta y) - f(x, y + \delta y)}{\delta x} \delta x + \frac{f(x, y + \delta y) - f(x, y)}{\delta y} \delta y$$

On taking limit when $\delta x \rightarrow 0$ and $\delta y \rightarrow 0$

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \quad \text{...(4) [Remember]}$$

dz is called as the total differential of z .

1.14 CHANGE OF TWO INDEPENDENT VARIABLES x AND y BY ANY OTHER VARIABLE t .

Differentiation of composite function

If $z = f(x, y)$

Where $x = \phi(t)$

$$y = \psi(t)$$

Here z is composite function of t .

Dividing (4) by dt , we have

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad \dots(5) \quad \text{[Remember]}$$

Then $\frac{dz}{dt}$ is called the total differential co-efficient of z .

1.15 CHANGE IN THE INDEPENDENT VARIABLES x AND y BY OTHER TWO VARIABLES u AND v .

Let $z = f(x, y)$

where $x = \phi(u, v)$

$y = \psi(u, v)$

Then from (5), we obtain

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \dots(6)$$

$$\text{and} \quad \frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial v} \quad \dots(7)$$

Example 30. If $u = x^3 + y^3$ where, $x = a \cos t$, $y = b \sin t$, find $\frac{du}{dt}$ and verify the result.

Solution. We have, $u = x^3 + y^3$

$$x = a \cos t$$

$$y = b \sin t$$

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} \\ &= (3x^2)(-a \sin t) + (3y^2)(b \cos t) \\ &= -3a^3 \cos^2 t \sin t + 3b^3 \sin^2 t \cos t \end{aligned} \quad \dots(1)$$

Verification. $u = x^3 + y^3$

$$= a^3 \cos^3 t + b^3 \sin^3 t$$

$$\frac{du}{dt} = -3a^3 \cos^2 t \sin t + 3b^3 \sin^2 t \cos t \quad \dots(2)$$

Results (1) and (2) are the same.

Verified.

Example 31. If $z = f(x, y)$ where $x = e^u \cos v$ and $y = e^u \sin v$, show that

$$(i) \quad y \frac{\partial z}{\partial u} + x \frac{\partial z}{\partial v} = e^{2u} \frac{\partial z}{\partial y} \quad (M.U. 2009; Nagpur University 2002)$$

$$(ii) \quad \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = e^{-2u} \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right] \quad (M.U. 2009)$$

Solution. (i) We have,

$$x = e^u \cos v, \quad y = e^u \sin v$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} = \frac{\partial z}{\partial x} e^u \cos v + \frac{\partial z}{\partial y} e^u \sin v = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$$

$$y \frac{\partial z}{\partial u} = xy \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} \quad \dots(1)$$

And

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$\begin{aligned}
 &= \frac{\partial z}{\partial x} (-e^u \sin v) + \frac{\partial z}{\partial y} (e^u \cos v) = -y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} \\
 x \frac{\partial z}{\partial v} &= -xy \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} \quad \dots(2)
 \end{aligned}$$

On adding (1) and (2), we get

$$\begin{aligned}
 y \frac{\partial z}{\partial u} + x \frac{\partial z}{\partial v} &= (x^2 + y^2) \frac{\partial z}{\partial y} = (e^{2u} \cos^2 v + e^{2u} \sin^2 v) \frac{\partial z}{\partial y} \\
 &= e^{2u} (\cos^2 v + \sin^2 v) \frac{\partial z}{\partial y} = e^{2u} \frac{\partial z}{\partial y} \quad \text{Proved.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \\
 &= \frac{\partial z}{\partial x} (e^u \cos v) + \frac{\partial z}{\partial y} e^u \sin v \\
 e^{-u} \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cos v + \frac{\partial z}{\partial y} \sin v
 \end{aligned}
 \quad \left| \begin{array}{l} x = e^u \cos v \\ \Rightarrow \frac{\partial x}{\partial u} = e^u \cos v \text{ and } \frac{\partial x}{\partial v} = -e^u \sin v \\ y = e^u \sin v \\ \Rightarrow \frac{\partial y}{\partial u} = e^u \sin v \text{ and } \frac{\partial y}{\partial v} = e^u \cos v \end{array} \right.$$

On squaring, we get

$$e^{-2u} \left(\frac{\partial z}{\partial u} \right)^2 = \left(\frac{\partial z}{\partial x} \right)^2 \cos^2 v + \left(\frac{\partial z}{\partial y} \right)^2 \sin^2 v + 2 \left(\frac{\partial z}{\partial x} \right) \left(\frac{\partial z}{\partial y} \right) \sin v \cos v \quad \dots(3)$$

Again

$$\begin{aligned}
 \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} \\
 &= \frac{\partial z}{\partial x} (-e^u \sin v) + \frac{\partial z}{\partial y} (e^u \cos v) \\
 e^{-u} \left(\frac{\partial z}{\partial v} \right) &= -\frac{\partial z}{\partial x} \sin v + \frac{\partial z}{\partial y} \cos v
 \end{aligned}$$

On squaring, we get

$$e^{-2u} \left(\frac{\partial z}{\partial v} \right)^2 = \left(\frac{\partial z}{\partial x} \right)^2 \sin^2 v + \left(\frac{\partial z}{\partial y} \right)^2 \cos^2 v - 2 \left(\frac{\partial z}{\partial x} \right) \left(\frac{\partial z}{\partial y} \right) \sin v \cos v \quad \dots(4)$$

On adding (3) and (4), we get

$$\begin{aligned}
 e^{-2u} \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right] &= \left(\frac{\partial z}{\partial x} \right)^2 (\sin^2 v + \cos^2 v) + \left(\frac{\partial z}{\partial y} \right)^2 (\sin^2 v + \cos^2 v) \\
 &= \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 \quad \text{Proved.}
 \end{aligned}$$

Example 32. If $u = u(y - z, z - x, x - y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

(Nagpur University, Winter 2002, U.P., I Sem., Winter 2002, A.M.I.E winter 2001)

Solution. Let $r = y - z, \quad s = z - x, \quad t = x - y$
so that $u = u(r, s, t)$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x}$$

$$= \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}(-1) + \frac{\partial u}{\partial t}(1) = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \dots(1)$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y} \\ &= \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t}(-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \quad \dots(2) \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z} \\ &= \frac{\partial u}{\partial r}(-1) + \frac{\partial u}{\partial s}(1) + \frac{\partial u}{\partial t}(0) = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \dots(3) \end{aligned}$$

Adding (1), (2) and (3), we get $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Proved.

Example 33. If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$, show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$
(U. P. I. Sem., Dec. 2004, Nagpur University, Summer 2000)

Solution. Here, we have $u = u\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right) = u(r, s)$

where

$$\begin{aligned} r &= \frac{y-x}{xy}, & \text{and} & & s &= \frac{z-x}{zx} \\ r &= \frac{1}{x} - \frac{1}{y} & \text{and} & & s &= \frac{1}{x} - \frac{1}{z} \\ \frac{\partial r}{\partial x} &= -\frac{1}{x^2} & \text{and} & & \frac{\partial s}{\partial x} &= -\frac{1}{x^2} \\ \frac{\partial r}{\partial y} &= \frac{1}{y^2} & \text{and} & & \frac{\partial s}{\partial z} &= \frac{1}{z^2} \\ \frac{\partial r}{\partial z} &= 0 & \text{and} & & \frac{\partial s}{\partial y} &= 0 \end{aligned}$$

We know that,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} \\ &= \frac{\partial u}{\partial r} \left(-\frac{1}{x^2}\right) + \frac{\partial u}{\partial s} \left(-\frac{1}{x^2}\right) = -\frac{1}{x^2} \frac{\partial u}{\partial r} - \frac{1}{x^2} \frac{\partial u}{\partial s} \end{aligned}$$

$$\Rightarrow x^2 \frac{\partial u}{\partial x} = -\frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} \quad \dots(1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} = \frac{\partial u}{\partial r} \frac{1}{y^2} + \frac{\partial u}{\partial s} \times 0 = \frac{1}{y^2} \frac{\partial u}{\partial r}$$

$$\Rightarrow y^2 \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \quad \dots(2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} = \frac{\partial u}{\partial r} \times 0 + \frac{\partial u}{\partial s} \times \frac{1}{z^2} = \frac{1}{z^2} \frac{\partial u}{\partial s}$$

$$\Rightarrow z^2 \frac{\partial u}{\partial z} = \frac{\partial u}{\partial s} \quad \dots(3)$$

On adding (1), (2) and (3), we get

$$x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0 \quad \text{Proved.}$$

Example 34. If $\phi(cx - az, cy - bz) = 0$ show that $ap + bq = c$:

$$\text{where } p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}$$

Solution. Here, we have

$$\phi(cx - az, cy - bz) = 0$$

$$\phi(r, s) = 0$$

where

$$r = cx - az, \quad s = cy - bz$$

$$\frac{\partial r}{\partial x} = c - a \frac{\partial z}{\partial x}, \quad \frac{\partial r}{\partial y} = -a \frac{\partial z}{\partial y}$$

$$\frac{\partial s}{\partial x} = -b \frac{\partial z}{\partial x}, \quad \frac{\partial s}{\partial y} = c - b \frac{\partial z}{\partial y}$$

We know that,

$$\frac{\partial \phi}{\partial r} = \frac{\partial \phi}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial \phi}{\partial s} \frac{\partial s}{\partial x}$$

$$0 = \frac{\partial \phi}{\partial r} \left(c - a \frac{\partial z}{\partial x} \right) + \frac{\partial \phi}{\partial s} \left(-b \frac{\partial z}{\partial x} \right)$$

$$\Rightarrow 0 = c \frac{\partial \phi}{\partial r} + \frac{\partial z}{\partial x} \left(-a \frac{\partial \phi}{\partial r} - b \frac{\partial \phi}{\partial s} \right)$$

$$c \frac{\partial \phi}{\partial r} = \frac{\partial z}{\partial x} \left(a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s} \right) \Rightarrow a \frac{\partial z}{\partial x} = \frac{ac \frac{\partial \phi}{\partial r}}{a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s}} \quad \dots(1)$$

Again

$$\frac{\partial \phi}{\partial y} = \frac{\partial \phi}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial \phi}{\partial s} \frac{\partial s}{\partial y}$$

$$0 = \frac{\partial \phi}{\partial r} \left(-a \frac{\partial z}{\partial y} \right) + \frac{\partial \phi}{\partial s} \left(c - b \frac{\partial z}{\partial y} \right)$$

$$0 = c \frac{\partial \phi}{\partial s} - \frac{\partial z}{\partial y} \left(a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s} \right) \Rightarrow c \frac{\partial \phi}{\partial s} = \frac{\partial z}{\partial y} \left(a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s} \right)$$

$$\Rightarrow b \frac{\partial z}{\partial y} = \frac{bc \frac{\partial \phi}{\partial s}}{a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s}} \quad \dots(2)$$

Adding (1) and (2), we get

$$a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{ac \frac{\partial \phi}{\partial r} + bc \frac{\partial \phi}{\partial s}}{a \frac{\partial \phi}{\partial r} + b \frac{\partial \phi}{\partial s}}$$

$$\Rightarrow a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = c \quad \Rightarrow ap + bq = c \quad \text{Proved.}$$

1.16 CHANGE IN BOTH THE INDEPENDENT AND DEPENDENT VARIABLES, (POLAR COORDINATES)

Example 35. If $w = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, show that

$$\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2$$

Solution. Here, $x = r \cos \theta$, $y = r \sin \theta$

$$\begin{array}{l|l} \frac{\partial x}{\partial r} = \cos \theta & \frac{\partial y}{\partial r} = \sin \theta \\ \frac{\partial x}{\partial \theta} = -r \sin \theta & \frac{\partial y}{\partial \theta} = r \cos \theta \end{array}$$

Now,

$$\begin{aligned} \frac{\partial w}{\partial r} &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r} \\ \frac{\partial w}{\partial r} &= \frac{\partial f}{\partial x} \cdot (\cos \theta) + \frac{\partial f}{\partial y} \cdot (\sin \theta) \end{aligned} \quad \dots(1)$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta} = \frac{\partial f}{\partial x} \cdot (-r \sin \theta) + \frac{\partial f}{\partial y} \cdot (r \cos \theta)$$

$$\Rightarrow \frac{1}{r} \frac{\partial w}{\partial \theta} = -\frac{\partial f}{\partial x} \sin \theta + \frac{\partial f}{\partial y} \cos \theta \quad \dots(2)$$

Squaring (1) and (2) and adding, we obtain

$$\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 \quad \text{Proved.}$$

Example 36. Transform the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar co-ordinates.

Solution. We have, $x = r \cos \theta$, $y = r \sin \theta$

$$\begin{aligned} r^2 &= x^2 + y^2, \quad \theta = \tan^{-1} \frac{y}{x} \\ \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{\partial u}{\partial r} \frac{x}{r} + \frac{\partial u}{\partial \theta} \frac{-y}{x^2 + y^2} = \frac{\partial u}{\partial r} \cos \theta - \frac{\partial u}{\partial \theta} \frac{\sin \theta}{r} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) \\ &= \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \cos \theta \frac{\partial}{\partial r} \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \cos \theta \left(\cos \theta \frac{\partial^2 u}{\partial r^2} + \frac{\sin \theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \right) \\ &\quad - \frac{\sin \theta}{r} \left(-\sin \theta \frac{\partial u}{\partial r} + \cos \theta \frac{\partial^2 u}{\partial \theta \partial r} - \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 u}{\partial \theta^2} \right) \\ &= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\ &\quad + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial \theta \partial r} + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \end{aligned}$$

$$\begin{aligned}
&= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \quad \dots(1) \\
&\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{\partial u}{\partial r} \frac{y}{r} + \frac{\partial u}{\partial \theta} \frac{x}{x^2 + y^2} = \frac{\partial u}{\partial r} \sin \theta + \frac{\partial u}{\partial \theta} \frac{\cos \theta}{r} \\
&\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) \\
&= \sin \theta \frac{\partial}{\partial r} \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) \\
&= \sin \theta \left[\sin \theta \frac{\partial^2 u}{\partial r^2} - \frac{\cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \right] \\
&\quad + \frac{\cos \theta}{r} \left[\cos \theta \frac{\partial u}{\partial r} + \sin \theta \frac{\partial^2 u}{\partial \theta \partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} + \frac{\cos \theta}{r} \frac{\partial^2 u}{\partial \theta^2} \right] \\
&= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
&\quad + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial \theta \partial r} - \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} \\
&\quad + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} \quad \dots(2)
\end{aligned}$$

Adding (1) and (2), we get

$$\begin{aligned}
\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= (\sin^2 \theta + \cos^2 \theta) \frac{\partial^2 u}{\partial r^2} + (\sin^2 \theta + \cos^2 \theta) \frac{1}{r} \frac{\partial u}{\partial r} + (\sin^2 \theta + \cos^2 \theta) \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\
&= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \quad \text{Ans.}
\end{aligned}$$

Example 37. If $u = f(r)$ and $x = r \cos \theta$, $y = r \sin \theta$, prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r). \quad (\text{Nagpur University, Winter 2004})$$

(A.M.I.E.T.E., Winter 2003, U.P. I Semester, Winter 2005, 2000)

Solution. Here, we have

$$\begin{aligned}
x &= r \cos \theta \\
y &= r \sin \theta
\end{aligned}$$

$$r^2 = x^2 + y^2 \text{ so that } \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$u = f(r)$$

$$\frac{\partial u}{\partial x} = \frac{df}{dr} \cdot \frac{\partial r}{\partial x} \quad \Rightarrow \quad \frac{\partial u}{\partial x} = \frac{df}{dr} \cdot \frac{x}{r}$$

Differentiating again w.r.t. x , we get

$$\frac{\partial^2 u}{\partial x^2} = \left(\frac{d^2 f}{dr^2} \cdot \frac{\partial r}{\partial x} \right) \cdot \frac{x}{r} + \frac{df}{dr} \cdot \left[\frac{r \cdot 1 - x \frac{\partial r}{\partial x}}{r^2} \right] = \left(\frac{d^2 f}{dr^2} \cdot \frac{x}{r} \right) \cdot \frac{x}{r} + \frac{df}{dr} \cdot \left[\frac{r \cdot 1 - x \cdot \frac{x}{r}}{r^2} \right]$$

$$= \frac{d^2 f}{dr^2} \frac{x^2}{r^2} + \frac{df}{dr} \cdot \frac{r^2 - x^2}{r^3} = \frac{d^2 f}{dr^2} \frac{x^2}{r^2} + \frac{df}{dr} \frac{y^2}{r^3} \quad \dots(1)$$

Similarly,

$$\frac{\partial^2 u}{\partial y^2} = \frac{d^2 f}{dr^2} \frac{y^2}{r^2} + \frac{df}{dr} \frac{x^2}{r^3} \quad \dots(2)$$

On adding (1) and (2), we get

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= \frac{d^2 f}{dr^2} \frac{x^2 + y^2}{r^2} + \frac{df}{dr} \frac{x^2 + y^2}{r^3} \\ &= \frac{d^2 f}{dr^2} + \frac{df}{dr} \frac{1}{r} = f''(r) + \frac{1}{r} f'(r) \end{aligned} \quad \text{Proved.}$$

Example 38. A function $f(x, y)$ is rewritten in terms of new variables

$$u = e^x \cos y, \quad v = e^x \sin y$$

$$\text{Show that (i) } \frac{\partial f}{\partial x} = u \frac{\partial f}{\partial u} + v \frac{\partial f}{\partial v} \quad \text{and} \quad \text{(ii) } \frac{\partial f}{\partial y} = -v \frac{\partial f}{\partial u} + u \frac{\partial f}{\partial v}$$

and hence deduce that

$$\text{(iii) } \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = (u^2 + v^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right)$$

$$\text{Solution.} \quad u = e^x \cos y, \quad \frac{\partial u}{\partial x} = e^x \cos y = u, \quad \Rightarrow \quad \frac{\partial u}{\partial y} = -e^x \sin y = -v$$

$$v = e^x \sin y, \quad \frac{\partial v}{\partial x} = e^x \sin y = v, \quad \frac{\partial v}{\partial y} = e^x \cos y = u$$

$$\text{(i) We know that } \frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} u + \frac{\partial f}{\partial v} v \quad \dots (1) \quad \text{Proved.}$$

$$\text{(ii) } \frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} = \frac{\partial f}{\partial u} (-v) + \frac{\partial f}{\partial v} u = -v \frac{\partial f}{\partial u} + u \frac{\partial f}{\partial v} \quad \dots (2) \quad \text{Proved.}$$

$$\begin{aligned} \text{(iii) } \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \\ &= \left(u \cdot \frac{\partial}{\partial u} + v \cdot \frac{\partial}{\partial v} \right) \left(u \cdot \frac{\partial f}{\partial u} + v \cdot \frac{\partial f}{\partial v} \right) \quad [\text{From (1)}] \\ &= u \frac{\partial}{\partial u} \left(u \frac{\partial f}{\partial u} + v \frac{\partial f}{\partial v} \right) + v \frac{\partial}{\partial v} \left(u \frac{\partial f}{\partial u} + v \frac{\partial f}{\partial v} \right) \\ &= u \frac{\partial}{\partial u} \left(u \frac{\partial f}{\partial u} \right) + u \frac{\partial}{\partial u} \left(v \frac{\partial f}{\partial v} \right) + v \frac{\partial}{\partial v} \left(u \frac{\partial f}{\partial u} \right) + v \frac{\partial}{\partial v} \left(v \frac{\partial f}{\partial v} \right) \\ &= u \left(u \frac{\partial^2 f}{\partial u^2} + \frac{\partial f}{\partial u} \right) + u \left(v \frac{\partial^2 f}{\partial u \partial v} \right) + v \left(u \frac{\partial^2 f}{\partial v \partial u} \right) + v \left(v \frac{\partial^2 f}{\partial v^2} + \frac{\partial f}{\partial v} \right) \\ &= u^2 \frac{\partial^2 f}{\partial u^2} + u \frac{\partial f}{\partial u} + uv \frac{\partial^2 f}{\partial v \partial u} + uv \frac{\partial^2 f}{\partial u \partial v} + v^2 \frac{\partial^2 f}{\partial v^2} + v \frac{\partial f}{\partial v} \quad \dots(3) \end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \left(-v \frac{\partial}{\partial u} + u \frac{\partial}{\partial v} \right) \left(-v \frac{\partial f}{\partial u} + u \frac{\partial f}{\partial v} \right) \quad [\text{From (2)}] \\
&= -v \frac{\partial}{\partial u} \left(-v \frac{\partial f}{\partial u} + u \frac{\partial f}{\partial v} \right) + u \frac{\partial}{\partial v} \left(-v \frac{\partial f}{\partial u} + u \frac{\partial f}{\partial v} \right) \\
&= -v \frac{\partial}{\partial u} \left(-v \frac{\partial f}{\partial u} \right) - v \frac{\partial}{\partial u} \left(u \frac{\partial f}{\partial v} \right) + u \frac{\partial}{\partial v} \left(-v \frac{\partial f}{\partial u} \right) + u \frac{\partial}{\partial v} \left(u \frac{\partial f}{\partial v} \right) \\
&= -v \left(-v \frac{\partial^2 f}{\partial u^2} \right) - v \left(u \frac{\partial^2 f}{\partial u \partial v} + \frac{\partial f}{\partial v} \right) + u \left(-v \frac{\partial^2 f}{\partial v \partial u} - \frac{\partial f}{\partial u} \right) + u \left(u \frac{\partial^2 f}{\partial v^2} \right) \\
&= v^2 \frac{\partial^2 f}{\partial u^2} - uv \frac{\partial^2 f}{\partial u \partial v} - v \frac{\partial f}{\partial v} - uv \frac{\partial^2 f}{\partial u \partial v} - u \frac{\partial f}{\partial u} + u^2 \frac{\partial^2 f}{\partial v^2} \quad \dots(4)
\end{aligned}$$

On adding (3) and (4), we obtain

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} &= u^2 \frac{\partial^2 f}{\partial u^2} + v^2 \frac{\partial^2 f}{\partial v^2} + v^2 \frac{\partial^2 f}{\partial u^2} + u^2 \frac{\partial^2 f}{\partial v^2} \\
&= (u^2 + v^2) \frac{\partial^2 f}{\partial u^2} + (u^2 + v^2) \frac{\partial^2 f}{\partial v^2} = (u^2 + v^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right) \quad \text{Proved.}
\end{aligned}$$

Example 39. If $x + y = 2 e^\theta \cos \phi$ and $x - y = 2 i e^\theta \sin \phi$

Show that : $\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial \phi^2} = 4 x y \frac{\partial^2 V}{\partial x \partial y}$ (A.M.I.E.T.E., Winter 2007)

(Nagpur University, Summer 2001, Winter 2000, U.P., I Semester, Winter 2001)

Solution. We have, $x + y = 2 e^\theta \cos \phi$... (1)

$x - y = 2 i e^\theta \sin \phi$... (2)

By adding and subtracting equations (1) and (2), we have

$$2x = 2e^\theta (\cos \phi + i \sin \phi) \Rightarrow x = e^{\theta+i\phi}$$

and

$$2y = 2e^\theta (\cos \phi - i \sin \phi) \Rightarrow y = e^{\theta-i\phi} \quad \dots(3)$$

It is clear that $V = f(x, y)$ and x, y are functions of θ and ϕ . Hence V is a composite function of θ and ϕ .

We want to convert V, θ, ϕ in V, x, y respectively.

From equation (3), we have

$$\frac{\partial x}{\partial \theta} = e^{\theta+i\phi} = x, \quad \frac{\partial x}{\partial \phi} = i e^{\theta+i\phi} = i x$$

$$\frac{\partial y}{\partial \theta} = e^{\theta-i\phi} = y, \quad \text{and} \quad \frac{\partial y}{\partial \phi} = -i e^{\theta-i\phi} = -i y$$

Now,
$$\frac{\partial V}{\partial \theta} = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial V}{\partial y} \cdot \frac{\partial y}{\partial \theta} = x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y}$$

and
$$\begin{aligned}
\frac{\partial^2 V}{\partial \theta^2} &= \frac{\partial}{\partial \theta} \left(\frac{\partial V}{\partial \theta} \right) = \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right) \left(x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} \right) \\
&= x \frac{\partial}{\partial x} \left[x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} \right] + y \frac{\partial}{\partial y} \left[x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} \right] \\
&= x \left[x \frac{\partial^2 V}{\partial x^2} + \frac{\partial V}{\partial x} + y \frac{\partial^2 V}{\partial x \partial y} \right] + y \left[x \frac{\partial^2 V}{\partial x \partial y} + y \frac{\partial^2 V}{\partial y^2} + \frac{\partial V}{\partial y} \right]
\end{aligned}$$

$$\Rightarrow \frac{\partial^2 V}{\partial \theta^2} = x^2 \frac{\partial^2 V}{\partial x^2} + y^2 \frac{\partial^2 V}{\partial y^2} + 2xy \frac{\partial^2 V}{\partial x \partial y} + \left(x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} \right) \quad \dots(4)$$

Again $\frac{\partial V}{\partial \phi} = \frac{\partial V}{\partial x} \frac{\partial x}{\partial \phi} + \frac{\partial V}{\partial y} \frac{\partial y}{\partial \phi} = i \left[x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right]$

$$\frac{\partial}{\partial \phi} = i \left[x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \right]$$

$$\frac{\partial^2 V}{\partial \phi^2} = \frac{\partial}{\partial \phi} \left(\frac{\partial V}{\partial \phi} \right) = i \left(x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} \right) i \left(x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right)$$

$$= ix \frac{\partial}{\partial x} \cdot i \left[x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right] - iy \frac{\partial}{\partial y} \cdot i \left[x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right]$$

$$= -x \frac{\partial}{\partial x} \left[x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right] + y \frac{\partial}{\partial y} \left[x \frac{\partial V}{\partial x} - y \frac{\partial V}{\partial y} \right]$$

$$= -x \left[\frac{\partial V}{\partial x} + x \frac{\partial^2 V}{\partial x^2} - y \frac{\partial^2 V}{\partial x \partial y} \right] + y \left[x \frac{\partial^2 V}{\partial x \partial y} - y \frac{\partial^2 V}{\partial y^2} - \frac{\partial V}{\partial y} \right]$$

$$\Rightarrow \frac{\partial^2 V}{\partial \phi^2} = - \left[x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} \right] + 2xy \frac{\partial^2 V}{\partial x \partial y} - \left[x^2 \frac{\partial^2 V}{\partial x^2} + y^2 \frac{\partial^2 V}{\partial y^2} \right] \quad \dots(5)$$

Adding (4) and (5), we get $\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial \phi^2} = 4xy \frac{\partial^2 V}{\partial x \partial y}$ **Proved.**

EXERCISE 1.5

1. If $z = u^2 + v^2$ and $u = at^2$, $v = 2at$, find $\frac{dz}{dt}$. **Ans.** $4a^2t(t^2 + 2)$

2. If $z = \sin^{-1}(x - y)$, $x = 3t$, $y = 4t^3$; show that $\frac{dz}{dt} = \frac{3}{\sqrt{1-t^2}}$.

3. If $w = f(u, v)$, where $u = x + y$ and $v = x - y$, show that $\frac{dw}{dx} + \frac{dw}{dy} = 2 \frac{dw}{du}$

4. If $u = xe^{yz}$, where $y = \sqrt{a^2 - x^2}$, $z = \sin^3 x$. Find $\frac{du}{dx}$ **Ans.** $e^{yz} \left(1 - \frac{x^2}{y} + 3x \cot x \right)$

5. If $u = x^2 + y^2 + z^2 - 2xyz = 1$, show that $\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}} = 0$

[Hint. $du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = 0$

$$= 2(x - yz) dx + 2(y - zx) dy + 2(z - xy) dz = 0$$

But $x^2 + y^2 + z^2 - 2xyz = 1$, $\Rightarrow y^2 - 2xyz = 1 - x^2 - z^2$
 $y^2 - 2xyz + x^2 z^2 = 1 + x^2 z^2 - x^2 - z^2 \Rightarrow (y - xz)^2 = (1 - x^2)(1 - z^2)$

6. If $z = z(u, v)$, $u = x^2 - 2xy - y^2$ and $v = y$. Show that

$$(x + y) \frac{\partial z}{\partial x} + (x - y) \frac{\partial z}{\partial y} = 0 \text{ is equivalent to } \frac{\partial z}{\partial v} = 0$$

7. If $u = f(x^2 + 2yz, y^2 + 2zx)$, prove that

$$(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$$

8. By changing the independent variables x and t to u and v by means of the relationships
 $u = x - at, \quad v = x + at$

Show that $a^2 \frac{\partial^2 y}{\partial x^2} - \frac{\partial^2 y}{\partial t^2} = 4a^2 \frac{\partial^2 y}{\partial u \partial v}$

9. If $x^2 = au + bv, y = au - bv$, prove that $\left(\frac{du}{dx}\right)_y \cdot \left(\frac{dx}{du}\right)_v = \frac{1}{2} = \left(\frac{\partial v}{\partial y}\right)_x \cdot \left(\frac{\partial y}{\partial v}\right)_u$

10. If $z = f(x, y)$ where $x = uv, y = \frac{u+v}{u-v}$, show that $2x \frac{\partial z}{\partial x} = u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}$.

11. If $u = x \cos \frac{y}{z}, x = 3r^2 + 2s, y = 4r - 2s^3, z = 2r^2 - 3s^2$ find $\frac{\partial u}{\partial r}, \frac{\partial u}{\partial s}$.

Ans. $\frac{\partial u}{\partial r} = 6r \cos \frac{y}{z} - \frac{4x}{z} \sin \frac{y}{z} + \frac{4xyr}{z^2} \sin \frac{y}{z}, \frac{\partial u}{\partial s} = 2 \cos \frac{y}{z} + \frac{6xs^2}{z} \sin \frac{y}{z} - \frac{6xys}{z^2} \sin \frac{y}{z}$

12. If $z = f(x, y)$ where $x = e^u \cos v, y = e^u \sin v$. Prove that

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = e^{-2u} \left[\left(\frac{\partial f}{\partial u}\right)^2 + \left(\frac{\partial f}{\partial v}\right)^2 \right]$$

13. If $x = \frac{\cos \theta}{u}, y = \frac{\sin \theta}{u}$ and $z = f(x, y)$, then show that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = u^4 \frac{\partial^2 z}{\partial u^2} + u^3 \frac{\partial z}{\partial u} + u^4 \frac{\partial^2 z}{\partial \theta^2}$$

14. If $x = z \sin^{-1} \frac{y}{x}$ where $x = 3r^2 + 2s, y = 4r - 2s^3, z = 2r^2 - 3s^2$, find $\frac{\partial u}{\partial r}, \frac{\partial u}{\partial s}$

Ans. $\frac{\partial u}{\partial r} = -\frac{6ryz}{x\sqrt{x^2-y^2}} + \frac{4z}{\sqrt{x^2-y^2}} + 4r \sin \frac{y}{x}, \frac{\partial u}{\partial s} = \frac{2yz}{x\sqrt{x^2-y^2}} - \frac{6s^2z}{\sqrt{x^2-y^2}} - 6s \sin \frac{y}{x}$

15. If $z = f(u, v)$ where $u = x \cos \theta - y \sin \theta, v = x \sin \theta + y \cos \theta$, show that

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = u \frac{\partial z}{\partial u} + v \frac{\partial z}{\partial v}, \theta \text{ being constant.}$$

16. Given the transformation $x = \cosh \xi \cos \eta, y = \sinh \xi \sin \eta$
 establish the following equation for the function u (function of x, y and also of ξ, η):

$$\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} = (\sinh^2 \xi + \sin^2 \eta) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

17. If $z = f(x, y)$, where $x = r \cos \theta, y = r \sin \theta$, prove that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 \quad \text{and} \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \left(\frac{\partial^2 z}{\partial \theta^2}\right) + \frac{1}{r} \left(\frac{\partial z}{\partial r}\right)$$

18. If by substituting $u = x^2 - y^2, v = 2xy, f(x, y) = \phi(u, v)$. Show that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 4(u^2 + v^2) \left(\frac{\partial^2 \phi}{\partial u^2} + \frac{\partial^2 \phi}{\partial v^2} \right)$$

19. If $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta, v = v(x, y, z)$, prove that

$$\left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2 = \left(\frac{\partial v}{\partial r}\right)^2 + \left(\frac{1}{r} \frac{\partial v}{\partial \theta}\right)^2 + \left(\frac{1}{r \sin \theta} \frac{\partial v}{\partial \phi}\right)^2$$

20. If v be a potential function such that $v = v(r)$ and $r^2 = x^2 + y^2 + z^2$, show that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = \frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr}$$

21. Given that $w = x + 2y + z^2$, $x = r/s$, $y = r^2 + e^s$, and $z = 2r$, show that $r \frac{\partial w}{\partial r} + s \frac{\partial w}{\partial s} = 12r^2 + 2se^s$.

22. Find $\frac{\partial w}{\partial v}$ when $u = 0$, $v = 0$

$$\text{If } w = (x^2 + y - 2)^4 + (x - y + 2)^3, \quad x = u - 2v + 1, \\ \text{and } y = 2u + v - 2$$

Ans. 99

23. If $x = u + v + w$, $y = vw + wu + uv$, $z = uvw$ and F is a function of x, y, z , then show that

$$u \frac{\partial F}{\partial u} + v \frac{\partial F}{\partial v} + w \frac{\partial F}{\partial w} = x \frac{\partial F}{\partial x} + 2y \frac{\partial F}{\partial y} + 3z \frac{\partial F}{\partial z}$$

24. If $u = x + ay$ and $v = x + by$, transform the equation

$$2 \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 3 \frac{\partial^2 z}{\partial y^2} = 0 \quad \text{into the equation } \frac{\partial^2 z}{\partial u \partial v} = 0, \text{ find the values of } a \text{ and } b.$$

$$(A.M.I.E.T.E., \text{ Summer } 2000) \quad \text{Ans. } \left(a = 1, b = \frac{2}{3}\right), \left(a = \frac{2}{3}, b = 1\right)$$

1.17 IMPORTANT DEDUCTIONS

Let $z = f(x, y)$, then

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

If $z = 0$, $dz = 0$

$$0 = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \quad \Rightarrow \quad \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = - \frac{\partial f}{\partial x}$$

$$\Rightarrow \quad \boxed{\frac{dy}{dx} = - \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}} \quad \text{[Remember] ... (1)}$$

We can find $\frac{d^2 y}{dx^2}$ by differentiating (1).

$$\text{Let } \frac{\partial f}{\partial x} = p, \quad \frac{\partial f}{\partial y} = q, \quad \frac{\partial^2 f}{\partial x^2} = r, \quad \frac{\partial^2 f}{\partial x \partial y} = s, \quad \frac{\partial^2 f}{\partial y^2} = t$$

$$\text{From (1)} \quad \frac{\partial y}{\partial x} = - \frac{p}{q}$$

$$\text{On differentiating again, we obtain } \frac{d^2 y}{dx^2} = - \frac{q \frac{dp}{dx} - p \frac{dq}{dx}}{q^2} \quad \dots (2)$$

$$\text{But } \frac{dp}{dx} = \frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} \frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dp}{dx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \cdot \frac{dy}{dx}$$

$$\frac{dp}{dx} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y \partial x} \frac{dy}{dx} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y \partial x} \left(- \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \right) = r - s \frac{p}{q} = \frac{qr - ps}{q}$$

$$\begin{aligned}\frac{dq}{dx} &= \frac{\partial q}{\partial x} + \frac{\partial q}{\partial y} \frac{dy}{dx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \left(-\frac{p}{q} \right) \\ &= \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y^2} \frac{p}{q} = s - \frac{tp}{q} = \frac{qs - tp}{q}\end{aligned}$$

Making substitutions in (2), we obtain

$$\begin{aligned}\frac{d^2 y}{dx^2} &= -\frac{q \frac{qr - ps}{q} - p \frac{qs - tp}{q}}{q^2} = -\frac{q^2 r - pqs - pqs + p^2 t}{q^3} \\ \Rightarrow \quad \boxed{\frac{d^2 y}{dx^2} = -\frac{q^2 r - 2pqs + p^2 t}{q^3}}\end{aligned}$$

Example 40. If $x^3 + 3x^2y + 6xy^2 + y^3 = 1$, find $\frac{dy}{dx}$.

Solution. Let $f(x, y) = x^3 + 3x^2y + 6xy^2 + y^3 - 1 = 0$

$$\frac{\partial f}{\partial x} = 3x^2 + 6xy + 6y^2$$

$$\frac{\partial f}{\partial y} = 3x^2 + 12xy + 3y^2$$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{3x^2 + 6xy + 6y^2}{3x^2 + 12xy + 3y^2} = -\frac{x^2 + 2xy + 2y^2}{x^2 + 4xy + y^2}$$

Ans.

Example 41. If $y^3 - 3ax^2 + x^3 = 0$, then prove that

$$\frac{d^2 y}{dx^2} = -\frac{2a^2 x^2}{y^5}$$

Solution. Let $f(x, y) = y^3 - 3ax^2 + x^3$... (1)

$$p = \frac{\partial f}{\partial x} = -6ax + 3x^2, \quad q = \frac{\partial f}{\partial y} = 3y^2$$

$$r = \frac{\partial^2 f}{\partial x^2} = -6a + 6x, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0, \quad t = \frac{\partial^2 f}{\partial y^2} = 6y$$

$$\frac{\partial^2 y}{\partial x^2} = -\frac{q^2 r - 2pqs + p^2 t}{q^3} \quad \dots (2) \text{ [Art 1.17]}$$

Putting the values of p, q, r, s and t in (2), we get

$$\begin{aligned}\frac{\partial^2 y}{\partial x^2} &= -\frac{(3y^2)^2 (-6a + 6x) - 2(-6ax + 3x^2)(3y^2)(0) + (-6ax + 3x^2)^2 (6y)}{(3y^2)^3} \\ &= -\frac{54y^4(-a+x) + 54(-2ax+x^2)^2 y}{27y^6}\end{aligned}$$

$$\frac{\partial^2 y}{\partial x^2} = -\frac{2y^3(-a+x) + 2(4a^2x^2 + x^4 - 4ax^3)}{y^5} \quad \dots (3)$$

Putting the value of $y^3 = 3ax^2 - x^3$ from (1) in (3), we get

$$\frac{\partial^2 y}{\partial x^2} = -\frac{2(3ax^2 - x^3)(-a+x) + 2(4a^2x^2 + x^4 - 4ax^3)}{y^5}$$

$$= -\frac{-6a^2x^2 + 6ax^3 + 2ax^3 - 2x^4 + 8a^2x^2 + 2x^4 - 8ax^3}{y^5}$$

$$\frac{\partial^2 y}{\partial x^2} = -\frac{2a^2x^2}{y^5}$$

Proved.

Example 42. If $x^3 + y^3 - 3axy = 0$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Solution. Let $f(x, y) = x^3 + y^3 - 3axy = 0$

$$p = \frac{\partial f}{\partial x} = 3x^2 - 3ay, \quad q = \frac{\partial f}{\partial y} = 3y^2 - 3ax$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x, \quad s = \frac{\partial^2 f}{\partial x \partial y} = -3a, \quad t = \frac{\partial^2 f}{\partial y^2} = 6y$$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{3x^2 - 3ay}{3y^2 - 3ax} = \frac{ay - x^2}{y^2 - ax}$$

$$\frac{d^2y}{dx^2} = -\frac{q^2r - 2pqst + p^2t}{q^3} \quad \dots(1) \text{ [Art. 1.17]}$$

Putting the values of p, q, r, s and t in (1), we get

$$\begin{aligned} \frac{d^2y}{dx^2} &= -\frac{(3y^2 - 3ax)^2 6x - 2(3x^2 - 3ay)(3y^2 - 3ax)(-3a) + (3x^2 - 3ay)^2 (6y)}{(3y^2 - 3ax)^3} \\ &= -\frac{2x(y^2 - ax)^2 + 2a(x^2 - ay)(y^2 - ax) + 2y(x^2 - ay)^2}{(y^2 - ax)^3} \end{aligned}$$

$$\frac{d^2y}{dx^2} = \frac{2a^3xy}{(ax - y^2)^3}$$

Ans.

Example 43. Find $\frac{dy}{dx}$ when $(\cos x)^y = (\sin y)^x$

Solution. Given equation can be written as : $(\cos x)^y - (\sin y)^x = 0$

Here $f(x, y) = (\cos x)^y - (\sin y)^x = 0$

$$\frac{\partial f}{\partial x} = y(\cos x)^{y-1}(-\sin x) - (\sin y)^x \log \sin y$$

$$= -[y \sin x (\cos x)^{y-1} + (\sin y)^x \log \sin y]$$

$$\frac{\partial f}{\partial y} = (\cos x)^y \log \cos x - x(\sin y)^{x-1} \cos y$$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

[Art. 1.17]

$$\frac{dy}{dx} = \frac{y \sin x (\cos x)^{y-1} + (\sin y)^x \log \sin y}{(\cos x)^y \log \cos x - x(\sin y)^{x-1} \cos y}$$

... (1)

In (1), put $(\cos x)^y$ for $(\sin y)^x$

$$\frac{dy}{dx} = \frac{y \sin x (\cos x)^{y-1} + (\cos x)^y \log \sin y}{(\cos x)^y \log \cos x - \frac{x(\cos x)^y}{\sin y} \cdot \cos y}$$

$$= \frac{(\cos x)^y [y \tan x + \log \sin y]}{(\cos x)^y [\log \cos x - x \cot y]} = \frac{y \tan x + \log \sin y}{\log \cos x - x \cot y} \quad \text{Ans.}$$

Example 44. If $f(x, y) = 0$ and $\phi(y, z) = 0$, show that $\frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y}$.

Solution. $f(x, y) = 0 \quad \dots(1)$

$\phi(y, z) = 0 \quad \dots(2)$

Differentiating (1) w.r.t. x , we get

$$0 = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \quad \dots(3)$$

Differentiating (2) w.r.t. ' y ', we get

$$0 = \frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dy} \Rightarrow \frac{dz}{dy} = -\frac{\frac{\partial \phi}{\partial y}}{\frac{\partial \phi}{\partial z}} \quad \dots(4)$$

Multiplying (3) and (4), we get

$$\begin{aligned} \frac{dy}{dx} \times \frac{dz}{dy} &= \left(-\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \right) \left(-\frac{\frac{\partial \phi}{\partial y}}{\frac{\partial \phi}{\partial z}} \right) \Rightarrow \frac{dz}{dx} = \frac{\frac{\partial f}{\partial x} \times \frac{\partial \phi}{\partial y}}{\frac{\partial f}{\partial y} \times \frac{\partial \phi}{\partial z}} \\ \Rightarrow \frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} &= \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y} \quad \text{Proved.} \end{aligned}$$

Example 45. If $u = x \log xy$ where $x^3 + y^3 + 3xy = 1$. Find $\frac{du}{dx}$

(U.P. I Sem., Dec. 2005, Com. 2002)

Solution. We have, $u = x \log xy$

$$\frac{\partial u}{\partial x} = x \left(\frac{1}{xy} \cdot y \right) + 1 \log xy$$

$$\Rightarrow \frac{\partial u}{\partial x} = 1 + \log xy \quad \dots(1)$$

$$\frac{\partial u}{\partial y} = x \frac{1}{xy} \cdot x = \frac{x}{y} \quad \dots(2)$$

$$x^3 + y^3 + 3xy = 1$$

On differentiating, we get

$$3x^2 + 3y^2 \frac{dy}{dx} + 3x \frac{dy}{dx} + 3y = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x^2 + y}{x + y^2} \quad \dots(3)$$

We know that $\frac{du}{dx} = \frac{\partial u}{\partial x} \frac{dx}{dx} + \frac{\partial u}{\partial y} \frac{dy}{dx}$

$$= (1 + \log xy) \cdot 1 + \frac{x}{y} \left(-\frac{x^2 + y}{x + y^2} \right) \quad [\text{From (1), (2), (3)}]$$

$$= 1 + \log xy - \frac{x}{y} \cdot \frac{x^2 + y}{x + y^2} \quad \text{Ans.}$$

EXERCISE 1.6

Find $\frac{dy}{dx}$ in the following cases :

1. $x \sin(x - y) - (x + y) = 0$

Ans. $[y + x^2 \cos(x - y)] / [x + x^2 \cos(x - y)]$

2. $x^y = y^x$

Ans. $y(y - x \log y) / x(x - y \log x)$

3. If $ax^2 + 2hxy + by^2 = 1$, find $\frac{d^2y}{dx^2}$

Ans. $\frac{h^2 - ab}{(hx + by)^3}$

4. If $u = x^2y + y^2z + z^2x$ and if z is defined implicitly as a function of x and y by the equation $x^2 + yz + z^3 = 0$

find $\frac{\partial u}{\partial x}$, where u is considered as a function of x and y alone.

Ans. $\frac{\partial u}{\partial x} = 2xy + z^2 - (y^2 + 2zx) \left(\frac{2x}{y + 3z^2} \right)$

5. Find $\frac{dy}{dx}$, if $\tan^{-1} \frac{x}{y} + y^3 + 1 = 0$

Ans. $\frac{y}{x - 3x^2y^2 - 3y^4}$

6. If $f(x, y, z) = 0$, prove that

$$\left(\frac{dx}{dy} \right)_z \cdot \left(\frac{dy}{dz} \right)_x \cdot \left(\frac{dz}{dx} \right)_y = -1$$

Hint. $\left(\frac{dx}{dy} \right)_z = -\frac{\partial f}{\partial y} \div \frac{\partial f}{\partial x}$

1.18 TYPICAL CASES

Example 46. If $x = f(u, v)$, $y = \phi(u, v)$, find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

Solution.

$$x = f(u, v) \quad \dots(1)$$

$$y = \phi(u, v) \quad \dots(2)$$

Differentiating (1), (2) w.r.t. x (treating y as constant), we obtain

$$1 = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} \quad \dots(3)$$

$$0 = \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial x} \quad \dots(4)$$

Solving the equations (3) and (4) for $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$, we obtain

$$\frac{\partial u}{\partial x} = \frac{\frac{\partial \phi}{\partial v}}{\frac{\partial f}{\partial u} \cdot \frac{\partial \phi}{\partial v} - \frac{\partial f}{\partial v} \cdot \frac{\partial \phi}{\partial u}} \quad \text{Ans.}$$

$$\frac{\partial v}{\partial x} = \frac{-\frac{\partial \phi}{\partial u}}{\frac{\partial f}{\partial u} \cdot \frac{\partial \phi}{\partial v} - \frac{\partial f}{\partial v} \cdot \frac{\partial \phi}{\partial u}} \quad \text{Ans.}$$

Similarly, differentiating (1) and (2) w.r.t. y , we get

$$0 = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} \quad \dots(5)$$

$$1 = \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} \quad \dots(6)$$

Solving the equations (5) and (6) for $\frac{\partial u}{\partial y}$ and $\frac{\partial v}{\partial y}$, we obtain

$$\frac{\partial u}{\partial y} = \frac{-\frac{\partial f}{\partial v}}{\frac{\partial f}{\partial u} \cdot \frac{\partial \phi}{\partial v} - \frac{\partial f}{\partial v} \cdot \frac{\partial \phi}{\partial u}} \quad \text{Ans.}$$

$$\frac{\partial v}{\partial y} = \frac{\frac{\partial f}{\partial u}}{\frac{\partial f}{\partial u} \cdot \frac{\partial \phi}{\partial v} - \frac{\partial f}{\partial v} \cdot \frac{\partial \phi}{\partial u}} \quad \text{Ans.}$$

Example 47. If $x = u^2 - v^2$ and $y = uv$, find

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y}, \quad \frac{\partial v}{\partial x} \quad \text{and} \quad \frac{\partial v}{\partial y} \quad \text{(Nagpur University, Winter 2003)}$$

Solution. Here, we have $x = u^2 - v^2$... (1)
 $y = uv$... (2)

$$\frac{\partial x}{\partial u} = 2u, \quad \frac{\partial y}{\partial u} = v$$

$$\frac{\partial x}{\partial v} = -2v, \quad \frac{\partial y}{\partial v} = u$$

Differentiating (1) w.r.t. x , we get

$$\Rightarrow 1 = 2u \frac{\partial u}{\partial x} - 2v \frac{\partial v}{\partial x} \quad \dots (3)$$

Similarly differentiating (2) w.r.t. x , we get

$$0 = v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} \quad \dots (4)$$

On solving (3) and (4), we get

$$\frac{\partial u}{\partial x} = \frac{u}{2u^2 + 2v^2}, \quad \frac{\partial v}{\partial x} = -\frac{v}{2u^2 + 2v^2} \quad \text{Ans.}$$

On differentiating (1) and (2) w.r.t. y , we get

$$0 = 2u \frac{\partial u}{\partial y} - 2v \frac{\partial v}{\partial y} \quad \dots (5)$$

$$1 = v \frac{\partial u}{\partial y} + u \frac{\partial v}{\partial y} \quad \dots (6)$$

On solving (5) and (6), we get

$$\frac{\partial v}{\partial y} = \frac{u}{u^2 + v^2}, \quad \text{and} \quad \frac{\partial u}{\partial y} = \frac{v}{u^2 + v^2} \quad \text{Ans.}$$

Example 48. Find p and q , if $x = \sqrt{a} (\sin u + \cos v)$

$$y = \sqrt{a} (\cos u - \sin v)$$

$$z = 1 + \sin (u - v)$$

where p and q mean $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ respectively.

Solution. We have, $x = \sqrt{a} (\sin u + \cos v)$... (1)

$$y = \sqrt{a} (\cos u - \sin v) \quad \dots (2)$$

$$z = 1 + \sin (u - v) \quad \dots (3)$$

Differentiating (3) w.r.t. x , we get

$$\frac{\partial z}{\partial x} = \cos (u - v) \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial x} \right] \quad \dots (4)$$

Differentiating (1) partially w.r.t. x , we get

$$1 = \sqrt{a} \left(\cos u \frac{\partial u}{\partial x} - \sin v \frac{\partial v}{\partial x} \right) \quad \dots(5)$$

Differentiating (2) partially w.r.t. x , we get

$$0 = \sqrt{a} \left(-\sin u \frac{\partial u}{\partial x} - \cos v \frac{\partial v}{\partial x} \right) \quad \dots(6)$$

Solving (5) and (6) for $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$, we get

$$\frac{\partial u}{\partial x} = \frac{1}{\sqrt{a}} \frac{\cos v}{\cos(u-v)}, \quad \frac{\partial v}{\partial x} = -\frac{\sin u}{\sqrt{a} \cos(u-v)}$$

Putting the values of $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$ in (4), we get

$$p = \frac{\partial z}{\partial x} = \cos(u-v) \left[\frac{1}{\sqrt{a}} \frac{\cos v}{\cos(u-v)} + \frac{\sin u}{\sqrt{a} \cos(u-v)} \right]$$

$$p = \frac{\partial z}{\partial x} = \frac{1}{\sqrt{a}} (\sin u + \cos v)$$

Ans.

Differentiating (3) w.r.t. y , we get

$$\frac{\partial z}{\partial y} = \cos(u-v) \left[\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} \right] \quad \dots(7)$$

Differentiating (1) and (2) partially w.r.t. y , we get

$$0 = \sqrt{a} \left(\cos u \frac{\partial u}{\partial y} - \sin v \frac{\partial v}{\partial y} \right) \quad \dots(8)$$

$$1 = \sqrt{a} \left(-\sin u \frac{\partial u}{\partial y} - \cos v \frac{\partial v}{\partial y} \right) \quad \dots(9)$$

Solving (8) and (9) for $\frac{\partial u}{\partial y}$ and $\frac{\partial v}{\partial y}$, we get

$$\frac{\partial u}{\partial y} = -\frac{\sin v}{\sqrt{a} \cos(u-v)}, \quad \text{and} \quad \frac{\partial v}{\partial y} = -\frac{\cos u}{\sqrt{a} \cos(u-v)}$$

Putting the values of $\frac{\partial u}{\partial y}$, and $\frac{\partial v}{\partial y}$ in (7), we have

$$q = \frac{\partial z}{\partial y} = \cos(u-v) \left[-\frac{\sin v}{\sqrt{a} \cos(u-v)} + \frac{\cos u}{\sqrt{a} \cos(u-v)} \right]$$

$$q = \frac{1}{\sqrt{a}} (-\sin v + \cos u)$$

Ans.

EXERCISE 1.7

1. Fill in the blanks

(i) If $f(x, y, z) = 0$, then $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x}$ is equal to.....

(ii) If $z = f(x, y)$, where $x = \varphi(t)$, $y = \psi(t)$, then $\frac{dz}{dt} = \dots\dots$

(iii) If $f(x, y) = 0$, then $\frac{dy}{dx} = \dots\dots$

(iv) If $u = x^2 + y^2$, $x = s + 3t$, $y = 2s - t$, then $\frac{du}{ds} =$

(v) If $f(x, y) = 0$ and $\phi(y, z) = 0$, then $\frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial x} = \dots\dots$

(vi) If $f(x, y) = 0$, then $\frac{d^2 y}{dx^2} = \dots\dots$ **Ans.** (i) -1 (ii) $\frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$ (iii) $-\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$

(iv) $2x + 4y$ (v) $\frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y}$ (vi) $-\frac{q^2 r - 2pq s + p^2 t}{q^3}$

1.19 GEOMETRICAL INTERPRETATION OF $\frac{\partial z}{\partial x}$ AND $\frac{\partial z}{\partial y}$

(Gujarat, I Semester, Jan. 2009)

Let $z = f(x, y)$ be a surface S .

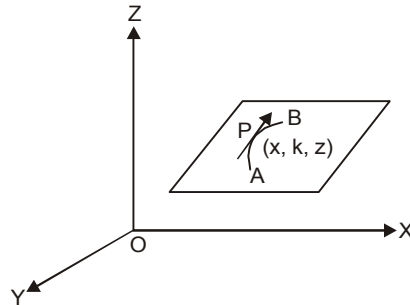
Let $y = k$ be a plane parallel to XZ -plane, passing through $P(x, k, z)$ cutting the surface $z = f(x, y)$ along the curve APB .

This section APB is a plane curve whose equations are

$$z = f(x, y)$$

$$y = k$$

The slope of the tangent to this curve is given by $\frac{\partial z}{\partial x}$.



Similarly, $\frac{\partial z}{\partial y}$ is the slope of the tangent to the curve

of intersection of the surface $z = f(x, y)$ with a plane parallel to YZ -plane.

1.20 TANGENT PLANE TO A SURFACE

Let $f(x, y, z) = 0$ be the equation of a surface S . Now we wish to find out the equation of a tangent plane to S at the point $P(x_1, y_1, z_1)$.

Let $Q(x_1 + \delta x_1, y_1 + \delta y_1, z_1 + \delta z_1)$ be a neighbouring point to P . Let the arc PQ be δs and the chord PQ be δc .

The direction cosines of PQ are

$$\frac{\delta x}{\delta c}, \frac{\delta y}{\delta c}, \frac{\delta z}{\delta c}$$

$$\Rightarrow \frac{\delta x}{\delta s} \cdot \frac{\delta s}{\delta c}, \frac{\delta y}{\delta s} \cdot \frac{\delta s}{\delta c}, \frac{\delta z}{\delta s} \cdot \frac{\delta s}{\delta c}$$

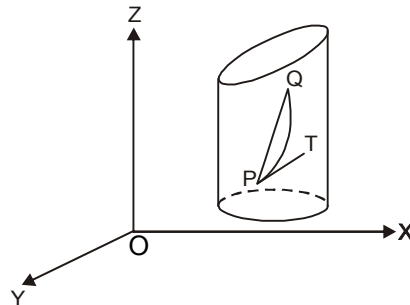
As $\delta s \rightarrow 0$, $Q \rightarrow P$ and PQ tends to a tangent line PT . The direction cosines of PT are

$$\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \quad \dots(1)$$

Differentiating $F(x, y, z) = 0$ w.r.t. 's', we get

$$\frac{\partial F}{\partial x} \frac{dx}{ds} + \frac{\partial F}{\partial y} \frac{dy}{ds} + \frac{\partial F}{\partial z} \frac{dz}{ds} = 0 \quad \dots(2)$$

From (1) and (2) it is clear that the tangent whose direction cosines are $\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}$ is



perpendicular to a line having direction ratios

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \quad \dots(3)$$

There are a number of tangent lines at P to the curves joining P and Q. All these tangents will be perpendicular to the line having direction ratios as given by (3).

Hence all these tangent lines will lie in a plane known as tangent plane.

Equation of tangent plane

$$(x - x_1) \frac{\partial F}{\partial x} + (y - y_1) \frac{\partial F}{\partial y} + (z - z_1) \frac{\partial F}{\partial z} = 0$$

Equation of the normal to the plane.

$$\frac{x - x_1}{\frac{\partial F}{\partial x}} = \frac{y - y_1}{\frac{\partial F}{\partial y}} = \frac{z - z_1}{\frac{\partial F}{\partial z}}$$

Example 49. Find the equation of the tangent plane and normal line to the surface

$$x^2 + 2y^2 + 3z^2 = 12 \text{ at } (1, 2, -1).$$

Solution. $F(x, y, z) = x^2 + 2y^2 + 3z^2 - 12$

$$\frac{\partial F}{\partial x} = 2x, \quad \frac{\partial F}{\partial y} = 4y, \quad \frac{\partial F}{\partial z} = 6z$$

$$\text{At the point } (1, 2, -1) \quad \frac{\partial F}{\partial x} = 2, \quad \frac{\partial F}{\partial y} = 8, \quad \frac{\partial F}{\partial z} = -6$$

Hence the equation of the tangent plane at $(1, 2, -1)$ is

$$2(x - 1) + 8(y - 2) - 6(z + 1) = 0$$

$$\Rightarrow 2x + 8y - 6z = 24 \Rightarrow x + 4y - 3z = 12$$

$$\text{Equation of normal is } \frac{x-1}{2} = \frac{y-2}{8} = \frac{z+1}{-6} \Rightarrow \frac{x-1}{1} = \frac{y-2}{4} = \frac{z+1}{-3}$$

Ans.

Example 50. Show that the surface $x^2 - 2yz + y^3 = 4$ is perpendicular to any number of the family of surfaces $x^2 + 1 = (2 - 4a)y^2 + az^2$ at the point of intersection $(1, -1, 2)$.

Solution. $f(x, y, z) = x^2 - 2yz + y^3 - 4 = 0 \quad \dots(1)$

$$F(x, y, z) = x^2 + 1 - (2 - 4a)y^2 - az^2 = 0 \quad \dots(2)$$

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = -2z + 3y^2, \quad \frac{\partial f}{\partial z} = -2y$$

Direction ratios to the normal of the tangent plane to (1) are

$$2x, -2z + 3y^2, -2y$$

DRs at the point $(1, -1, 2)$ are $2, -1, 2$.

Now differentiating (2), we get

$$\frac{\partial F}{\partial x} = 2x, \quad \frac{\partial F}{\partial y} = -2(2 - 4a)y, \quad \frac{\partial F}{\partial z} = -2az.$$

Direction ratios to the normal of the tangent plane to (2) are

$$2x, (-4 + 8a)y, -2az.$$

DRs at the point $(1, -1, 2)$ are $2, 4 - 8a, -4a$

Now

$$\begin{aligned} l_1 l_2 + m_1 m_2 + n_1 n_2 &= (2)(2) + (-1)(4 - 8a) + 2(-4a) \\ &= 4 - 4 + 8a - 8a = 0. \end{aligned}$$

Hence, the given surfaces are perpendicular at $(1, -1, 2)$.

Ans.

EXERCISE 1.8

1. Find the equation of tangent plane and the normal line to the surface

$$x y z = 6 \text{ at } (1, 2, 3). \quad \text{Ans. } 6x + 3y + 2z = 18, \quad \frac{x-1}{6} = \frac{y-2}{3} = \frac{z-3}{2}$$

2. Find the equations of the tangent plane and the normal to the surface $z^2 = 4(1 + x^2 + y^2)$ at

$$(2, 2, 6). \quad \text{Ans. } 4x + 4y - 3z + 2 = 0, \quad \frac{x-2}{4} = \frac{y-2}{4} = \frac{z-6}{-3}$$

3. Find the equations of the tangent plane and the normal to the surface

$$\frac{x^2}{2} - \frac{y^2}{3} = z \text{ at } (2, 3, -1) \quad \text{Ans. } 2x - 2y - z + 1 = 0, \quad \frac{x-2}{2} = \frac{y-3}{-2} = \frac{z+1}{-1}$$

4. Show that the plane $3x + 12y - 6z - 17 = 0$, touches the conicoid $3x^2 - 6y^2 + 9z^2 + 17 = 0$.

$$\text{Find also the point of contact.} \quad \text{Ans. } \left(1, 2, \frac{2}{3}\right)$$

5. Show that the plane $ax + by + cz + d = 0$ touches the surface $px^2 + qy^2 + 2z = 0$,

$$\text{if } \frac{a^2}{p} + \frac{b^2}{q} + 2cd = 0.$$

Applications of differential Calculus

(Error, Jacobians, Taylor's Series, Maxima and Minima)

1.21 ERROR DETERMINATION

We know that

$$\lim_{\delta x \rightarrow 0} \frac{\partial y}{\partial x} = \frac{dy}{dx}$$

$$\frac{\partial y}{\partial x} = \frac{dy}{dx} \text{ approximately } \Rightarrow \delta y = \left(\frac{dy}{dx}\right) \cdot \delta x \text{ approximately}$$

Definitions:

- (i) δx is known as *absolute error* in x . (ii) $\frac{\delta x}{x}$ is known as *relative error* in x .
 (iii) $\left(\frac{\delta x}{x}\right) \times 100$ is known as *percentage error* in x .

Example 51. The power dissipated in a resistor is given by $P = \frac{E^2}{R}$. Find by using Calculus the approximate percentage change in P when E is increased by 3% and R is decreased by 2%.
 (A.M.I.E., Summer 2001)

Solution. Here, we have $P = \frac{E^2}{R} \Rightarrow \log P = 2 \log E - \log R$

On differentiating, we get

$$\frac{\delta P}{P} = \frac{2}{E} \delta E - \frac{\delta R}{R} \Rightarrow 100 \frac{\delta P}{P} = 2 \times \frac{100 \delta E}{E} - \frac{100 \delta R}{R}$$

$$100 \frac{\delta P}{P} = 2(3) - (-2) = 8 \quad \left[\text{Given, } \frac{100 \delta E}{E} = 2\%, \frac{100 \delta R}{R} = -2\% \right]$$

Percentage change in $P = 8\%$

Ans.

Example 52. The diameter and altitude of a can in the shape of a right circular cylinder are measured as 40 and 64 cm respectively. The possible error in each measurement is $\pm 5\%$. Find approximately the maximum possible error in the computed value for the volume and the lateral surface. Find the corresponding percentage error.

Solution. Here we have, Diameter of the can (D) = 40 cm

$$\frac{100 \delta D}{D} = \frac{100 \delta h}{h} = \pm 5\%$$

$$V = \pi r^2 h = \frac{\pi (D^2) h}{4} = \frac{\pi}{4} D^2 h$$

$$\log V = \log \frac{\pi}{4} + 2 \log D + \log h$$

$$\frac{\delta V}{V} = 0 + \frac{2 \delta D}{D} + \frac{\delta h}{h}$$

$$\frac{\delta V}{V} 100 = \frac{2 \delta D}{D} 100 + \frac{\delta h}{h} 100 = 2(\pm 5) + (\pm 5) = \pm 15 \quad \text{Ans.}$$

Again

$$S = 2 \pi r l = \pi D h$$

$$\log S = \log \pi + \log D + \log h$$

$$\frac{\delta S}{S} = 0 + \frac{\delta D}{D} + \frac{\delta h}{h}$$

$$\frac{\delta S}{S} 100 = \frac{\delta D}{D} 100 + \frac{\delta h}{h} 100 = (\pm 5) + (\pm 5) = \pm 10 \quad \text{Ans.}$$

Example 53. The period T of a simple pendulum is

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Find the maximum error in T due to possible errors upto 1% in l and 2% in g .
(U.P. I semester winter 2003)

Solution. We have, $T = 2\pi \sqrt{\frac{l}{g}}$

$$\Rightarrow \log T = \log 2\pi + \frac{1}{2} \log l - \frac{1}{2} \log g$$

Differentiating, we get

$$\frac{\partial T}{T} = 0 + \frac{1}{2} \frac{\partial l}{l} - \frac{1}{2} \frac{\partial g}{g}$$

$$\Rightarrow \left(\frac{\partial T}{T} \right) \times 100 = \frac{1}{2} \left[\left(\frac{\partial l}{l} \right) \times 100 - \left(\frac{\partial g}{g} \right) \times 100 \right]$$

$$\text{But } \frac{\partial l}{l} \times 100 = 1, \frac{\partial g}{g} \times 100 = 2$$

$$\therefore \left(\frac{\partial T}{T} \right) \times 100 = \frac{1}{2} [1 \pm 2] = \frac{3}{2}$$

Maximum error in $T = 1.5\%$

Ans.

Example 54. A balloon is in the form of right circular cylinder of radius 1.5 m and length 4 m and is surmounted by hemispherical ends. If the radius is increased by 0.01 m and the length by 0.05 m, find the percentage change in the volume of the balloon.

(U.P. I Sem., Dec., 2005, Comp 2002)

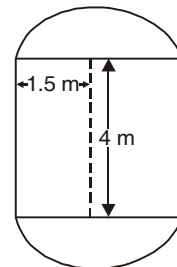
Solution. Radius of the cylinder (r) = 1.5 m

Length of the cylinder (h) = 4 m

Volume of the balloon = Volume of cylinder + Volume of two hemispheres

$$\text{Volume } (V) = \pi r^2 h + \frac{2}{3} \pi r^3 + \frac{2}{3} \pi r^3 = \pi r^2 h + \frac{4}{3} \pi r^3$$

$$\delta V = \pi 2r \delta r \cdot h + \pi r^2 \cdot \delta h + \frac{4}{3} \pi 3r^2 \cdot \delta r$$



$$\begin{aligned}
 \frac{\delta V}{V} &= \frac{\pi r [2 \delta r \cdot h + r \cdot \delta h + 4 r \delta r]}{\pi r^2 h + \frac{4}{3} \pi r^3} = \frac{2 \cdot \delta r \cdot h + r \cdot \delta h + 4 r \cdot \delta r}{r h + \frac{4}{3} r^2} \\
 &= \frac{2 \times 0.01 \times 4 + 1.5 \times 0.05 + 4 \times 1.5 \times 0.01}{1.5 \times 4 + \frac{4}{3} (1.5)^2} \\
 &= \frac{0.08 + 0.075 + 0.06}{6 + 3} = \frac{0.215}{9} \\
 100 \frac{\delta V}{V} &= \frac{100 \times 0.215}{9} = \frac{21.5}{9} = 2.389\%
 \end{aligned}$$

Ans.

Example 55. In estimating the number of bricks in a pile which is measured to be $(5m \times 10m \times 5m)$, count of bricks is taken as 100 bricks per m^3 . Find the error in the cost when the tape is stretched 2% beyond its standard length. The cost of bricks is ₹ 2,000 per thousand bricks.
(U.P., I Semester, Winter 2000)

Solution. Volume $V = x y z$

$$\log V = \log x + \log y + \log z$$

Differentiating, we get

$$\begin{aligned}
 \frac{\delta V}{V} &= \frac{\delta x}{x} + \frac{\delta y}{y} + \frac{\delta z}{z} \\
 100 \frac{\delta V}{V} &= \frac{100 \delta x}{x} + \frac{100 \delta y}{y} + \frac{100 \delta z}{z} = 2 + 2 + 2 \\
 \frac{100 \delta V}{V} &= 6 \\
 \delta V &= \frac{6V}{100} = \frac{6(5 \times 10 \times 5)}{100} = 15 \text{ cubicmetre.}
 \end{aligned}$$

Number of bricks in $\delta V = 15 \times 100 = 1500$

$$\text{Error in cost} = \frac{1500 \times 2000}{1000} = 3000$$

Thus error in cost, a loss to the seller of bricks = ₹ 3000.

Ans.

Example 56. The angles of a triangle are calculated from the sides a, b, c . If small changes $\delta a, \delta b, \delta c$ are made in the sides, show that approximately

$$\delta A = \frac{a}{2\Delta} [\delta a - \delta b \cos C - \delta c \cos B]$$

where Δ is the area of the triangle and A, B, C are the angles opposite to a, b, c respectively.

Verify that $\delta A + \delta B + \delta C = 0$ (U.P., I Sem., Winter 2001, A.M.I.E.T.E., 2001)

Solution. We know that

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow a^2 = b^2 + c^2 - 2bc \cos A \quad \dots(1)$$

Differentiating both sides of (1), we get

$$2a \delta a = 2b \delta b + 2c \delta c - 2b \delta c \cos A - 2\delta b c \cos A + 2bc \sin A \delta A \quad (\text{approx.})$$

$$a \delta a = b \delta b + c \delta c - b \delta c \cos A - \delta b c \cos A + bc \sin A \delta A$$

$$\Rightarrow bc \sin A \delta A = a \delta a - (b - c \cos A) \delta b - (c - b \cos A) \delta c$$

$$\Rightarrow 2\Delta \delta A = a \delta a - (a \cos C + c \cos A - c \cos A) \delta b - (a \cos B + b \cos A - b \cos A) \delta c$$

$$\left[\begin{aligned} &\because \Delta = \frac{1}{2} bc \sin A \\ &b \cos C + c \cos B = a \end{aligned} \right]$$

$$2 \Delta \delta A = a \delta a - a \delta b \cos C - a \delta c \cos B \\ = a (\delta a - \delta b \cos C - \delta c \cos B)$$

$$\Rightarrow \delta A = \frac{a}{2 \Delta} [\delta a - \delta b \cos C - \delta c \cos B] \quad \dots(2)$$

Similarly,

$$\delta B = \frac{b}{2 \Delta} [\delta b - \delta c \cos A - \delta a \cos C] \quad \dots(3)$$

$$\delta C = \frac{c}{2 \Delta} [\delta c - \delta a \cos B - \delta b \cos A] \quad \dots(4)$$

On adding (2), (3) and (4), we get

$$[\delta A + \delta B + \delta C] = \frac{1}{2 \Delta} [(a - b \cos C - c \cos B) \delta a + (b - a \cos C - c \cos A) \delta b \\ + (c - a \cos B - b \cos A) \delta c] \\ = \frac{1}{2 \Delta} [(a - a) \delta a + (b - b) \delta b + (c - c) \delta c] \\ = 0 \quad [\because b \cos C + c \cos B = a] \text{ Verified.}$$

Example 57. The height h and semi-vertical angle α of a cone are measured, and from there A , the total area of the cone, including the base, is calculated. If h and α are in error by small quantities δh and $\delta \alpha$ respectively, find the corresponding error in the area. Show further that,

if $\alpha = \frac{\pi}{6}$, an error of + 1 per cent in h will be approximately compensated by an error of - 19.8' in α . (A.M.I.E.T.E., Summer 2003)

Solution. Let l be the slant height of the cone and r its radius

$$l = h \sec \alpha$$

$$r = h \tan \alpha$$

$$A = \pi r^2 + \pi r l$$

$$= \pi h^2 \tan^2 \alpha + \pi (h \tan \alpha) (h \sec \alpha)$$

$$= \pi h^2 [\tan^2 \alpha + \tan \alpha \sec \alpha]$$

$$\delta A = 2\pi h \delta h [\tan^2 \alpha + \tan \alpha \sec \alpha]$$

$$+ \pi h^2 [2 \tan \alpha \sec^2 \alpha \delta \alpha + \sec^2 \alpha \cdot \delta \alpha \cdot \sec \alpha + \tan \alpha \sec \alpha \tan \alpha \delta \alpha]$$

$$\delta A = 2\pi h [\tan \alpha + \sec \alpha] \tan \alpha \cdot \delta h + \pi h^2 [2 \tan \alpha \sec \alpha + \sec^2 \alpha + \tan^2 \alpha] \cdot \sec \alpha \cdot \delta \alpha$$

$$\delta A = 2\pi h [\tan \alpha + \sec \alpha] \tan \alpha \cdot \delta h + \pi h^2 [\tan \alpha + \sec \alpha]^2 \sec \alpha \cdot \delta \alpha$$

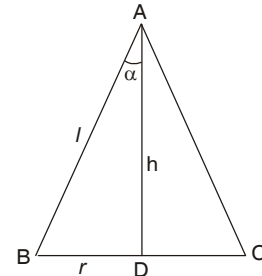
$$\delta A = \pi h^2 [\tan \alpha + \sec \alpha] \left[2 \tan \alpha \frac{\delta h}{h} + (\tan \alpha + \sec \alpha) \sec \alpha \cdot \delta \alpha \right]$$

On putting $\delta A = 0, \Rightarrow \alpha = \frac{\pi}{6}, \frac{\delta h}{h} \times 100 = 1$, we get

$$\Rightarrow 0 = \pi h^2 \left[\tan \frac{\pi}{6} + \sec \frac{\pi}{6} \right] \left[\left(2 \tan \frac{\pi}{6} \right) \frac{1}{100} + \left(\tan \frac{\pi}{6} + \sec \frac{\pi}{6} \right) \sec \frac{\pi}{6} \delta \alpha \right]$$

$$\Rightarrow 0 = \left[\left(2 \tan \frac{\pi}{6} \right) \frac{1}{100} + \left(\tan \frac{\pi}{6} + \sec \frac{\pi}{6} \right) \sec \frac{\pi}{6} \delta \alpha \right]$$

$$\Rightarrow 0 = \frac{2}{\sqrt{3}} \frac{1}{100} + \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \right) \frac{2}{\sqrt{3}} \delta \alpha \Rightarrow \frac{2}{\sqrt{3}} \frac{1}{100} - \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \right) \frac{2}{\sqrt{3}} \delta \alpha$$



$$\Rightarrow \frac{1}{100} = -\left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right) \delta\alpha \Rightarrow \frac{1}{100} = \frac{-3}{\sqrt{3}} \delta\alpha \Rightarrow \delta\alpha = -\frac{1}{100\sqrt{3}}$$

$$\Rightarrow \delta\alpha = -\frac{1}{100\sqrt{3}} \frac{180}{\pi} \text{ degree} = \left(-\frac{9}{5\sqrt{3}}\right) \frac{60}{\pi} \text{ minutes} = -19.8 \text{ minutes} \quad \text{Ans.}$$

Example 58. Find the possible percentage error in computing the parallel resistance r of three resistances r_1, r_2, r_3 from the formula $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$ if r_1, r_2, r_3 are each in error by plus 1.2%.

Solution. Here, $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \quad \dots(1)$

Differentiating, we get

$$-\frac{1}{r^2} dr = -\frac{1}{r_1^2} dr_1 - \frac{1}{r_2^2} dr_2 - \frac{1}{r_3^2} dr_3$$

$$\Rightarrow \frac{1}{r} \left(\frac{100 dr}{r}\right) = \frac{1}{r_1} \left(\frac{100 dr_1}{r_1}\right) + \frac{1}{r_2} \left(\frac{100 dr_2}{r_2}\right) + \frac{1}{r_3} \left(\frac{100 dr_3}{r_3}\right)$$

$$= \frac{1}{r_1} (1.2) + \frac{1}{r_2} (1.2) + \frac{1}{r_3} (1.2) = (1.2) \left[\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}\right]$$

$$= 1.2 \left(\frac{1}{r}\right) \quad \text{[From (1).]}$$

$$\Rightarrow \frac{100 dr}{r} = 1.2\% \quad \text{Ans.}$$

Example 59. If the sides and angles of a plane triangle vary in such a way that its circumradius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$, where da, db, dc are small increments in the sides, a, b, c respectively.

Solution. From the sine rule,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

We know that

$$R = \frac{a}{2 \sin A}, \quad \dots(1)$$

Differentiating, we get

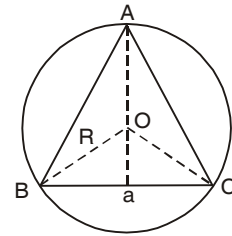
$$\frac{\partial R}{\partial A} = -\frac{a \cos A}{2 \sin^2 A}$$

$$\frac{\partial R}{\partial a} = \frac{1}{2 \sin A}$$

By total differentiation $dR = \frac{\partial R}{\partial A} dA + \frac{\partial R}{\partial a} da$

$$0 = -\frac{a \cos A}{2 \sin^2 A} \cdot dA + \frac{1}{2 \sin A} \cdot da, \quad R \text{ being constant}$$

$$\Rightarrow \frac{\cos A}{\sin A} dA = \frac{1}{\sin A} da$$



$$\Rightarrow \frac{da}{\cos A} = \frac{a}{\sin A} \cdot dA = 2R \cdot dA \quad [\text{Using (1)}]$$

$$\Rightarrow \frac{da}{\cos A} = 2R \, dA \quad \dots(1)$$

$$\text{Similarly,} \quad \frac{db}{\cos B} = 2R \, dB \quad \dots(2)$$

$$\text{and} \quad \frac{dc}{\cos C} = 2R \, dC \quad \dots(3)$$

Adding (1), (2) and (3), we have

$$\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R [dA + dB + dC] \quad \dots(4)$$

But in any triangle ABC, $A + B + C = \pi$

Hence, $dA + dB + dC = 0$

Putting value of $dA + dB + dC = 0$ in (4), we get

$$\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R (0) = 0 \Rightarrow \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0 \quad \text{Proved.}$$

Example 60. Compute an approximate value of $(1.04)^{3.01}$.

Solution. Let $f(x, y) = x^y$

$$\text{We have} \quad \frac{\partial f}{\partial x} = y x^{y-1}, \quad \frac{\partial f}{\partial y} = x^y \log x$$

$$\begin{aligned} \text{Here, let} \quad x &= 1, \delta x = 0.04, \\ y &= 3, \delta y = 0.01 \end{aligned} \quad \dots(1)$$

$$\begin{aligned} \text{Now} \quad df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \\ &= y x^{y-1} dx + x^y \log x dy \end{aligned} \quad \dots(2)$$

Substituting the values from (1) in (2), we get

$$\begin{aligned} df &= (3) (1)^{3-1} (0.04) + (1)^3 \log (1) (0.01) = 0.12 \\ (1.04)^{3.01} &= f(1, 3) + df = 1 + 0.12 = 1.12 \end{aligned} \quad \text{Ans.}$$

Example 61. Find $[(3.82)^2 + 2(2.1)^3]^{\frac{1}{5}}$

Solution. Let $f(x, y) = (x^2 + 2y^3)^{\frac{1}{5}}$

$$\begin{aligned} \text{Taking} \quad x &= 4, \delta x = 3.82 - 4 = -0.18 \\ y &= 2, \delta y = 2.1 - 2 = 0.1 \end{aligned}$$

$$\frac{\partial f}{\partial x} = \frac{1}{5} [x^2 + 2y^3]^{-\frac{4}{5}} (2x) = \frac{2}{5} (4) [16 + 2(2)^3]^{-\frac{4}{5}} = \frac{8}{5} \left(\frac{1}{16} \right) = \frac{1}{10}$$

$$\frac{\partial f}{\partial y} = \frac{1}{5} [x^2 + 2y^3]^{-\frac{4}{5}} (6y^2) = \frac{6}{5} (2)^2 [16 + 2(2)^3]^{-\frac{4}{5}} = \frac{24}{5} \times \frac{1}{16} = \frac{3}{10}$$

By total differentiation, we get

$$df = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y = \frac{1}{10} (-0.18) + \frac{3}{10} (0.1) = -0.018 + 0.03 = 0.012$$

$$\begin{aligned}
 [(3.82)^2 + 2(2.1)^3]^{\frac{1}{5}} &= f(4, 2) + df \\
 &= [(4)^2 + 2(2)^3]^{\frac{1}{5}} + 0.012 = 2 + 0.012 = 2.012 \quad \text{Ans.}
 \end{aligned}$$

EXERCISE 1.9

1. If the density ρ of a body be inferred from its weights W , ω in air and water respectively, show that the relative error in ρ due to errors δW , $\delta \omega$ in W , ω is $\frac{\delta \rho}{\rho} = \frac{-\omega}{W-\omega} \cdot \frac{\delta W}{W} + \frac{\delta \omega}{W-\omega}$.

2. The period of oscillation of a pendulum is computed by the formula

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Show that the percentage error in $T = \frac{1}{2}$ [% error in l – % error in g]

If $l = 6$ cm and relative error in g is equal to $\frac{1}{160}$, find the error in the determination of T .

(Given $g = 981$ cm/sec²)

Ans. – 0.00153

3. The indicated horse power I of an engine is calculated from the formula.

$$I = \frac{PLAN}{33000}$$

where $A = \frac{\pi}{4} d^2$. Assuming that errors of r percent may have been made in measuring P , L , N and d . Find the greatest possible error in I . Ans. 5 r %

4. The dimensions of a cone are radius 4 cm, height 6 cm. What is the error in its volume if the scale used in taking the measurement is short by 0.01 cm per cm. Ans. 0.96π cm³.

5. The work that must be done to propel a ship of displacement D for a distance s in time t is proportional to $s^2 D^{2/3} t^2$.

Find approximately the percentage increase of work necessary when the displacement is increased by 1%, the time is diminished by 1% and the distance is increased by 3%. Ans. $\frac{14}{3}$ %

6. The power P required to propel a ship of length l moving with a velocity V is given by $P = kV^3 l^2$. Find the percentage increase in power if increase in velocity is 3% and increase in length is 4%. Ans. 17%

7. In estimating the cost of a pile of bricks measured as $2\text{m} \times 15\text{m} \times 1.2\text{m}$, the tape is stretched 1% beyond the standard length if the count is 450 bricks to 1m^3 and bricks cost ₹ 1300 per 1000, find the approximate error in the cost. Ans. ₹ 631.80

8. In estimating the cost of a pile of bricks measured as $6' \times 50' \times 4'$, the tape is stretched 1% beyond the standard length. If the count is 12 bricks to ft^3 , and bricks cost ₹ 100 per 1000, find the approximate error in the cost. (U.P. I Sem., Dec. 2004) Ans. 720 bricks, ₹ 25.20

9. The sides of a triangle are measured as 12 cm and 15 cm and the angle included between them as 60° . If the lengths can be measured within 1% accuracy while the angle can be measured within 2% accuracy. Find the percentage error in determining (i) area of the triangle (ii) length of opposite side of the triangle. (A.M.I.E.T.E., Winter 2002)

10. The voltage V across a resistor is measured with error h , and the resistance R is measured with an error k . Show that the error in calculating the power $W(V, R) = \frac{V^2}{R}$ generated in the resistor is

$\frac{V}{R^2} (2Rh - V k)$. If V can be measured to an accuracy of 0.5 p.c. and R to an accuracy of 1 p.c., what is the approximate possible percentage error in W ? Ans. Zero percent

11. Find the possible percentage error in computing the parallel resistance r of two resistance r_1 and r_2 from the formula $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$, where r_1 and r_2 are both is error by + 2% each. **Ans.** 2%
12. In the manufacture of closed cylindrical boxes with specified sides a, b, c ($a \neq b \neq c$), small changes of $A\%, B\%, C\%$ occurred in a, b, c , respectively from box to box from the specified dimension. However, the volume and surface area of all boxes were according to specification, show that:

$$\frac{A}{a(b-c)} = \frac{B}{b(c-a)} = \frac{C}{c(a-b)}$$
13. Find the percentage error in calculating the area of ellipse $x^2/a^2 + y^2/b^2 = 1$, when error of + 1% is made in measuring the major and minor axes. **Ans.** 2%
(U.P., I Sem, Jan 2011)
14. If $f = x^2 y^2 z^{\frac{1}{10}}$, find the approximate value of f , when $x = 1.99, y = 3.01$ and $z = 0.98$. **Ans.** 107.784
15. A diameter and altitude of a can in the form of right circular cylinder are measured as 4 cm and 6 cm respectively. The possible error in each measurement is 0.1 cm. Find approximately the maximum possible error in the value computed for the volume and lateral surface. **Ans.** 5.0336 cm³, 3.146 cm²
(A.M.I.E., Summer 2001)
16. Prove that the relative error of a quotient does not exceed the sum of the relative errors of the dividend and the divisor. **Ans.** 2%
(A.M.I.E., Winter 2001)

1.22 JACOBIANS

If u and v are functions of the two independent variables x and y , then the determinant

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

is called the jacobian of u, v with respect to x, y and is written as

$$\frac{\partial (u, v)}{\partial (x, y)} \text{ or } J \left(\frac{u, v}{x, y} \right)$$

Similarly, the jacobian of u, v, w with respect to x, y, z is

$$\frac{\partial (u, v, w)}{\partial (x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Example 62. If $x = r \cos \theta, y = r \sin \theta$; evaluate $\frac{\partial (x, y)}{\partial (r, \theta)}$, and $\frac{\partial (r, \theta)}{\partial (x, y)}$

Solution. We have,

$$\begin{array}{l|l} x = r \cos \theta, & y = r \sin \theta \\ \frac{\partial x}{\partial r} = \cos \theta, & \frac{\partial y}{\partial r} = \sin \theta \\ \frac{\partial x}{\partial \theta} = -r \sin \theta, & \frac{\partial y}{\partial \theta} = r \cos \theta \end{array}$$

$$\frac{\partial (x, y)}{\partial (r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta = r (\cos^2 \theta + \sin^2 \theta) = r$$

Now,

$$r^2 = x^2 + y^2, \quad \theta = \tan^{-1} \frac{y}{x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial \theta}{\partial x} = \frac{-y}{x^2 + y^2} = -\frac{y}{r^2}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2} = \frac{x}{r^2}$$

$$\frac{\partial (r, \theta)}{\partial (x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{vmatrix} = \frac{x^2}{r^3} + \frac{y^2}{r^3} = \frac{x^2 + y^2}{r^3} = \frac{r^2}{r^3} = \frac{1}{r} \quad \text{Ans.}$$

Note : $\frac{\partial (x, y)}{\partial (r, \theta)} \times \frac{\partial (r, \theta)}{\partial (x, y)} = r \cdot \frac{1}{r} = 1$

Example 63. If $x = a \cosh \xi \cos \eta$, $y = a \sinh \xi \sin \eta$, show that

$$\frac{\partial (x, y)}{\partial (\xi, \eta)} = \frac{a^2}{2} (\cosh 2\xi - \cos 2\eta)$$

Solution. Here, we have, $x = a \cosh \xi \cos \eta$
 $y = a \sinh \xi \sin \eta$

$$\begin{aligned} \frac{\partial (x, y)}{\partial (\xi, \eta)} &= \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{vmatrix} = \begin{vmatrix} a \sinh \xi \cos \eta & -a \cosh \xi \sin \eta \\ a \cosh \xi \sin \eta & a \sinh \xi \cos \eta \end{vmatrix} \\ &= a^2 \begin{vmatrix} \sinh \xi \cos \eta & -\cosh \xi \sin \eta \\ \cosh \xi \sin \eta & \sinh \xi \cos \eta \end{vmatrix} = a^2 [\sinh^2 \xi \cos^2 \eta + \cosh^2 \xi \sin^2 \eta] \\ &= a^2 [\sinh^2 \xi (1 - \sin^2 \eta) + (1 + \sinh^2 \xi) \sin^2 \eta] \\ &= a^2 [\sinh^2 \xi - \sinh^2 \xi \sin^2 \eta + \sin^2 \eta + \sinh^2 \xi \sin^2 \eta] \\ &= a^2 [\sinh^2 \xi + \sin^2 \eta] = \frac{a^2}{2} [\cosh 2\xi - 1 + 1 - \cos 2\eta] = \frac{a^2}{2} [\cosh 2\xi - \cos 2\eta] \quad \text{Proved.} \end{aligned}$$

Example 64. If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$.

Show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4.

(U.P. I Sem. Jan 2011; 2004, Comp. 2002, A.M.I.E., Summer 2002, 2000, Winter 2001)

Solution. Here, we have $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$

$$\begin{aligned} \frac{\partial (y_1, y_2, y_3)}{\partial (x_1, x_2, x_3)} &= \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix} = \begin{vmatrix} -\frac{x_2 x_3}{x_1^2} & \frac{x_3}{x_1} & \frac{x_2}{x_1} \\ \frac{x_3}{x_2} & -\frac{x_3 x_1}{x_2^2} & \frac{x_1}{x_2} \\ \frac{x_2}{x_3} & \frac{x_1}{x_3} & -\frac{x_1 x_2}{x_3^2} \end{vmatrix} \\ &= \frac{1}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -x_2 x_3 & x_3 x_1 & x_1 x_2 \\ x_2 x_3 & -x_3 x_1 & x_1 x_2 \\ x_2 x_3 & x_3 x_1 & -x_1 x_2 \end{vmatrix} = \frac{x_1^2 x_2^2 x_3^2}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \\ &= -1(1-1) - 1(-1-1) + 1(1+1) = 0 + 2 + 2 = 4 \quad \text{Proved.} \end{aligned}$$

Example 65. If $x = r \sin \theta \cos \phi$,
 $y = r \sin \theta \sin \phi$,
 $z = r \cos \theta$,

Show that $\frac{\partial (x, y, z)}{\partial (r, \theta, \phi)} = r^2 \sin \theta$.

(U.P., I Semester, Winter 2000)

Solution. We have, $x = r \sin \theta \cos \phi$,

$y = r \sin \theta \sin \phi$,

$z = r \cos \theta$

$$\frac{\partial x}{\partial r} = \sin \theta \cos \phi,$$

$$\frac{\partial y}{\partial r} = \sin \theta \sin \phi,$$

$$\frac{\partial z}{\partial r} = \cos \theta$$

$$\frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi,$$

$$\frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi,$$

$$\frac{\partial z}{\partial \theta} = -r \sin \theta$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi,$$

$$\frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi,$$

$$\frac{\partial z}{\partial \phi} = 0$$

$$\frac{\partial (x, y, z)}{\partial (r, \theta, \phi)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} = \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix}$$

$$= r^2 \sin \theta \begin{vmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{vmatrix}$$

$$\begin{aligned} &= r^2 \sin \theta [\sin \theta \cos \phi (0 + \sin \theta \cos \phi) - \cos \theta \cos \phi (0 - \cos \phi \cos \theta) - \sin \phi (-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi)] \\ &= r^2 \sin \theta [\sin^2 \theta \cos^2 \phi + \cos^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta \sin^2 \phi] \\ &= r^2 \sin \theta [(\sin^2 \theta + \cos^2 \theta) \cos^2 \phi + (\sin^2 \theta + \cos^2 \theta) \sin^2 \phi] \\ &= r^2 \sin \theta [\cos^2 \phi + \sin^2 \phi] = r^2 \sin \theta \end{aligned}$$

Ans.

EXERCISE 1.10

1. If $u = x^2$, $v = y^2$, find $\frac{\partial (u, v)}{\partial (x, y)}$

Ans. $4xy$

2. If $u = \frac{y-x}{1+xy}$ and $v = \tan^{-1} y - \tan^{-1} x$, find $\frac{\partial (u, v)}{\partial (x, y)}$

Ans. 0

3. If $u = xyz$, $v = xy + yz + zx$, $w = x + y + z$, compute $\frac{\partial (u, v, w)}{\partial (x, y, z)}$

Ans. $(x-y)(y-z)(z-x)$

4. If $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ find $\frac{\partial (x, y, z)}{\partial (r, \theta, z)}$

5. If $u_1 = \frac{x_1}{x_n}$, $u_2 = \frac{x_2}{x_n}$, ..., $u_{n-1} = \frac{x_{n-1}}{x_n}$ and $x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2 = 1$ find $\frac{\partial (u_1, u_2, \dots, u_{n-1})}{\partial (x_1, x_2, \dots, x_{n-1})}$

Ans. $\frac{1}{x_n^{n-1}}$

6. Find the value of $\frac{\partial (y_1, y_2, y_3)}{\partial (x_1, x_2, x_3)}$, $y_1 = (1 - x_1)$, $y_2 = x_1(1 - x_2)$, $y_3 = x_1x_2(1 - x_3)$ **Ans.** $-x_1^2x_2$

7. If $u = \frac{x}{y-z}$, $v = \frac{y}{z-x}$, $w = \frac{z}{x-y}$ then show that $\frac{\partial (u, v, w)}{\partial (x, y, z)} = 0$

8. Fill in the blanks

(i) If $x = r \cos \theta$, $y = r \sin \theta$, then the value of Jacobian $\frac{\partial (x, y)}{\partial (r, \theta)}$ is

Ans. r

(ii) If $u = x(1 - y)$, $v = xy$, then the value of the Jacobian $\frac{\partial(u,v)}{\partial(x,y)} = \dots\dots\dots$ **Ans. x**

1.23 PROPERTIES OF JACOBIANS

(1) First Property

If u and v are the functions of x and y , then

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = 1 \quad (U. P. I Semester Dec. 2005)$$

Proof.

$$\text{Let } u = f(x, y) \quad \dots(1)$$

$$v = \phi(x, y) \quad \dots(2)$$

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

On interchanging the rows and columns of second determinant

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v} \end{vmatrix} \quad \dots(3)$$

On differentiating (1) and (2) w. r. t. u and v , we get

$$\left. \begin{aligned} \frac{\partial u}{\partial u} &= 1 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \frac{\partial u}{\partial v} &= 0 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial u} &= 0 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \frac{\partial v}{\partial v} &= 1 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v} \end{aligned} \right\} \quad \dots(4)$$

On making substitutions from (4) in (3), we get

$$\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad \text{Proved.}$$

Example 66. If $x = uv$, $y = \frac{u+v}{u-v}$, find $\frac{\partial(u,v)}{\partial(x,y)}$.

Solution. Here it is easy to find $\frac{\partial x}{\partial u}, \frac{\partial x}{\partial v}, \frac{\partial y}{\partial u}, \frac{\partial y}{\partial v}$. But to find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ is

comparatively difficult. So we first find $\frac{\partial(x,y)}{\partial(u,v)}$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -2v & 2u \end{vmatrix} = \frac{uv}{(u-v)^2} \begin{vmatrix} 1 & 1 \\ -2 & 2 \end{vmatrix} = \frac{uv}{(u-v)^2} (2+2) = \frac{4uv}{(u-v)^2}$$

$$\text{But } \frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} = 1 \Rightarrow \frac{\partial(u,v)}{\partial(x,y)} \times \frac{4uv}{(u-v)^2} = 1 \Rightarrow \frac{\partial(u,v)}{\partial(x,y)} = \frac{(u-v)^2}{4uv} \quad \text{Ans.}$$

Example 67. If $u = xyz$, $v = x^2 + y^2 + z^2$, $w = x + y + z$, find $J = \frac{\partial(x, y, z)}{\partial(u, v, w)}$.

Solution. Since u, v, w are explicitly given, so first we evaluate (U.P. I Sem., Winter 2002)

$$J' = \frac{\partial(u, v, w)}{\partial(x, y, z)}$$

$$J' = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} yz & zx & xy \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

$$\begin{aligned} &= yz(2y - 2z) - zx(2x - 2z) + xy(2x - 2y) = 2[yz(y - z) - zx(x - z) + xy(x - y)] \\ &= 2[x^2y - x^2z - xy^2 + xz^2 + y^2z - yz^2] = 2[x^2(y - z) - x(y^2 - z^2) + yz(y - z)] \\ &= 2(y - z)[x^2 - x(y + z) + yz] = 2(y - z)[y(z - x) - x(z - x)] \\ &= 2(y - z)(z - x)(y - x) = -2(x - y)(y - z)(z - x) \end{aligned}$$

Hence, by $JJ' = 1$, we have

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{-1}{2(x - y)(y - z)(z - x)} \quad \text{Ans.}$$

EXERCISE 1.11

1. Given $u = x^2 - y^2$, $v = 2xy$, calculate $\frac{\partial(x, y)}{\partial(u, v)}$ Ans. $\frac{1}{4(x^2 + y^2)}$

2. If $x = uv$, $y = \frac{u+v}{u-v}$, find $\frac{\partial(u, v)}{\partial(x, y)}$ Ans. $\frac{(u-v)^2}{4uv}$

3. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, find $\frac{\partial(r, \theta, \phi)}{\partial(x, y, z)}$ Ans. $\frac{1}{r^2 \sin \theta}$

4. Verify $JJ' = 1$, if $x = uv$, $y = \frac{u}{v}$

5. Verify $JJ' = 1$, if $x = e^v \sec u$, $y = e^v \tan u$.

6. Verify $JJ' = 1$, if $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$

(2) Second Property (Chain Rule)

If u, v are the functions of r, s where r, s are functions of x, y , then

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)} \quad (\text{U.P. I Sem. Jan 2011})$$

Proof.
$$\frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix}$$

On interchanging the columns and rows in second determinant

$$= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \times \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial s}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial s}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial(u, v)}{\partial(x, y)} \quad \text{Proved.}$$

Similarly,
$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{\partial(u, v, w)}{\partial(r, s, t)} \times \frac{\partial(r, s, t)}{\partial(x, y, z)}$$

Example 68. Find the value of the Jacobian $\frac{\partial(u, v)}{\partial(r, \theta)}$, where $u = x^2 - y^2$, $v = 2xy$ and

$$x = r \cos \theta, y = r \sin \theta.$$

Solution. $u = x^2 - y^2$, $v = 2xy$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2) = 4r^2$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

$$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)} = 4r^2 \cdot r = 4r^3 \quad \text{Ans.}$$

EXERCISE 1.12

1. If $u = e^x \cos y$, $v = e^x \sin y$, where $x = lr + sm$, $y = mr - sl$, verify $\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, s)} = \frac{\partial(u, v)}{\partial(r, s)}$

2. If $u = x(1 - r^2)^{-\frac{1}{2}}$, $v = y(1 - r^2)^{-\frac{1}{2}}$

$$w = z(1 - r^2)^{-\frac{1}{2}} \quad \text{where } r^2 = x^2 + y^2 + z^2$$

Show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = (1 - r^2)^{-\frac{5}{2}}$ (Q. Bank, U. P. 2001)

3. If $u = x + y + z$, $u^2 v = y + z$, $u^3 w = z$, show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = u^{-5}$

Hint. Put $r = x + y + z$, $s = y + z$, $t = z$

$$u = r \quad u^2 v = s \quad u^3 w = t$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{\partial(u, v, w)}{\partial(r, s, t)} \times \frac{\partial(r, s, t)}{\partial(x, y, z)} = \frac{1}{\left(\frac{\partial(r, s, t)}{\partial(u, v, w)} \right)} \times \left(\frac{\partial(r, s, t)}{\partial(x, y, z)} \right)$$

4. If $u = x + y + z$, $uv = y + z$, $uvw = z$. Evaluate $\frac{\partial(x, y, z)}{\partial(u, v, w)}$ (U. P. I Sem. Winter 2003) **Ans.** $u^2 v$

5. If $u^3 + v^3 = x + y$, $u^2 + v^2 = x^3 + y^3$, show that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{2} \frac{(y^2 - x^2)}{uv(u - v)}$

(3) Third Property

If functions u, v, w of three independent variables x, y, z are not independent, then

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$$

Proof. As u, v, w are not independent, then $f(u, v, w) = 0$... (1)

Differentiating (1) w.r.t x, y, z , we get

$$\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial x} = 0 \quad \dots (2)$$

$$\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial y} = 0 \quad \dots (3)$$

$$\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial z} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial z} = 0 \quad \dots (4)$$

Eliminating $\frac{\partial f}{\partial u}, \frac{\partial f}{\partial v}, \frac{\partial f}{\partial w}$ from (2), (3) and (4), we have

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial y} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial z} & \frac{\partial w}{\partial z} \end{vmatrix} = 0$$

On interchanging rows and columns, we get

$$\Rightarrow \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = 0$$

$$\Rightarrow \frac{\partial(u, v, w)}{\partial(x, y, z)} = 0 \quad \text{Proved.}$$

Converse (The sufficient condition)

If it is given that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$ and u, v, w are not independent of one another then they are connected by a relation $f(u, v, w) = 0$.

Example 69. If $u = x y + y z + z x$, $v = x^2 + y^2 + z^2$ and $w = x + y + z$, determine whether there is a functional relationship between u, v, w and if so, find it.

Solution. We have, $u = x y + y z + z x$, $v = x^2 + y^2 + z^2$, $w = x + y + z$

$$\begin{aligned} \frac{\partial(u, v, w)}{\partial(x, y, z)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} y+z & z+x & x+y \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2 \begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 2 \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} \quad R_1 \rightarrow R_1 + R_2 \\ &= 2 \begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 2 \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} \quad R_1 \rightarrow R_1 + R_2 \end{aligned}$$

$$= 2(x + y + z) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 0 \quad (R_1 = R_3)$$

Hence, the functional relationship exists between u , v and w .

Now, $w^2 = (x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$

$$w^2 = v + 2u$$

$$w^2 - v - 2u = 0$$

which is the required relationship.

Ans.

Example 70. Verify whether the following functions are functionally dependent, and if so, find the relation between them.

$$u = \frac{x + y}{1 - xy}, \quad v = \tan^{-1} x + \tan^{-1} y$$

Solution.
$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1+y^2}{(1-xy)^2} & \frac{1+x^2}{(1-xy)^2} \\ \frac{1}{1+x^2} & \frac{1}{1+y^2} \end{vmatrix} = \frac{1}{(1-xy)^2} - \frac{1}{(1-xy)^2} = 0$$

Hence u , v are functionally related.

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x + y}{1 - xy}$$

$$v = \tan^{-1} u$$

$\Rightarrow$

$$u = \tan v.$$

Ans.

EXERCISE 1.13

1. Verify whether $u = \frac{x - y}{x + y}$, $v = \frac{x + y}{x}$ are

functionally dependent, and if so, find the relation between them.

Ans. $u = \frac{2 - v}{v}$

2. Determine functional dependence and find relation between

$$u = \frac{x - y}{x + y}, \quad v = \frac{xy}{(x + y)^2}$$

Ans. $4v = 1 - u^2$

3. Are $x + y - z$, $x - y + z$, $x^2 + y^2 + z^2 - 2yz$ functionally dependent? If so, find a relation between them.

Ans. $u^2 + v^2 = 2w$

4. If $u = x + y + z$, $v = x^2 + y^2 + z^2$, $w = x^3 + y^3 + z^3 - 3xyz$, prove that u , v , w are not independent and find the relation between them.

Ans. $2w = u(3v - u^2)$

5. Are the following two functions of x , y , z functionally dependent? If so find the relation between them.

$$u = \frac{x - y}{x + z}, \quad v = \frac{x + z}{y + z}$$

Ans. $v = \frac{1}{1 - u}$

6. If $u = \frac{x + y}{z}$, $v = \frac{y + z}{x}$, $w = \frac{y(x + y + z)}{xz}$, show that u , v , w are not independent and find the

relation between them.

(U.P., 1st Semester, 2009)

Ans. $uv - w = 1$

1.24 JACOBIAN OF IMPLICIT FUNCTIONS

The variables x , y , u , v are connected by implicit functions

$$f_1(x, y, u, v) = 0 \quad \dots (1)$$

$$f_2(x, y, u, v) = 0 \quad \dots (2)$$

where u , v are implicit functions of x , y .

Differentiating (1) and (2) w.r.t. x and y , we get

$$\frac{\partial f_1}{\partial x} + \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial x} = 0 \quad \dots (3)$$

$$\frac{\partial f_1}{\partial y} + \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial y} = 0 \quad \dots (4)$$

$$\frac{\partial f_2}{\partial x} + \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial x} = 0 \quad \dots (5)$$

$$\frac{\partial f_2}{\partial y} + \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial y} = 0 \quad \dots (6)$$

Now, we have

$$\begin{aligned} \frac{\partial (f_1, f_2)}{\partial (u, v)} \times \frac{\partial (u, v)}{\partial (x, y)} &= \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{vmatrix} \times \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \\ &= \begin{vmatrix} \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial x} & \frac{\partial f_1}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial y} \\ \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial x} & \frac{\partial f_2}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} -\frac{\partial f_1}{\partial x} & -\frac{\partial f_1}{\partial y} \\ -\frac{\partial f_2}{\partial x} & -\frac{\partial f_2}{\partial y} \end{vmatrix} \\ &\quad \text{[From (3), (4), (5), (6)]} \\ &= (-1)^2 \frac{\partial (f_1, f_2)}{\partial (x, y)} \\ \frac{\partial (u, v)}{\partial (x, y)} &= (-1)^2 \frac{\partial (f_1, f_2) / \partial (x, y)}{\partial (f_1, f_2) / \partial (u, v)} \end{aligned}$$

In general, the variables $x_1, x_2, \dots, x_n$ are connected with $u_1, u_2, \dots, u_n$ implicitly as $f_1(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_n) = 0, f_2(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_n) = 0 \dots \dots \dots f_n(x_1, x_2, \dots, x_n, u_1, u_2, \dots, u_n) = 0$.
Then we have

$$\frac{\partial (u_1, u_2, \dots, u_n)}{\partial (x_1, x_2, \dots, x_n)} = (-1)^n \frac{\partial (f_1, f_2, \dots, f_n) / \partial (x_1, x_2, \dots, x_n)}{\partial (f_1, f_2, \dots, f_n) / \partial (u_1, u_2, \dots, u_n)}$$

Example 71. If $x^2 + y^2 + u^2 - v^2 = 0$ and $uv + xy = 0$, prove that $\frac{\partial (u, v)}{\partial (x, y)} = \frac{x^2 - y^2}{u^2 + v^2}$

Solution. Let

$$f_1 = x^2 + y^2 + u^2 - v^2, \quad f_2 = uv + xy$$

$$\frac{\partial (f_1, f_2)}{\partial (x, y)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & 2y \\ y & x \end{vmatrix} = 2(x^2 - y^2)$$

$$\frac{\partial (f_1, f_2)}{\partial (u, v)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ v & u \end{vmatrix} = 2(u^2 + v^2)$$

But

$$\frac{\partial (u, v)}{\partial (x, y)} = (-1)^2 \frac{\frac{\partial (f_1, f_2)}{\partial (x, y)}}{\frac{\partial (f_1, f_2)}{\partial (u, v)}} = \frac{2(x^2 - y^2)}{2(u^2 + v^2)} = \frac{x^2 - y^2}{u^2 + v^2}$$

Proved.

Example 72. If $u^3 + v + w = x + y^2 + z^2$, $u + v^3 + w = x^2 + y + z^2$, $u + v + w^3 = x^2 + y^2 + z$, prove

$$\text{that } \frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{1 - 4(xy + yz + zx) + 16xyz}{2 - 3(u^2 + v^2 + w^2) + 27u^2 v^2 w^2}$$

Solution. Let

$$f_1 = u^3 + v + w - x - y^2 - z^2$$

$$f_2 = u + v^3 + w - x^2 - y - z^2$$

$$f_3 = u + v + w^3 - x^2 - y^2 - z$$

$$\text{Now, } \frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} -1 & -2y & -2z \\ -2x & -1 & -2z \\ -2x & -2y & -1 \end{vmatrix} = -1 + 4(yz + zx + xy) - 16xyz$$

$$\text{and } \frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} 3u^2 & 1 & 1 \\ 1 & 3v^2 & 1 \\ 1 & 1 & 3w^2 \end{vmatrix} = 2 - 3(u^2 + v^2 + w^2) + 27u^2 v^2 w^2$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)/\partial(x, y, z)}{\partial(f_1, f_2, f_3)/\partial(u, v, w)} = \frac{1 - 4(yz + zx + xy) + 16xyz}{2 - 3(u^2 + v^2 + w^2) + 27u^2 v^2 w^2} \quad \text{Proved.}$$

Example 73. If $x + y + z = u$, $y + z = uv$, $z = uvw$, show that $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$

Solution. Let

$$f_1 = x + y + z - u$$

$$f_2 = y + z - uv$$

$$f_3 = z - uvw$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial w} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial w} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial w} \end{vmatrix} = \begin{vmatrix} -1 & 0 & 0 \\ -v & -u & 0 \\ -vw & -uw & -uv \end{vmatrix} = -u^2 v$$

$$\text{But } \frac{\partial(x, y, z)}{\partial(u, v, w)} = (-1)^3 \frac{\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)}}{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}} = -\frac{-u^2 v}{1} = u^2 v \quad \text{Proved.}$$

Example 74. If u, v, w are the roots of the equation $(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$ in λ find

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} \quad \text{(U.P. I Sem. Jan, 2011; Winter 2001)}$$

Solution. $(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$

$$\Rightarrow 3\lambda^3 - 3(x+y+z)\lambda^2 + 3(x^2+y^2+z^2)\lambda - (x^3+y^3+z^3) = 0$$

$$\text{Sum of the roots} = u + v + w = x + y + z \quad \dots (1)$$

$$\text{Product of the roots} = uv + vw + wu = x^2 + y^2 + z^2 \quad \dots (2)$$

$$uvw = \frac{1}{3}(x^3 + y^3 + z^3) \quad \dots (3)$$

Equations (1), (2) and (3) can be rewritten as

$$f_1 = u + v + w - x - y - z$$

$$f_2 = uv + vw + wu - x^2 - y^2 - z^2$$

$$f_3 = uvw - \frac{1}{3}(x^3 + y^3 + z^3)$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{vmatrix} = \begin{vmatrix} -1 & -1 & -1 \\ -2x & -2y & -2z \\ -x^2 & -y^2 & -z^2 \end{vmatrix}$$

$$= (-1)(-2)(-1) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = -2 \begin{vmatrix} 0 & 0 & 1 \\ x-y & y-z & z \\ x^2-y^2 & y^2-z^2 & z^2 \end{vmatrix} \begin{matrix} C_1 \rightarrow C_1 - C_2 \\ C_2 \rightarrow C_2 - C_3 \end{matrix}$$

$$= -2(x-y)(y-z) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & z \\ x+y & y+z & z^2 \end{vmatrix} = -2(x-y)(y-z)(y+z-x-y) \\ = -2(x-y)(y-z)(z-x)$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} & \frac{\partial f_1}{\partial w} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} & \frac{\partial f_2}{\partial w} \\ \frac{\partial f_3}{\partial u} & \frac{\partial f_3}{\partial v} & \frac{\partial f_3}{\partial w} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ v+w & u+w & u+v \\ vw & wu & uv \end{vmatrix} \\ = \begin{vmatrix} 0 & 0 & 1 \\ v-u & w-v & u+v \\ w(v-u) & u(w-v) & uv \end{vmatrix} \begin{matrix} C_1 \rightarrow C_1 - C_2 \\ C_2 \rightarrow C_2 - C_3 \end{matrix}$$

$$= (v-u)(w-v) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & u+v \\ w & u & uv \end{vmatrix} = (v-u)(w-v)(u-w) \\ = -(u-v)(v-w)(w-u)$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = -\frac{\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)}}{\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)}} = -\frac{-2(x-y)(y-z)(z-x)}{-(u-v)(v-w)(w-u)} = -\frac{2(x-y)(y-z)(z-x)}{(u-v)(v-w)(w-u)}$$

Ans.

EXERCISE 1.14

1. If $u^3 + v^3 = x + y$, $u^2 + v^2 = x^3 + y^3$, then prove that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{2} \frac{y^2 - x^2}{uv(u - v)}$
2. If $u^3 + v^3 + w^3 = x + y + z$, $u^2 + v^2 + w^2 = x^3 + y^3 + z^3$, $u + v + w = x^2 + y^2 + z^2$, show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{(x - y)(y - z)(z - x)}{(u - v)(v - w)(w - u)}$
3. If $u = \frac{x}{\sqrt{1 - r^2}}$, $v = \frac{y}{\sqrt{1 - r^2}}$, $w = \frac{z}{\sqrt{1 - r^2}}$, show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{1}{(1 - r^2)^{5/2}}$ where $r^2 = x^2 + y^2 + z^2$
4. If $u_1 = x_1 + x_2 + x_3 + x_4$, $u_1 u_2 = x_2 + x_3 + x_4$, $u_1 u_2 u_3 = x_3 + x_4$, $u_1 u_2 u_3 u_4 = x_4$ show that $\frac{\partial(x_1, x_2, x_3, x_4)}{\partial(u_1, u_2, u_3, u_4)} = u_1^3 \cdot u_2^2 \cdot u_3$
5. If u, v, w are the roots of the equation in λ and $\frac{x}{a + \lambda} + \frac{y}{b + \lambda} + \frac{z}{c + \lambda} = 1$, then find $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.
Ans. $\frac{(u - v)(v - w)(w - u)}{(a - b)(b - c)(c - a)}$

1.25 PARTIAL DERIVATIVES OF IMPLICIT FUNCTIONS BY JACOBIAN

Given $f_1(x, y, u, v) = 0$, $f_2(x, y, u, v) = 0$

$$\frac{\partial f_1}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_1}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial f_1}{\partial x} \cdot 1 = 0 \quad \dots (1)$$

$$\frac{\partial f_2}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f_2}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial f_2}{\partial x} \cdot 1 = 0 \quad \dots (2)$$

Solving (1) and (2), we get

$$\begin{aligned} \frac{\frac{\partial u}{\partial x}}{\frac{\partial f_1}{\partial v} \cdot \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial v}} &= \frac{\frac{\partial v}{\partial x}}{\frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial u} - \frac{\partial f_1}{\partial u} \cdot \frac{\partial f_2}{\partial x}} = \frac{1}{\frac{\partial f_1}{\partial u} \cdot \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \cdot \frac{\partial f_2}{\partial u}} \\ \frac{\partial u}{\partial x} &= \frac{\frac{\partial f_1}{\partial v} \cdot \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial v}}{\frac{\partial f_1}{\partial u} \cdot \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \cdot \frac{\partial f_2}{\partial u}} = - \frac{\frac{\partial(f_1, f_2)}{\partial(x, v)}}{\frac{\partial(f_1, f_2)}{\partial(u, v)}} \\ \frac{\partial v}{\partial x} &= \frac{\frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial u} - \frac{\partial f_1}{\partial u} \cdot \frac{\partial f_2}{\partial x}}{\frac{\partial f_1}{\partial u} \cdot \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \cdot \frac{\partial f_2}{\partial u}} = - \frac{\frac{\partial(f_1, f_2)}{\partial(x, u)}}{\frac{\partial(f_1, f_2)}{\partial(v, u)}} \end{aligned}$$

and if, $f_1(x, y, z, u, v, w) = 0$, $f_2(x, y, z, u, v, w) = 0$
 $f_3(x, y, z, u, v, w) = 0$

$$\frac{\partial x}{\partial u} = - \frac{\partial(f_1, f_2, f_3) / \partial(u, y, z)}{\partial(f_1, f_2, f_3) / \partial(x, y, z)}$$

and so on.

Note. First we write the Jacobian in the denominator and then we write the Jacobian in the numerator by replacing x by u .

Example 75. Use Jacobians to find $\left(\frac{\partial u}{\partial x}\right)_v$ if:

$$u^2 + xv^2 - xy = 0 \text{ and } u^2 + xyv + v^2 = 0$$

Solution. Let $f_1 = u^2 + xv^2 - xy$, $f_2 = u^2 + xyv + v^2$

$$\frac{\partial(f_1, f_2)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & 2xv \\ 2u & xy + 2v \end{vmatrix} = 2uxy + 4uv - 4uxv$$

$$\begin{aligned} \frac{\partial(f_1, f_2)}{\partial(x, v)} &= \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial v} \end{vmatrix} = \begin{vmatrix} v^2 - y & 2xv \\ yv & xy + 2v \end{vmatrix} \\ &= x y v^2 + 2v^3 - xy^2 - 2yv - 2xyv^2 = -xyv^2 + 2v^3 - xy^2 - 2yv \end{aligned}$$

$$\frac{\partial u}{\partial x} = - \frac{\partial(f_1, f_2) / \partial(x, v)}{\partial(f_1, f_2) / \partial(u, v)} = \frac{xyv^2 - 2v^3 + xy^2 + 2yv}{2uxy + 4uv - 2xuv}$$

Proved.

Example 76. If $u = x + y^2$, $v = y + z^2$, $w = z + x^2$, prove that

$$(i) \frac{\partial x}{\partial u} = \frac{1}{1 + 8xyz} \quad (ii) \text{ Also find } \frac{\partial^2 x}{\partial u^2}.$$

Solution. (i) Here $f_1 \equiv u - x - y^2$, $f_2 \equiv v - y - z^2$
 $f_3 \equiv w - z - x^2$.

$$\text{Now} \quad \frac{\partial(f_1, f_2, f_3)}{\partial(u, y, z)} = \begin{vmatrix} 1 & -2y & 0 \\ 0 & -1 & -2z \\ 0 & 0 & -1 \end{vmatrix} = 1;$$

$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} -1 & -2y & 0 \\ 0 & -1 & -2z \\ -2x & 0 & -1 \end{vmatrix} = -1(1+0) + 2y(0-4zx) = -1-8xyz$$

$$\frac{\partial x}{\partial u} = - \frac{\partial(f_1, f_2, f_3) / \partial(u, y, z)}{\partial(f_1, f_2, f_3) / \partial(x, y, z)}$$

$$\frac{\partial x}{\partial u} = - \left(\frac{1}{-1-8xyz} \right) = \frac{1}{1+8xyz}$$

Proved.

$$\begin{aligned} (ii) \quad \frac{\partial^2 x}{\partial u^2} &= \frac{\partial}{\partial u} \left(\frac{\partial x}{\partial u} \right) = \frac{\partial}{\partial u} \left(\frac{1}{1+8xyz} \right) = - \frac{1}{(1+8xyz)^2} \frac{\partial}{\partial u} (1+8xyz) \\ &= - \frac{1}{(1+8xyz)^2} \left[0 + 8 \left(\frac{\partial x}{\partial u} yz + \frac{\partial y}{\partial u} zx + \frac{\partial z}{\partial u} xy \right) \right] \\ &= \frac{-8}{(1+8xyz)^2} \left[\frac{\partial x}{\partial u} yz + \frac{\partial y}{\partial u} zx + \frac{\partial z}{\partial u} xy \right] \quad \dots (1) \end{aligned}$$

$$\text{We have,} \quad \frac{\partial(f_1, f_2, f_3)}{\partial(x, u, z)} = \begin{vmatrix} -1 & 1 & 0 \\ 0 & 0 & -2z \\ -2x & 0 & -1 \end{vmatrix} = 4zx;$$

$$\frac{\partial (f_1, f_2, f_3)}{\partial (x, y, u)} = \begin{vmatrix} -1 & -2y & 1 \\ 0 & -1 & 0 \\ -2x & 0 & 0 \end{vmatrix} = -2x.$$

Now,
$$\frac{\partial y}{\partial u} = -\frac{\partial (f_1, f_2, f_3) / \partial (x, u, z)}{\partial (f_1, f_2, f_3) / \partial (x, y, z)}$$

$$\therefore \frac{\partial y}{\partial u} = -\frac{4zx}{-1-8xyz} = \frac{4zx}{1+8xyz}$$

$$\frac{\partial z}{\partial u} = -\frac{\partial (f_1, f_2, f_3) / \partial (x, y, u)}{\partial (f_1, f_2, f_3) / \partial (x, y, z)}$$

$$\therefore \frac{\partial z}{\partial u} = -\frac{-2x}{-1-8xyz} = \frac{-2x}{1+8xyz}$$

Substituting in (1), we have

$$\begin{aligned} \frac{\partial^2 x}{\partial u^2} &= \frac{-8}{(1+8xyz)^2} \left[\frac{yz}{1+8xyz} + \frac{4z^2x^2}{1+8xyz} + \frac{-2x^2y}{1+8xyz} \right] \\ &= \frac{-8(yz + 4z^2x^2 - 2x^2y)}{(1+8xyz)^3} \end{aligned} \quad \text{Ans.}$$

Example 77. Given, $x = u + v + w$, $y = u^2 + v^2 + w^2$, $z = u^3 + v^3 + w^3$

show that
$$\frac{\partial u}{\partial x} = \frac{vw}{(u-v)(u-w)}$$

Solution. Let

$$\begin{aligned} f_1 &= u + v + w - x = 0 \\ f_2 &= u^2 + v^2 + w^2 - y = 0 \\ f_3 &= u^3 + v^3 + w^3 - z = 0 \\ \frac{\partial u}{\partial x} &= -\frac{\partial (f_1, f_2, f_3) / \partial (x, v, w)}{\partial (f_1, f_2, f_3) / \partial (u, v, w)} \end{aligned} \quad \dots (1)$$

$$\frac{\partial (f_1, f_2, f_3)}{\partial (x, v, w)} = \begin{vmatrix} -1 & 1 & 1 \\ 0 & 2v & 2w \\ 0 & 3v^2 & 3w^2 \end{vmatrix} = -6vw \begin{vmatrix} 1 & 1 \\ v & w \end{vmatrix} = 6vw(v-w) \quad \dots (2)$$

$$\frac{\partial (f_1, f_2, f_3)}{\partial (u, v, w)} = \begin{vmatrix} 1 & 1 & 1 \\ 2u & 2v & 2w \\ 3u^2 & 3v^2 & 3w^2 \end{vmatrix} = 6(v-u)(w-u)(w-v) \quad \dots (3)$$

Thus from (1), (2) and (3), we get

$$\frac{\partial u}{\partial x} = -\frac{6vw(v-w)}{6(v-u)(w-u)(w-v)} = \frac{vw}{(u-v)(u-w)} \quad \text{Proved.}$$

EXERCISE 1.15

1. If $u^2 + xv^2 - uxy = 0$, $v^2 - xy^2 + 2uv + u^2 = 0$ find $\frac{\partial u}{\partial x}$. Ans. $-\frac{(v^2 - uy)(u + v) + xvy^2}{(u + v)(2u - xy - 2xv)}$

2. If $x = u + e^{-v} \sin u$, $y = v + e^{-v} \cos u$, find $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$. Ans. $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{e^{-v} \sin u}{1 - e^{-3v}}$

3. If $x = u^2 - v^2$, $y = 2uv$, find $\frac{\partial u}{\partial x}$; $\frac{\partial u}{\partial y}$; $\frac{\partial v}{\partial x}$; $\frac{\partial v}{\partial y}$ and $\frac{\partial (u, v)}{\partial (x, y)}$

Ans. $\frac{u}{2(u^2 + v^3)}$; $\frac{v}{2(u^2 + v^3)}$; $\frac{-v}{2(u^2 + v^2)}$; $\frac{u}{2(u^2 + v^2)}$; $\frac{1}{4(u^2 + v^2)}$

4. If $u^3 + xv^2 - uy = 0$, $u^2 + xyv + v^2 = 0$, find $\frac{\partial u}{\partial x}$ Ans. $\frac{xyv^2 - 2v^3}{2xyu^2 - xy^2 + 6u^2v - 2v^2y - 4xuv}$
5. If $u^2 + xv^2 = x + y$, $v^2 + yu^2 = x - y$, find $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y}$ Ans. $\frac{1-x-v^2}{2u(1-xy)}, \frac{1+y+u^2}{-2v(1-xy)}$
6. If $u = xyz$, $v = x^2 + y^2 + z^2$, $w = x + y + z$ find $\frac{\partial x}{\partial u}$ Ans. $\frac{1}{(x-y)(x-z)}$
7. If $u = x^2 + y^2 + z^2$, $v = xyz$, find $\frac{\partial x}{\partial u}$ Ans. $\frac{x}{2(2x^2 - y^2)}$

1.26 TAYLOR'S SERIES OF TWO VARIABLES

If $f(x, y)$ and all its partial derivatives upto the n th order are finite and continuous for all points (x, y) , where

$$a \leq x \leq a + h, b \leq y \leq b + k$$

$$\text{Then } f(a+h, b+k) = f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f + \frac{1}{3!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^3 f + \dots$$

Proof. Suppose that $f(x+h, y+k)$ is a function of one variable only, say x where y is assumed as constant. Expanding by Taylor's Theorem for one variable, we have

$$f(x+\delta x, y+\delta y) = f(x, y+\delta y) + \delta x \frac{\partial}{\partial x} f(x, y+\delta y) + \frac{(\delta x)^2}{2!} \frac{\partial^2}{\partial x^2} f(x, y+\delta y) + \dots$$

Now expanding for y , we get

$$\begin{aligned} &= \left[f(x, y) + \delta y \frac{\partial}{\partial y} f(x, y) + \frac{(\delta y)^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \right] + \delta x \cdot \frac{\partial}{\partial x} \left[f(x, y) + \delta y \frac{\partial}{\partial y} f(x, y) + \dots \right] \\ &\quad + \frac{(\delta x)^2}{2!} \frac{\partial^2}{\partial x^2} \left[f(x, y) + \delta y \frac{\partial}{\partial y} f(x, y) + \dots \right] + \dots \\ &= \left[f(x, y) + \delta y \frac{\partial}{\partial y} f(x, y) + \frac{(\delta y)^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \right] + \\ &\quad + \delta x \left[\frac{\partial}{\partial x} f(x, y) + \delta y \frac{\partial^2}{\partial x \partial y} f(x, y) \right] + \frac{(\delta x)^2}{2!} \left[\frac{\partial^2}{\partial x^2} f(x, y) + \dots \right] + \dots \\ &= f(x, y) + \left[\delta x \frac{\partial}{\partial x} f(x, y) + \delta y \frac{\partial}{\partial y} f(x, y) \right] + \frac{1}{2!} \left[(\delta x)^2 \frac{\partial^2}{\partial x^2} f(x, y) + 2 \delta x \delta y \frac{\partial^2}{\partial x \partial y} f(x, y) \right. \\ &\quad \left. + (\delta y)^2 \frac{\partial^2}{\partial y^2} f(x, y) \right] + \dots \end{aligned}$$

$$\Rightarrow f(a+h, b+k) = f(a, b) + \left[h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right] + \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right] + \dots$$

$$\Rightarrow f(a+h, b+k) = f(a, b) + \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right] f + \frac{1}{2!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right]^2 f + \dots$$

On putting $a = 0$, $b = 0$, $h = x$, $k = y$, we get

$$f(x, y) = f(0, 0) + \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) + \frac{1}{2!} \left(x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} \right) + \dots$$

Example 78. Expand $e^x \sin y$ in powers of x and y , $x = 0, y = 0$ as far as terms of third degree.

Solution.

		$x = 0, y = 0$
$f(x, y)$	$e^x \sin y$	0
$f_x(x, y)$	$e^x \sin y$	0
$f_y(x, y)$	$e^x \cos y$	1
$f_{xx}(x, y)$	$e^x \sin y$	0
$f_{xy}(x, y)$	$e^x \cos y$	1
$f_{yy}(x, y)$	$-e^x \sin y$	0
$f_{xxx}(x, y)$	$e^x \sin y$	0
$f_{xxy}(x, y)$	$e^x \cos y$	1
$f_{xyy}(x, y)$	$-e^x \sin y$	0
$f_{yyy}(x, y)$	$-e^x \cos y$	-1

By Taylor's theorem

$$\begin{aligned}
 f(x, y) &= f(0, 0) + \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right) f(0, 0) + \frac{1}{2!} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^2 f(0, 0) \\
 &\quad + \frac{1}{3!} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^3 f(0, 0) + \dots \\
 &= f(0, 0) + x f_x(0, 0) + y f_y(0, 0) + \frac{x^2}{2!} f_{xx}(0, 0) + \frac{2xy}{2!} f_{xy}(0, 0) + \frac{y^2}{2!} f_{yy}(0, 0) \\
 &\quad + \frac{1}{3!} x^3 f_{xxx}(0, 0) + \frac{3x^2 y}{3!} f_{xxy}(0, 0) + \frac{3}{3!} x y^2 f_{xyy}(0, 0) + \frac{1}{3!} y^3 f_{yyy}(0, 0) + \dots \\
 e^x \sin y &= 0 + x(0) + y(1) + \frac{x^2}{2}(0) + xy(1) + \frac{y^2}{2}(0) + \frac{x^3}{6}(0) + \frac{3x^2 y}{6}(1) + \frac{3xy^2}{6}(0) + \frac{y^3}{6}(-1) + \dots \\
 &= y + xy + \frac{x^2 y}{2} - \frac{y^3}{6} + \dots
 \end{aligned}$$

Ans.

Example 79. Find the expansion for $\cos x \cos y$ in powers of x, y upto fourth order terms.

Solution.

By Taylor's Series

$$\begin{aligned}
 f(x, y) &= f(0, 0) + x f_x(0, 0) + y f_y(0, 0) + \frac{1}{2!} [x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)] \\
 &\quad + \frac{1}{3!} [x^3 f_{xxx}(0, 0) + 3x^2 y f_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)] \\
 &\quad + \frac{1}{4!} [x^4 f_{xxxx}(0, 0) + 4x^3 y f_{xxxy}(0, 0) + 6x^2 y^2 f_{xxyy}(0, 0) + 4xy^3 f_{xyyy}(0, 0) + y^4 f_{yyyy}(0, 0)] + \dots \\
 \cos x \cos y &= 1 + 0 + 0 + \frac{1}{2}(-x^2 + 0 - y^2) + \frac{1}{6}(0 + 0 + 0 + 0) + \frac{1}{24}(x^4 + 0 + 6x^2 y^2 + 0 + y^4) \\
 &= 1 - \frac{x^2}{2} - \frac{y^2}{2} + \frac{x^4}{24} + \frac{x^2 y^2}{4} + \frac{y^4}{24} + \dots
 \end{aligned}$$

Ans.

		$x = 0, y = 0$
$f(x, y)$	$\cos x \cos y,$	1
f_x	$-\sin x \cos y,$	0
f_y	$-\cos x \sin y,$	0
f_{xx}	$-\cos x \cos y,$	-1
f_{xy}	$\sin x \sin y,$	0
f_{yy}	$-\cos x \cos y,$	-1
f_{xxx}	$\sin x \cos y,$	0
$f_{xx y}$	$\cos x \sin y,$	0
$f_{x yy}$	$\sin x \cos y,$	0
f_{yyy}	$\cos x \sin y,$	0
f_{xxxx}	$\cos x \cos y,$	1
$f_{xxx y}$	$-\sin x \sin y,$	0
$f_{xx yy}$	$\cos x \cos y,$	1
$f_{x yyy}$	$-\sin x \sin y,$	0
f_{yyyy}	$\cos x \cos y,$	1

Example 80. Find the first six terms of the expansion of the function $e^x \log(1 + y)$ in a Taylor's series in the neighbourhood of the point $(0, 0)$.

Solution.

Taylor's series is

$$\begin{aligned}
 f(x, y) &= f(0, 0) + \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right) \\
 &+ \frac{1}{2!} \left(x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} \right) + \dots \\
 \Rightarrow e^x \log(1 + y) &= 0 + (x \times 0 + y \times 1) \\
 &+ \frac{1}{2!} [x^2 \times (0) + 2xy \times 1 + y^2 \times (-1)] + \dots \\
 \Rightarrow e^x \log(1 + y) &= y + xy - \frac{y^2}{2} \quad \text{Ans.}
 \end{aligned}$$

		$x = 0, y = 0$
$f(x, y)$	$e^x \log(1 + y)$	0
$\frac{\partial f}{\partial x}$	$e^x \log(1 + y)$	0
$\frac{\partial f}{\partial y}$	$\frac{e^x}{1 + y}$	1
$\frac{\partial^2 f}{\partial x^2}$	$e^x \log(1 + y)$	0
$\frac{\partial^2 f}{\partial y^2}$	$-\frac{e^x}{(1 + y)^2}$	-1
$\frac{\partial^2 f}{\partial x \partial y}$	$\frac{e^x}{(1 + y)}$	1

EXERCISE 1.16

- Expand $e^x \cos y$ at $(0, 0)$ upto three terms. Ans. $1 + x + \frac{1}{2}(x^2 - y^2) + \dots$
- Expand $z = e^{2x} \cos 3y$ in power series of x and y upto quadratic terms. (AMIE Summer 2004)
Ans. $1 + 2x + 2x^2 - \frac{9}{2}y^2 + \dots$
- Show that $e^y \log(1 + x) = x + xy - \frac{x^2}{2}$ approximately.
- Verify $\sin(x + y) = x + y - \frac{(x + y)^3}{3} + \dots$

Example 81. Expand $\sin(xy)$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{2}\right)$ as far as the terms of second degree.
(Nagpur University, Summer 2003)

Solution. We have, $f(x, y) = \sin(xy)$

Here $\begin{cases} a + h = x \text{ and } h = x - 1 \\ \Rightarrow a + (x - 1) = x \Rightarrow a = 1 \end{cases}$

$\begin{cases} b + k = y \text{ and } k = y - \frac{\pi}{2} \\ \Rightarrow b + y - \frac{\pi}{2} = y \Rightarrow b = \frac{\pi}{2} \end{cases}$

By Taylor's theorem for a function of two variables, we have

$$f(a + h, b + k) = f(a, b) + hf_x(a, b) + kf_y(a, b)$$

$$+ \frac{1}{2!} \{h^2 f_{xx}(a, b) + 2hkf_{xy}(a, b) + k^2 f_{yy}(a, b)\}$$

$$\Rightarrow f(x, y) = f\left(1, \frac{\pi}{2}\right) + (x-1) f_x\left(1, \frac{\pi}{2}\right) + \left(y - \frac{\pi}{2}\right) f_y\left(1, \frac{\pi}{2}\right)$$

$$+ \frac{1}{2!} \left\{ (x-1)^2 f_{xx}\left(1, \frac{\pi}{2}\right) + 2(x-1)\left(y - \frac{\pi}{2}\right) f_{xy}\left(1, \frac{\pi}{2}\right) + \left(y - \frac{\pi}{2}\right)^2 f_{yy}\left(1, \frac{\pi}{2}\right) \right\}$$

$$\Rightarrow \sin(xy) = 1 + (x-1) \cdot 0 + \left(y - \frac{\pi}{2}\right) \cdot 0 +$$

$$\frac{1}{2!} \left\{ (x-1)^2 \left(-\frac{\pi^2}{4}\right) + 2(x-1)\left(y - \frac{\pi}{2}\right) \left(-\frac{\pi}{2}\right) + \left(y - \frac{\pi}{2}\right)^2 (-1) \right\} + \dots$$

$$\Rightarrow \sin(xy) = 1 - \frac{\pi^2}{8} (x-1)^2 - \frac{\pi}{2} (x-1) \left(y - \frac{\pi}{2}\right) - \frac{1}{2} \left(y - \frac{\pi}{2}\right)^2 + \dots$$

Ans.

Example 82. Expand $e^x \cos y$ near the point $\left(1, \frac{\pi}{4}\right)$ by Taylor's Theorem.
(U.P., I Semester Dec. 2007)

Solution. $f(x + h, y + k) = f(x, y) + \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right] f$

$$+ \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f + \frac{1}{3!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^3 f + \dots$$

$$e^x \cos y = f(x, y) = f\left[1 + (x-1), \frac{\pi}{4} + \left(y - \frac{\pi}{4}\right)\right]$$

$$\text{where } h = x - 1, k = y - \frac{\pi}{4} = f\left(1 + h, \frac{\pi}{4} + k\right)$$

Putting these values in Taylor's Theorem, we get

$$e^x \cos y = \frac{e}{\sqrt{2}} + \left[(x-1) \frac{e}{\sqrt{2}} + \left(y - \frac{\pi}{4}\right) \left(\frac{-e}{\sqrt{2}}\right)\right]$$

		$x = 1, y = \frac{\pi}{2}$
$f(x, y)$	$\sin(xy)$	1
$f_x(x, y)$	$y \cos(xy)$,	0
$f_y(x, y)$	$x \cos(xy)$,	0
$f_{xx}(x, y)$	$-y^2 \sin(xy)$,	$-\frac{\pi^2}{4}$
$f_{xy}(x, y)$	$\cos(xy) - xy \sin(xy)$,	$-\frac{\pi}{2}$
$f_{yy}(x, y)$	$-x^2 \sin(xy)$,	-1

		$x = 1, y = \frac{\pi}{4}$
$f(x, y)$	$e^x \cos y$	$\frac{e}{\sqrt{2}}$
$\frac{\partial f}{\partial x}$	$e^x \cos y$,	$\frac{e}{\sqrt{2}}$
$\frac{\partial f}{\partial y}$	$-e^x \sin y$,	$\frac{-e}{\sqrt{2}}$
$\frac{\partial^2 f}{\partial x^2}$	$e^x \cos y$,	$\frac{e}{\sqrt{2}}$
$\frac{\partial^2 f}{\partial y^2}$	$-e^x \cos y$,	$\frac{-e}{\sqrt{2}}$
$\frac{\partial^2 f}{\partial x \partial y}$	$-1 e^x \sin y$,	$\frac{-e}{\sqrt{2}}$

$$\begin{aligned}
& + \frac{1}{2!} \left[(x-1)^2 \frac{e}{\sqrt{2}} + 2(x-1) \left(y - \frac{\pi}{4} \right) \left(\frac{-e}{\sqrt{2}} \right) + \left(y - \frac{\pi}{4} \right)^2 \left(\frac{-e}{\sqrt{2}} \right) \right] + \dots \\
& = \frac{e}{\sqrt{2}} \left[1 + (x-1) - \left(y - \frac{\pi}{4} \right) + \frac{(x-1)^2}{2} - (x-1) \left(y - \frac{\pi}{4} \right) - \left(y - \frac{\pi}{4} \right)^2 + \dots \right] \quad \text{Ans.}
\end{aligned}$$

Example 83. If $f(x, y) = \tan^{-1}(xy)$, compute an approximate value of $f(0.9, -1.2)$.

Solution. We have,

$$f(x, y) = \tan^{-1}(xy)$$

Let us expand $f(x, y)$ near the point $(1, -1)$

$$\begin{aligned}
f(0.9, -1.2) &= f(1 - 0.1, -1 - 0.2) \\
&= f(1, -1) + \left[(-0.1) \frac{\partial f}{\partial x} + (-0.2) \frac{\partial f}{\partial y} \right] + \frac{1}{2!} \left[(-0.1)^2 \frac{\partial^2 f}{\partial x^2} \right. \\
&\quad \left. + 2(-0.1)(-0.2) \frac{\partial^2 f}{\partial x \partial y} + (-0.2)^2 \frac{\partial^2 f}{\partial y^2} \right] + \dots \quad \dots(1)
\end{aligned}$$

		$x = 1, \quad y = -1$
$f(x, y)$	$\tan^{-1}(xy)$	$-\frac{\pi}{4}$
$\frac{\partial f}{\partial x}$	$\frac{y}{1+x^2y^2}$,	$-\frac{1}{2}$
$\frac{\partial f}{\partial y}$	$\frac{x}{1+x^2y^2}$,	$\frac{1}{2}$
$\frac{\partial^2 f}{\partial x^2}$	$-\frac{(2x)y}{(1+x^2y^2)^2}$,	$\frac{1}{2}$
$\frac{\partial^2 f}{\partial y \partial x}$	$\frac{1+x^2y^2 - x(2xy^2)}{(1+x^2y^2)^2} = \frac{1-x^2y^2}{(1+x^2y^2)^2}$	0
$\frac{\partial^2 f}{\partial y^2}$	$\frac{-x(2x^2y)}{(1+x^2y^2)^2}$,	$\frac{1}{2}$

Substituting the values of $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ etc. in (1), we get

$$\begin{aligned}
f(0.9, -1.2) &= -\frac{\pi}{4} + (-0.1) \left(-\frac{1}{2} \right) + (-0.2) \left(\frac{1}{2} \right) + \frac{1}{2} \left[(-0.1)^2 \left(\frac{1}{2} \right) \right. \\
&\quad \left. + 2(-0.1)(-0.2) 0 + (-0.2)^2 \left(\frac{1}{2} \right) \right] + \dots \\
&= -\frac{22}{28} + 0.05 - 0.1 + \frac{1}{2} (0.005 + 0.02) \\
&= -0.786 + 0.05 - 0.1 + 0.0125 = -0.8235 \quad \text{Ans.}
\end{aligned}$$

Example 84. Obtain Taylor's expansion of $\tan^{-1} \frac{y}{x}$ about $(1, 1)$ upto and including the second degree terms. Hence compute $f(1.1, 0.9)$. (U.P., I Sem. Winter 2005, 2002)

Solution.

		$x = 1, y = 1$
$f(x, y)$	$\tan^{-1} \frac{y}{x}$	$\frac{\pi}{4}$
$\frac{\partial f}{\partial x}$	$\frac{1}{1 + \frac{y^2}{x^2}} \left(-\frac{y}{x^2} \right) = \frac{-y}{x^2 + y^2},$	$-\frac{1}{2}$
$\frac{\partial f}{\partial y}$	$\frac{1}{1 + \frac{y^2}{x^2}} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2},$	$\frac{1}{2}$
$\frac{\partial^2 f}{\partial x^2}$	$\frac{y(2x)}{(x^2 + y^2)^2} = \frac{2xy}{(x^2 + y^2)^2},$	$\frac{1}{2}$
$\frac{\partial^2 f}{\partial y^2}$	$\frac{-x(2y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2},$	$-\frac{1}{2}$
$\frac{\partial^2 f}{\partial y \partial x}$	$\frac{(x^2 + y^2) - (x)(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2},$	0

By Taylor's Theorem

$$f(x, y) = f(a, b) + \left[(x-a) \frac{\partial f}{\partial x} + (y-b) \frac{\partial f}{\partial y} \right] + \frac{1}{2!} \left[(x-a)^2 \frac{\partial^2 f}{\partial x^2} + 2(x-a)(y-b) \frac{\partial^2 f}{\partial x \partial y} + (y-b)^2 \frac{\partial^2 f}{\partial y^2} \right] + \dots$$

Here,

$$\begin{aligned} a = 1, b = 1 \\ \tan^{-1} \frac{y}{x} = \frac{\pi}{4} + (x-1) \left(-\frac{1}{2} \right) + (y-1) \frac{1}{2} + \frac{1}{2!} \left[(x-1)^2 \left(\frac{1}{2} \right) \right. \\ \left. + 2(x-1)(y-1)(0) + (y-1)^2 \left(-\frac{1}{2} \right) \right] + \dots \end{aligned}$$

$$\tan^{-1} \frac{y}{x} = \frac{\pi}{4} - \frac{1}{2}(x-1) + \frac{1}{2}(y-1) + \frac{1}{4}(x-1)^2 - \frac{1}{4}(y-1)^2 + \dots \quad \dots(1)$$

Putting $(x-1) = 1.1 - 1 = 0.1$, $(y-1) = 0.9 - 1 = -0.1$ in (1), we get

$$f(1.1, 0.9) = \frac{\pi}{4} - \frac{1}{2}(0.1) - \frac{1}{2}(-0.1) + \frac{1}{4}(0.1)^2 - \frac{1}{4}(-0.1)^2$$

$$= 0.786 - 0.05 + 0.05 + 0.0025 - 0.0025 = 0.786$$

Ans.

Example 85. Expand $\frac{(x+h)(y+k)}{x+h+y+k}$ in powers of h, k upto and inclusive of the second degree terms. (A.M.I.E., Summer 2001)

Solution. $f(x+h, y+k) = \frac{(x+h)(y+k)}{x+h+y+k}$

$$f(x, y) = \frac{xy}{x+y}$$

$$\frac{\partial f}{\partial x} = \frac{(x+y)y - xy}{(x+y)^2} = \frac{y^2}{(x+y)^2}$$

$$\frac{\partial f}{\partial y} = \frac{(x+y)x - xy}{(x+y)^2} = \frac{x^2}{(x+y)^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{-2y^2}{(x+y)^3}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{(x+y)^2 2x - 2(x+y)x^2}{(x+y)^4} = \frac{(x+y)2x - 2x^2}{(x+y)^3} = \frac{2xy}{(x+y)^3}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{-2x^2}{(x+y)^3}$$

$$f(x+h, y+k) = f(x, y) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f(x, y) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(x, y) + \dots$$

$$\begin{aligned} \frac{(x+h)(y+k)}{x+h+y+k} &= \frac{xy}{x+y} + h \frac{y^2}{(x+y)^2} + k \frac{x^2}{(x+y)^2} \\ &\quad + \frac{h^2}{2!} \frac{(-2y^2)}{(x+y)^3} + \frac{1}{2!} 2hk \frac{2xy}{(x+y)^3} + \frac{1}{2!} k^2 \frac{(-2x^2)}{(x+y)^3} + \dots \\ &= \frac{xy}{x+y} + \frac{hy^2}{(x+y)^2} + \frac{kx^2}{(x+y)^2} - \frac{h^2y^2}{(x+y)^3} + \frac{2hkxy}{(x+y)^3} - \frac{k^2x^2}{(x+y)^3} + \dots \quad \text{Ans.} \end{aligned}$$

Example 86. Expand $x^2y + 3y - 2$ in powers of $x - 1$ and $y + 2$ using Taylor's Theorem.

(A.M.I.E.T.E., Winter 2003, A.M.I.E., Summer 2004, 2003)

Solution. $f(x, y) = x^2y + 3y - 2$

Here $a + h = x$ and $h = x - 1$, so $a = 1$
 $b + k = y$ and $k = y + 2$ so $b = -2$

		$x = 1, y = -2$
$f(x, y)$	$x^2y + 3y - 2,$	-10
$f_x(x, y)$	$2xy,$	-4
$f_y(x, y)$	$x^2 + 3,$	4
$f_{xx}(x, y)$	$2y,$	-4
$f_{xy}(x, y)$	$2x,$	2
$f_{yy}(x, y)$	$0,$	0
$f_{xxx}(x, y)$	$0,$	0
$f_{xxy}(x, y)$	$2,$	2
$f_{xyy}(x, y)$	$0,$	0
$f_{yyy}(x, y)$	$0,$	0

Now Taylor's Theorem is

$$f(a+h, b+k) = f(a, b) + \left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right)_{(a,b)} + \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right]_{(a,b)} \\ + \frac{1}{3!} \left[h^3 \frac{\partial^3 f}{\partial x^3} + 3h^2k \frac{\partial^3 f}{\partial x^2 \partial y} + 3hk^2 \frac{\partial^3 f}{\partial x \partial y^2} + k^3 \frac{\partial^3 f}{\partial y^3} \right] + \dots$$

Putting the values of $f(a, b)$ etc. in Taylor's Theorem, we get

$$x^2y + 3y - 2 = -10 + [(x-1)(-4) + (y+2)(4)] \\ + \frac{1}{2!} [(x-1)^2(-4) + 2(x-1)(y+2)(2) + (y+2)^2(0)] \\ + \frac{1}{3!} [(x-1)^3(0) + 3(x-1)^2(y+2)(2) + 3(x-1)(y+2)^2(0) + (y+2)^3(0)] \\ x^2y + 3y - 2 = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2) \text{Ans.}$$

EXERCISE 1.17

1. Expand e^{xy} at $(1, 1)$ upto three terms.

$$\text{Ans. } e[1 + (x-1) + (y-1) + \frac{1}{2!}[(x-1)^2 + 4(x-1)(y-1) + (y-1)^2]]$$

2. Expand y^x at $(1, 1)$ upto second term

$$\text{Ans. } 1 + (y-1) + (x-1)(y-1) + \dots$$

3. Expand $e^{ax} \sin by$ in powers of x and y as far as the terms of third degree. (U.P. I sem. Jan 2011)

$$\text{Ans. } by + abxy + \frac{1}{3!}(3a^2bx^2y - b^3y^3) + \dots$$

4. Expand $(x^2y + \sin y + e^x)$ in powers of $(x-1)$ and $(x-\pi)$.

$$\text{Ans. } \pi + e + (x-1)(2\pi + e) + \frac{1}{2}(x-1)^2(2\pi + e) + 2(x-1)(y-\pi).$$

5. Expand $(1+x+y^2)^{1/2}$ at $(1, 0)$.

$$\text{Ans. } \sqrt{2} \left[1 + \frac{x-1}{4} - \frac{(x-1)^2}{32} + \frac{y^2}{4} + \dots \right]$$

6. Obtain the linearised form $T(x, y)$ of the function $f(x, y) = x^2 - xy + \frac{1}{2}y^2 + 3$ at the point $(3, 2)$, using the Taylor's series expansion. Find the maximum error in magnitude in the approximation $f(x, y) \approx T(x, y)$ over the rectangle R: $|x-3| < 0.1, |y-2| < 0.1$.

$$\text{Ans. } 8 + 4(x-3) - (y-2), \text{ Error } 0.04.$$

7. Expand $\sin(x+h)(y+k)$ by Taylor's Theorem.

$$\text{Ans. } \sin xy + h(x+y) \cos xy + hk \cos xy - \frac{1}{2}h^2(x+y)^2 \sin xy + \dots$$

8. Fill in the blank:

$$f(x, y) = f(2, 3) + \dots \text{Ans. } \left[(x-2) \frac{\partial}{\partial x} + (y-3) \frac{\partial}{\partial y} \right] f + \frac{1}{2!} \left[(x-2) \frac{\partial}{\partial x} + (y-3) \frac{\partial}{\partial y} \right]^2 f + \dots$$

9. If $f(x) = f(0) + kf_1(0) + \frac{k^2}{2!}f_2(0k), 0 < \theta < 1$ then the value of θ when $k=1$ and $f(x) = (1-x)^{3/2}$ is given as (U.P. Ist Semester, Dec 2008)

1.27 MAXIMUM VALUE

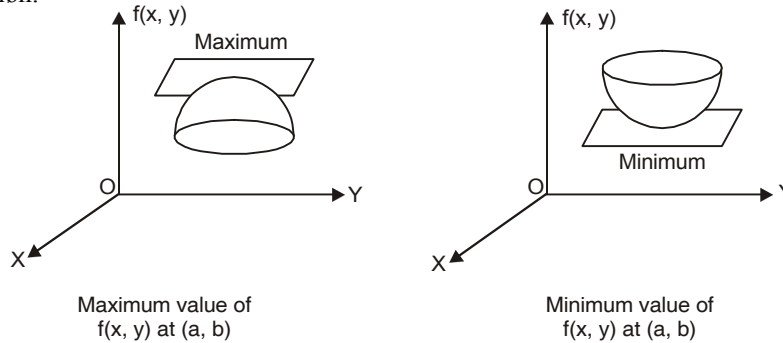
A function $f(x, y)$ is said to have a maximum value at $x = a, y = b$, if there exists a small neighbourhood of (a, b) such that,

$$f(a, b) > f(a+h, b+k)$$

Minimum Value. A function $f(x, y)$ is said to have a minimum value for $x = a, y = b$, if there exists a small neighbourhood of (a, b) such that

$$f(a, b) < f(a + h, b + k)$$

The maximum and minimum values of a function are also called extreme or extremum values of the function.



Saddle point or Minimax. It is a point where a function is neither maximum nor minimum.

Geometrical Interpretation. Such a surface (looks like the leather seat on the back of a horse) forms a ridge rising in one direction and falling in another direction.

1.28 CONDITIONS FOR EXTREMUM VALUES

If $f(a + h, b + k) - f(a, b)$ remains of the same sign for all values (positive or negative) of h, k then $f(a, b)$ is said to be extremum value of $f(x, y)$ at (a, b)

(i) If $f(a + h, b + k) - f(a, b) < 0$, then $f(a, b)$ is maximum.

(ii) If $f(a + h, b + k) - f(a, b) > 0$, then $f(a, b)$ is minimum.

By Taylor's Theorem

$$\begin{aligned} f(a + h, b + k) &= f(a, b) + \left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right)_{(a,b)} + \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right] + \dots \\ \Rightarrow f(a + h, b + k) - f(a, b) &= \left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right)_{(a,b)} + \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right] \dots (1) \\ \Rightarrow f(a + h, b + k) - f(a, b) &= \left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right)_{(a,b)} \dots (2) \end{aligned}$$

For small values of h, k , the second and higher order terms are still smaller and hence may be neglected.

The sign of L.H.S. of (2) is governed by $h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y}$ which may be positive or negative depending on h, k .

Hence, the necessary condition for $f(a, b)$ to be a maximum or minimum is that

$$\left(h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right) = 0 \Rightarrow \frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$$

By solving the equations, we get, point $x = a, y = b$ which may be maximum or minimum value.

Then from (1)

$$f(a + h, b + k) - f(a, b) = \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right]$$

$$= \frac{1}{2!} [h^2 r + 2 h k s + k^2 t] \quad \dots(3)$$

$$\text{where } r = \frac{\partial^2 f}{\partial x^2}, s = \frac{\partial^2 f}{\partial x \partial y}, t = \frac{\partial^2 f}{\partial y^2} \text{ at } (a, b)$$

Now the sign of L.H.S. of (3) is sign of $[rh^2 + 2hks + k^2 t]$

$$= \text{sign of } \frac{1}{r} [r^2 h^2 + 2hkrs + k^2 rt] = \text{sign of } \frac{1}{r} [(r^2 h^2 + 2hkrs + k^2 s^2) + (-k^2 s^2 + k^2 rt)]$$

$$= \text{sign of } \frac{1}{r} [(hr + ks)^2 + k^2 (rt - s^2)]$$

$$= \text{sign of } \frac{1}{r} [(always +ve) + k^2 (rt - s^2)] \quad [(hr + ks)^2 = +ve]$$

$$= \text{sign of } \frac{1}{r} [k^2 (rt - s^2)] = \text{sign of } r \text{ if } rt - s^2 > 0$$

Hence, if $rt - s^2 > 0$, then $f(x, y)$ has a maximum or minimum at (a, b) , according as $r < 0$ or $r > 0$.

Note: (i) If $rt - s^2 < 0$, then L.H.S. will change with h and k hence there is no maximum or minimum at (a, b) , i.e., it is a saddle point.

$$(ii) \text{ If } rt - s^2 = 0, \text{ then } rh^2 + 2shk + tk^2 = \frac{1}{r} [(rh + sk)^2 + k^2 (rt - s^2)]$$

$$= \frac{1}{r} (rh + sk)^2 \text{ which is zero for values of } h, k, \text{ such that}$$

$$\Rightarrow \frac{h}{k} = -\frac{s}{r}$$

This is, therefore, a doubtful case, further investigation is required.

1.29 WORKING RULE TO FIND EXTREMUM VALUES

$$(i) \text{ Differentiate } f(x, y) \text{ and find out } \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y^2}$$

$$(ii) \text{ Put } \frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0 \text{ and solve these equations for } x \text{ and } y. \text{ Let } (a, b) \text{ be the values of } (x, y).$$

$$(iii) \text{ Evaluate } r = \frac{\partial^2 f}{\partial x^2}, s = \frac{\partial^2 f}{\partial x \partial y}, t = \frac{\partial^2 f}{\partial y^2} \text{ for these values } (a, b).$$

$$(iv) \text{ If } rt - s^2 > 0 \text{ and}$$

$$(a) r < 0, \text{ then } f(x, y) \text{ has a maximum value.}$$

$$(b) r > 0, \text{ then } f(x, y) \text{ has a minimum value.}$$

$$(v) \text{ If } rt - s^2 < 0, \text{ then } f(x, y) \text{ has no extremum value at the point } (a, b).$$

$$(vi) \text{ If } rt - s^2 = 0, \text{ then the case is doubtful and needs further investigation.}$$

Note: The point (a, b) which are the roots of $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$, are called stationary points.

Example 87. Discuss the maximum and minimum of $x^2 + y^2 + 6x + 12$.

Solution. We have, $f(x, y) = x^2 + y^2 + 6x + 12$

$$\frac{\partial f}{\partial x} = 2x + 6, \frac{\partial f}{\partial y} = 2y, \frac{\partial^2 f}{\partial x^2} = 2, \frac{\partial^2 f}{\partial y^2} = 2, \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\text{For maxima and minima, } \frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0$$

$$\begin{aligned} \Rightarrow 2x + 6 &= 0, \text{ and } 2y = 0 \\ \Rightarrow x &= -3, \text{ and } y = 0 \\ \text{At } (-3, 0) \quad rt - s^2 &= 2 \times 2 - 0 = 4 > 0 \\ r &= 2 > 0 \end{aligned}$$

Hence $f(x, y)$ is minimum when $x = -3$ and $y = 0$

$$\text{Minimum value} = f(-3, 0) = 9 + 0 - 18 + 12 = 3$$

Ans.

Example 88. Find the absolute maximum and minimum values of

$$f(x, y) = 2 + 2x + 2y - x^2 - y^2$$

on triangular plate in the first quadrant, bounded by the lines $x = 0$, $y = 0$ and $y = 9 - x$.

(Gujarat, I semester, Jan. 2009)

Solution. We have, $f(x, y) = 2 + 2x + 2y - x^2 - y^2$

$$\frac{\partial f}{\partial x} = 2 - 2x, \quad \frac{\partial f}{\partial y} = 2 - 2y$$

$$\frac{\partial^2 f}{\partial x^2} = -2, \quad \frac{\partial^2 f}{\partial x \partial y} = 0, \quad \frac{\partial^2 f}{\partial y^2} = -2$$

For maxima and minima,

$$\frac{\partial f}{\partial x} = 0 \Rightarrow 2 - 2x = 0 \Rightarrow x = 1$$

$$\frac{\partial f}{\partial y} = 0 \Rightarrow 2 - 2y = 0 \Rightarrow y = 1$$

$$\text{At } (1, 1) \quad rt - s^2 = (-2)(-2) - 0 = 4 > 0$$

Hence $f(x, y)$ is maximum at $(1, 1)$.

$$\text{Maximum value of } f(x, y) = 2 + 2 + 2 - 1 - 1 = 4$$

Ans.

Example 89. Examine $f(x, y) = x^3 + y^3 - 3axy$ for maximum and minimum values.

(U.P. I Sem., Dec. 2004), (M.U. 2004, 2003)

Solution. We have, $f(x, y) = x^3 + y^3 - 3axy$

$$p = \frac{\partial f}{\partial x} = 3x^2 - 3ay, \quad q = \frac{\partial f}{\partial y} = 3y^2 - 3ax$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x, \quad s = \frac{\partial^2 f}{\partial x \partial y} = -3a, \quad t = \frac{\partial^2 f}{\partial y^2} = 6y$$

For maxima and minima

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

$$3x^2 - 3ay = 0 \quad \left| \quad 3y^2 - 3ax = 0 \right.$$

$$\Rightarrow x^2 = ay \Rightarrow y = \frac{x^2}{a} \quad \dots(1) \quad \left| \quad \Rightarrow y^2 = ax \quad \dots(2) \right.$$

Putting the value of y from (1) in (2), we get

$$x^4 = a^3x \Rightarrow x(x^3 - a^3) = 0$$

$$\Rightarrow x(x - a)(x^2 + ax + a^2) = 0$$

$$\Rightarrow x = 0, a$$

Putting $x = 0$ in (1), we get $y = 0$,

Putting $x = a$ in (1), we get $y = a$,

Stationary pairs	(0, 0)	(a, a)
r	0	$6a$
s	$-3a$	$-3a$
t	0	$6a$
$rt - s^2$	$-9a^2 < 0$	$27a^2 > 0$

At (0, 0) there is no extremum value, since $rt - s^2 < 0$.

At (a, a), $rt - s^2 > 0$, $r > 0$

Therefore (a, a) is a point of minimum value.

The minimum value of $f(a, a) = a^3 + a^3 - 3a^3 = -a^3$

Ans.

Example 90. Show that the function

$$f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$$

is maximum at $(-7, -7)$ and minimum at $(3, 3)$.

Solution. We have, $f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$... (1)

$$\begin{aligned} \frac{\partial f}{\partial x} &= 3x^2 - 63 + 12y, & \frac{\partial f}{\partial y} &= 3y^2 - 63 + 12x \\ \frac{\partial^2 f}{\partial x^2} &= 6x, & \frac{\partial^2 f}{\partial x \partial y} &= 12, & \frac{\partial^2 f}{\partial y^2} &= 6y \end{aligned}$$

For extremum, we have

$$p = \frac{\partial f}{\partial x} = 3x^2 - 63 + 12y = 0 \Rightarrow x^2 + 4y - 21 = 0 \quad \dots (2)$$

$$q = \frac{\partial f}{\partial y} = 3y^2 - 63 + 12x = 0 \Rightarrow y^2 + 4x - 21 = 0 \quad \dots (3)$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x \Rightarrow s = \frac{\partial^2 f}{\partial x \partial y} = 12 \Rightarrow t = \frac{\partial^2 f}{\partial y^2} = 6y$$

We have to solve (2) and (3) for x and y .

On subtracting (3) from (2), we have

$$\begin{aligned} x^2 - y^2 - 4(x - y) &= 0 \Rightarrow (x - y)(x + y - 4) = 0 \\ x &= y \text{ and } x + y = 4 \end{aligned} \quad \dots (4)$$

If $x = y$ then (2) becomes, $x^2 + 4x - 21 = 0$, $(x + 7)(x - 3) = 0$

$$x = -7, \text{ and } x = 3$$

$$y = -7, \text{ and } y = 3$$

Two stationary points are $(-7, -7)$ and $(3, 3)$.

On solving (2) and (4), we get

$$\begin{aligned} x^2 + 4(4 - x) - 21 &= 0, \Rightarrow x^2 - 4x - 5 = 0 \\ \Rightarrow (x - 5)(x + 1) &= 0 \\ x &= -1, x = 5 \\ y &= 5, y = -1 \end{aligned}$$

Two more stationary points are $(-1, 5)$ and $(5, -1)$.

Hence four possible extremum points of $f(x, y)$ are $(-7, -7)$, $(3, 3)$, $(-1, 5)$ and $(5, -1)$ may be.

Stationary pairs		$(-7, -7)$	$(3, 3)$	$(-1, 5)(5, -1)$
$r = 6x$	-42	$+18$	-6	30
$s = 12$	12	12	12	12
$t = 6y$	-42	18	30	-6
$rt - s^2$	$+1620$	$+180$	-324	-324

At $(-7, -7)$

$$r = -\text{ve, and } rt - s^2 = +\text{ve}$$

Hence, $f(x, y)$ is maximum at $(-7, -7)$,

At $(3, 3)$

$$r = +\text{ve, and } rt - s^2 = +\text{ve}$$

Hence, $f(x, y)$ is minimum at $(3, 3)$.

Proved.

Example 91. Find the extreme values of $u = x^2 y^2 - 5x^2 - 8xy - 5y^2$.

Solution. We have,

$$\begin{aligned} u = x^2 y^2 - 5x^2 - 8xy - 5y^2 &\Rightarrow p = \frac{\partial u}{\partial x} = 2xy^2 - 10x - 8y \\ q = \frac{\partial u}{\partial y} = 2x^2 y - 8x - 10y &\Rightarrow r = \frac{\partial^2 u}{\partial x^2} = 2y^2 - 10 \\ s = \frac{\partial^2 u}{\partial x \partial y} = 4xy - 8 &\Rightarrow t = \frac{\partial^2 u}{\partial y^2} = 2x^2 - 10 \end{aligned}$$

For extreme values of u , $\frac{\partial u}{\partial x} = 0$, $\frac{\partial u}{\partial y} = 0$

$$2xy^2 - 10x - 8y = 0 \Rightarrow x = \frac{8y}{2y^2 - 10}$$

$$2x^2 y - 8x - 10y = 0$$

$$\Rightarrow 2 \left(\frac{8y}{2y^2 - 10} \right)^2 y - 8 \left(\frac{8y}{2y^2 - 10} \right) - 10y = 0$$

$$\Rightarrow \frac{128 y^2}{(2y^2 - 10)^2} - \frac{64}{2y^2 - 10} - 10 = 0, \quad y = 0 \text{ then } x = 0 \Rightarrow \frac{16 y^2}{(y^2 - 5)^2} - \frac{16}{y^2 - 5} - 5 = 0$$

$$\Rightarrow 16 y^2 - 16 (y^2 - 5) - 5 (y^2 - 5)^2 = 0$$

$$\Rightarrow 16 y^2 - 16 y^2 + 80 - 5 (y^2 - 5)^2 = 0$$

$$\Rightarrow (y^2 - 5)^2 = 16 \Rightarrow y^2 - 5 = \pm 4 \Rightarrow y^2 = 9 \text{ and } 1 \text{ and } \Rightarrow y = \pm 3, \pm 1$$

$$\text{Now, } x = \frac{4y}{y^2 - 5}$$

$$\text{If } y = 1 \text{ then } x = -1;$$

$$\text{If } y = -1 \text{ then } x = 1$$

$$\text{If } y = 3 \text{ then } x = 3;$$

$$\text{If } y = -3 \text{ then } x = -3$$

Stationary pairs	$(0, 0)$	$(1, -1)$	$(-1, 1)$	$(3, 3)$	$(-3, -3)$
$r = 2y^2 - 10$	-10	-8	-8	8	8
$s = 4xy - 8$	-8	-12	-12	28	28
$t = 2x^2 - 10$	-10	-8	-8	8	8
$rt - s^2$	$+36$	-80	-80	-720	-720

At (0, 0), r is - ve.

Origin (0, 0) is the only point at which $r t - s^2 > 0$.

Hence, the function u is maximum at origin.

Ans.

Example 92. A rectangular box, open at the top, is to have a volume of 32 c.c. Find the dimensions of the box requiring least material for its construction.

(M.U. 2009; U.P. I semester Jan. 2011; Dec. 2005, A.M.I.E Summer 2001)

Solution. Let l , b and h be the length, breadth, and height of the box respectively and S its surface area and V the volume.

$$V = 32 \text{ c.c.}$$

$$\Rightarrow \quad l b h = 32 \Rightarrow b = \frac{32}{lh}$$

$$S = 2(l + b)h + lb \quad \dots(1)$$

Putting the value of b in (1), we get

$$S = 2 \left(l + \frac{32}{lh} \right) h + l \left(\frac{32}{lh} \right)$$

$$S = 2lh + \frac{64}{l} + \frac{32}{h} \quad \dots(2)$$

Differentiating (2) partially w.r.t. l , we get

$$\frac{\partial S}{\partial l} = 2h - \frac{64}{l^2} \quad \dots(3)$$

Differentiating (2) partially w.r.t. h , we get

$$\frac{\partial S}{\partial h} = 2l - \frac{32}{h^2} \quad \dots(4)$$

For maximum and minimum S , we get

$$\frac{\partial S}{\partial l} = 0 \Rightarrow 2h - \frac{64}{l^2} = 0 \Rightarrow h = \frac{32}{l^2} \quad \dots(5)$$

$$\frac{\partial S}{\partial h} = 0 \Rightarrow 2l - \frac{32}{h^2} = 0 \Rightarrow l = \frac{16}{h^2} \quad \dots(6)$$

From (5) and (6), $l = 4$, $h = 2$ and $b = 4$

$$\frac{\partial^2 S}{\partial l^2} = \frac{128}{l^3} = \frac{128}{64} = 2$$

$$\frac{\partial^2 S}{\partial l \partial h} = 2$$

$$\frac{\partial^2 S}{\partial h^2} = \frac{64}{h^3} = \frac{64}{8} = 8$$

$$\frac{\partial^2 S}{\partial l^2} \cdot \frac{\partial^2 S}{\partial h^2} - \left(\frac{\partial^2 S}{\partial l \partial h} \right)^2 = (2)(8) - (2)^2 = +12$$

$$\frac{\partial^2 S}{\partial l^2} = +2, \text{ so } S \text{ is minimum for } l = 4, b = 4, h = 2$$

Ans.

EXERCISE 1.18

Find the stationary points of the following functions

1. $f(x, y) = y^2 + 4xy + 3x^2 + x^3$

Ans. $\left(\frac{2}{3}, -\frac{4}{3} \right)$, Minimum

2. $f(x, y) = x^3 y^2 (1 - x - y)$ [A.M.I.E., Summer 2004] **Ans.** $\left(\frac{1}{2}, \frac{1}{3}\right)$, Maximum
3. $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^3 + 72x$. (M.U. 2007, 2005, 2004) **Ans.** (6, 0), (4, 0)
4. $f(x, y) = x^2 + 2xy + 2y^2 + 2x + 3y$ such that $x^2 - y = 1$. **Ans.** $\left(-\frac{3}{4}, -\frac{7}{16}\right), -\frac{155}{128}$.
5. $xy e^{-(2x+3y)}$ (A.M.I.E., Winter 2000)
6. Find the extreme value of the function $f(x, y) = x^2 + y^2 + xy + x - 4y + 5$.
State whether this value is a relative maximum or a relative minimum.
Ans. Minimum value of $f(x, y)$ at $(-2, 3) = -2$.
7. Find the values of x and y for which $x^2 + y^2 + 6x = 12$ has a minimum value and find this minimum value. **Ans.** $(-3, 0)$, 3.
8. Find a point within a triangle such that the sum of the square of its distances from the three angular points is a minimum.
9. A tapering log has a square cross-section whose side varies uniformly and is equal to a at the top and $b\left(b > \frac{3a}{2}\right)$ at the bottom. Show that the volume of the greatest conical frustum that can be obtained from the log is $\frac{\pi b^3 l}{27(b-a)}$, where l is the length of the log.
10. A tree trunk of length l metres has the shape of a frustum of a circular cone with radii of its ends a and b metres where $a > b$. Find the length of a beam of uniform square cross section which can be cut from the tree trunk so that the beam has the greatest volume. **Ans.** $\frac{8a^3 l}{27(a-b)}$

1.30 LAGRANGE METHOD OF UNDETERMINED MULTIPLIERS

Let $f(x, y, z)$ be a function of three variables x, y, z and the variables be connected by the relation.

$$\phi(x, y, z) = 0 \quad \dots(1)$$

$\Rightarrow f(x, y, z)$ to have stationary values,

$$\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0, \frac{\partial f}{\partial z} = 0 \Rightarrow \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = 0 \quad \dots(2)$$

By total differentiation of (1), we get $\frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0 \quad \dots(3)$

Multiplying (3) by λ and adding to (2), we get

$$\left(\frac{\partial f}{\partial x} dx + \lambda \frac{\partial \phi}{\partial x} dx\right) + \left(\frac{\partial f}{\partial y} dy + \lambda \frac{\partial \phi}{\partial y} dy\right) + \left(\frac{\partial f}{\partial z} dz + \lambda \frac{\partial \phi}{\partial z} dz\right) = 0$$

$$\left(\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y}\right) dy + \left(\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z}\right) dz = 0$$

This equation will hold good if

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \quad \dots(4)$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \quad \dots(5)$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \quad \dots(6)$$

On solving (1), (4), (5), (6), we can find the values of x , y , z and λ for which $f(x, y, z)$ has stationary value.

Draw Back in Lagrange method is that the nature of stationary point cannot be determined.

Example 93. Find the point upon the plane $ax + by + cz = p$ at which the function

$$f = x^2 + y^2 + z^2 \quad (\text{Nagpur University, Winter 2000})$$

has a minimum value and find this minimum f .

Solution. We have, $f = x^2 + y^2 + z^2$... (1)

$$ax + by + cz = p \Rightarrow \phi = ax + by + cz - p \quad \dots (2)$$

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \Rightarrow 2x + \lambda a = 0 \Rightarrow x = \frac{-\lambda a}{2}$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \Rightarrow 2y + \lambda b = 0 \Rightarrow y = \frac{-\lambda b}{2}$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \Rightarrow 2z + \lambda c = 0 \Rightarrow z = \frac{-\lambda c}{2}$$

Substituting the values of x , y , z in (2), we get

$$a\left(\frac{-\lambda a}{2}\right) + b\left(\frac{-\lambda b}{2}\right) + c\left(\frac{-\lambda c}{2}\right) = p$$

$$\lambda (a^2 + b^2 + c^2) = -2p \Rightarrow \lambda = \frac{-2p}{a^2 + b^2 + c^2}$$

$$\therefore x = \frac{ap}{a^2 + b^2 + c^2}, y = \frac{bp}{a^2 + b^2 + c^2}, z = \frac{cp}{a^2 + b^2 + c^2}$$

$$\begin{aligned} \text{The minimum value of } f &= \frac{a^2 p^2}{(a^2 + b^2 + c^2)^2} + \frac{b^2 p^2}{(a^2 + b^2 + c^2)^2} + \frac{c^2 p^2}{(a^2 + b^2 + c^2)^2} \\ &= \frac{p^2 (a^2 + b^2 + c^2)}{(a^2 + b^2 + c^2)^2} = \frac{p^2}{a^2 + b^2 + c^2} \end{aligned} \quad \text{Ans.}$$

Example 94. Find the maximum value of $u = x^p y^q z^r$ when the variables x , y , z are subject to the condition $ax + by + cz = p + q + r$.

Solution. Here, we have $u = x^p y^q z^r$... (1)

If $\log u = p \log x + q \log y + r \log z$... (2)

$$\frac{1}{u} \frac{\partial u}{\partial x} = \frac{p}{x} \Rightarrow \frac{\partial u}{\partial x} = \frac{pu}{x}$$

$$\frac{1}{u} \frac{\partial u}{\partial y} = \frac{q}{y} \Rightarrow \frac{\partial u}{\partial y} = \frac{qu}{y}$$

$$\frac{1}{u} \frac{\partial u}{\partial z} = \frac{r}{z} \Rightarrow \frac{\partial u}{\partial z} = \frac{ru}{z}$$

$$ax + by + cz = p + q + r$$

$$\phi(x, y, z) = ax + by + cz - p - q - r$$

$$\frac{\partial \phi}{\partial x} = a, \quad \frac{\partial \phi}{\partial y} = b, \quad \frac{\partial \phi}{\partial z} = c$$

Lagranges equations are

$$\frac{\partial u}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \Rightarrow \frac{pu}{x} + \lambda a = 0 \Rightarrow x = -\frac{pu}{\lambda a}$$

$$\frac{\partial u}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \Rightarrow \frac{qu}{y} + \lambda b = 0 \Rightarrow y = -\frac{qu}{\lambda b}$$

$$\frac{\partial u}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \Rightarrow \frac{ru}{z} + \lambda c = 0 \Rightarrow z = -\frac{ru}{\lambda c}$$

Putting in (2), we have

$$-\frac{pu}{\lambda} - \frac{qu}{\lambda} - \frac{ru}{\lambda} = p + q + r$$

$$-\frac{u}{\lambda}(p + q + r) = p + q + r \Rightarrow -\frac{u}{\lambda} = 1 \Rightarrow \lambda = -u$$

$$x = -\frac{pu}{\lambda a} = \frac{-pu}{-ua} = \frac{p}{a}$$

$$y = -\frac{qu}{\lambda b} = \frac{-qu}{-ub} = \frac{q}{b}$$

$$z = -\frac{ru}{\lambda c} = \frac{-ru}{-uc} = \frac{r}{c}$$

Putting in (1), we have

$$\text{Maximum value of } u = \left(\frac{p}{a}\right)^p \left(\frac{q}{b}\right)^q \left(\frac{r}{c}\right)^r$$

Ans.

Example 95. Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.

Solution. Let $2x$, $2y$, $2z$ be the length, breadth and height of the rectangular solid.

Let R be the radius of the sphere.

$$\text{Volume of solid } V = 8x \cdot y \cdot z \quad \dots(1)$$

$$x^2 + y^2 + z^2 = R^2 \quad \dots(2)$$

$$\Rightarrow \phi(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0$$

$$\frac{\partial V}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \Rightarrow 8yz + \lambda(2x) = 0 \quad \dots(3)$$

$$\frac{\partial V}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \Rightarrow 8xz + \lambda(2y) = 0 \quad \dots(4)$$

$$\frac{\partial V}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \Rightarrow 8xy + \lambda(2z) = 0 \quad \dots(5)$$

$$\text{From (3)} \quad 2\lambda x = -8yz \Rightarrow 2\lambda x^2 = -8xyz$$

$$\text{From (4)} \quad 2\lambda y = -8xz \Rightarrow 2\lambda y^2 = -8xyz$$

$$\text{From (5)} \quad 2\lambda z = -8xy \Rightarrow 2\lambda z^2 = -8xyz$$

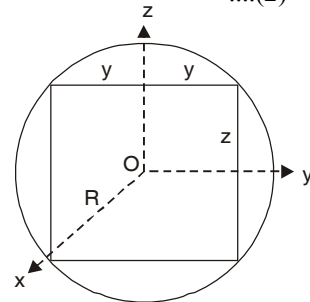
$$2\lambda x^2 = 2\lambda y^2 = 2\lambda z^2$$

$$\Rightarrow x^2 = y^2 = z^2$$

$$\Rightarrow x = y = z$$

Hence, rectangular solid is a cube.

Proved.



Example 96. A rectangular box, which is open at the top, has a capacity of 256 cubic feet. Determine the dimensions of the box such that the least material is required for the construction of the box. Use Lagrange's method of multipliers to obtain the solution.

Solution. Let x, y, z be the length, breadth and height of the box.

$$\Rightarrow \text{Volume} = xyz = 256 \Rightarrow xyz - 256 = 0 \quad \dots(1)$$

$$\Rightarrow \phi(x, y, z) = xyz - 256$$

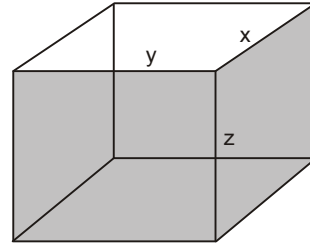
Let S be the material surface of the box.

$$S = xy + 2yz + 2zx$$

$$\frac{\partial S}{\partial x} = y + 2z \quad \text{and} \quad \frac{\partial \phi}{\partial x} = yz$$

$$\frac{\partial S}{\partial y} = x + 2z \quad \text{and} \quad \frac{\partial \phi}{\partial y} = xz$$

$$\frac{\partial S}{\partial z} = 2y + 2x \quad \text{and} \quad \frac{\partial \phi}{\partial z} = xy$$



By Lagrange's method of multiplier, we have

$$\frac{\partial S}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \Rightarrow y + 2z + \lambda yz = 0 \quad \dots(2)$$

$$\frac{\partial S}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \Rightarrow x + 2z + \lambda xz = 0 \quad \dots(3)$$

$$\frac{\partial S}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \Rightarrow 2y + 2x + \lambda xy = 0 \quad \dots(4)$$

Multiplying (2) by x , we get

$$xy + 2xz + \lambda xyz = 0$$

$$\Rightarrow xy + 2xz + 256\lambda = 0 \quad (xyz = 256)$$

$$\Rightarrow xy + 2xz = -256\lambda \quad \dots(5)$$

Multiplying (3) by y , we get

$$xy + 2yz + \lambda xyz = 0$$

$$\Rightarrow xy + 2yz + 256\lambda = 0$$

$$\Rightarrow xy + 2yz = -256\lambda \quad \dots(6)$$

Multiplying (4) by z , we get

$$2yz + 2xz + \lambda xyz = 0 \Rightarrow 2yz + 2xz + 256\lambda = 0$$

$$\Rightarrow 2yz + 2xz = -256\lambda \quad \dots(7)$$

From (5) and (6), we have

$$xy + 2xz = xy + 2yz \Rightarrow 2xz = 2yz \Rightarrow x = y$$

From (6) and (7), we have

$$xy + 2yz = 2yz + 2xz \Rightarrow xy = 2xz \Rightarrow y = 2z$$

From (1)

$$xyz = 256$$

$$\Rightarrow (y)(y)\left(\frac{y}{2}\right) = 256 \Rightarrow y^3 = 512 \Rightarrow y = 8$$

$$x = 8, y = 8, z = 4$$

Hence, length = breadth = 8', height = 4'.

Ans.

Example 97. Use the method of the Lagrange's multipliers to find the volume of the largest rectangular parallelopiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

(Nagpur Univesity, Summer 2008, Winter 2003)

(A.M.I.E.T.E., Summer 2004, U.P., I Semester, Winter 2002, 2000)

Solution. Here, we have $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$$\Rightarrow \phi(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \quad \dots(1)$$

Let $2x$, $2y$, $2z$ be the length, breadth and height of the rectangular parallelopiped inscribed in the ellipsoid.

$$\begin{aligned} V &= (2x)(2y)(2z) = 8xyz \\ \frac{\partial V}{\partial x} &= 8yz; \quad \frac{\partial V}{\partial y} = 8xz, \quad \frac{\partial V}{\partial z} = 8xy \\ \frac{\partial \phi}{\partial x} &= \frac{2x}{a^2}, \quad \frac{\partial \phi}{\partial y} = \frac{2y}{b^2}, \quad \frac{\partial \phi}{\partial z} = \frac{2z}{c^2} \end{aligned}$$

Lagrange's equations are

$$\frac{\partial V}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \quad \Rightarrow \quad 8yz + \lambda \frac{2x}{a^2} = 0 \quad \dots(1)$$

$$\frac{\partial V}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \quad \Rightarrow \quad 8xz + \lambda \frac{2y}{b^2} = 0 \quad \dots(2)$$

$$\frac{\partial V}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \quad \Rightarrow \quad 8xy + \lambda \frac{2z}{c^2} = 0 \quad \dots(3)$$

Multiplying (1), (2) and (3) by x , y , z respectively and adding, we get

$$\begin{aligned} 24xyz + 2\lambda \left[\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right] &= 0 \quad \left[\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right] \\ \Rightarrow 24xyz + 2\lambda(1) &= 0 \quad \Rightarrow \quad \lambda = -12xyz \end{aligned}$$

Putting the value of λ in (1), we get

$$8yz + (-12xyz) \frac{2x}{a^2} = 0 \quad \Rightarrow \quad 1 - \frac{3x^2}{a^2} = 0 \quad \Rightarrow \quad x = \frac{a}{\sqrt{3}}$$

Similarly from (2) and (3), we have

$$y = \frac{b}{\sqrt{3}}, \quad z = \frac{c}{\sqrt{3}}$$

Volume of the largest rectangular parallelopiped = $8xyz$

$$= 8 \left(\frac{a}{\sqrt{3}} \right) \left(\frac{b}{\sqrt{3}} \right) \left(\frac{c}{\sqrt{3}} \right) = \frac{8abc}{3\sqrt{3}} \quad \text{Ans.}$$

Example 98. The shape of a hole pored by a drill is a cone surmounted by cylinder. If the cylinder be of height h and radius r and the semi-vertical angle of the cone be α , where

$\tan \alpha = \frac{h}{r}$ show that for a total height H of the hole, the volume removed is maximum if

$$h = H(\sqrt{7} + 1)/6. \quad (R.G.P.V., Bhopal I sem. 2003)$$

Solution. Let $ABCD$ be the given cylinder of height ' h ' and radius ' r ' and $DPCO$ be the cone of course, of radius r .

Now, since α is the semi-vertical angle of the cone.

$$\therefore \tan \alpha = \frac{PC}{OP} = \frac{r}{OP} \quad \dots(1)$$

$$\text{but, given that } \tan \alpha = \frac{h}{r} \quad \dots(2)$$

$$\text{From (1) and (2), we have } \frac{h}{r} = \frac{r}{OP} \Rightarrow OP = \frac{r^2}{h} \quad \dots(3)$$

$$\text{Total height of the hole} = H \\ \Rightarrow H = h + OP \Rightarrow OP = H - h \quad \dots(4)$$

$$\text{From (3) and (4), we have} \\ \frac{r^2}{h} = H - h \quad \dots(5)$$

$$\text{Again, let } \phi = H - h - \frac{r^2}{h} \quad \dots(6)$$

In drilling a hole, the volume of the removed portion

$$\frac{\partial \phi}{\partial r} = -\frac{2r}{h}, \quad \frac{\partial \phi}{\partial h} = -1 + \frac{r^2}{h^2}$$

V = Volume of the cylinder + Volume of the cone.

$$= \pi r^2 h + \frac{1}{3} \pi r^2 (OP) = \pi r^2 h + \frac{1}{3} \pi r^2 \cdot \frac{r^2}{h} \quad [\text{From (3)}]$$

$$V = \pi r^2 h + \frac{\pi r^4}{3h},$$

$$\frac{\partial V}{\partial r} = 2\pi r h + \frac{4\pi r^3}{3h} \quad \dots(7)$$

By Lagrange's Method

$$\frac{\partial V}{\partial r} + \lambda \frac{\partial \phi}{\partial r} = 0 \quad \Rightarrow \quad 2\pi r h + \frac{4\pi r^3}{3h} + \lambda \left(\frac{-2r}{h} \right) = 0 \quad \dots(8)$$

$$\frac{\partial V}{\partial h} + \lambda \frac{\partial \phi}{\partial h} = 0 \quad \Rightarrow \quad \pi r^2 - \frac{\pi r^4}{3h^2} + \lambda \left(-1 + \frac{r^2}{h^2} \right) = 0 \quad \dots(9)$$

Multiplying (8) by r and (9) by $2h$, we get

$$2\pi r^2 h + \frac{4\pi r^4}{3h} + \lambda \left(\frac{-2r^2}{h} \right) = 0$$

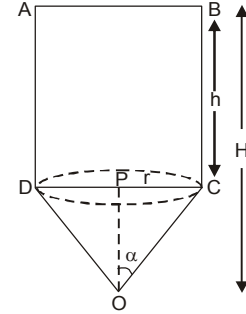
$$2\pi r^2 h - \frac{2\pi r^4}{3h} + 2\lambda \left(-h + \frac{r^2}{h} \right) = 0$$

On subtracting, we get

$$\frac{6\pi r^4}{3h} + \lambda \left(\frac{-2r^2}{h} + 2h - \frac{2r^2}{h} \right) = 0 \quad \Rightarrow \quad \frac{6\pi r^4}{3h} + \lambda \left(\frac{-4r^2}{h} + 2h \right) = 0$$

$$2\pi r^4 + \lambda (-4r^2 + 2h^2) = 0$$

$$\Rightarrow \quad \pi r^4 + \lambda (-2r^2 + h^2) = 0 \quad \Rightarrow \quad \lambda = \frac{\pi r^4}{-h^2 + 2r^2}$$



Putting the value of λ in (8), we get

$$2\pi r h + \frac{4\pi r^3}{3h} + \left(\frac{\pi r^4}{-h^2 + 2r^2} \right) \left(-\frac{2r}{h} \right) = 0 \Rightarrow h + \frac{2r^2}{3h} - \frac{r^4}{h(h^2 - 2r^2)} = 0 \left[\frac{r^2}{h} = H - h \right]$$

$$h + \frac{2}{3}(H - h) + \frac{h^2(H - h)^2}{h[h^2 - 2h(H - h)]} = 0 \Rightarrow h + \frac{2}{3}(H - h) + \frac{(H^2 + h^2 - 2hH)}{h - 2H + 2h} = 0$$

$$h + \frac{2}{3}(H - h) + \frac{H^2 + h^2 - 2hH}{3h - 2H} = 0$$

$$3h^2 - 2Hh + \frac{2}{3}(H - h)(3h - 2H) + H^2 + h^2 - 2hH = 0$$

$$9h^2 - 6Hh + 6Hh - 4H^2 - 6h^2 + 4Hh + 3H^2 + 3h^2 - 6hH = 0$$

$$6h^2 - 2hH - H^2 = 0$$

$$h = \frac{2H \pm \sqrt{4H^2 + 24H^2}}{12}$$

$$h = \frac{H \pm H\sqrt{7}}{6} = H \frac{[\sqrt{7} + 1]}{6} \text{ (-ve is not possible)}$$

Proved.

Example 99. A tent of a given volume has a square base of side $2a$, has its four-side vertical of length b and is surmounted by a regular pyramid of height h . Find the values of a and b in terms of h such that the canvas required for its construction is minimum.

Solution. Let V be the volume and S be the surface of the tent.

$$V = 4a^2b + \frac{1}{3}(4a^2)h \quad [\text{Volume of pyramid} = \frac{1}{3} \text{ Area of the base} \times \text{height}]$$

$$S = 8ab + 4a\sqrt{a^2 + h^2} \quad [\text{Surface Area of pyramid} = \frac{1}{2} \text{ perimeter} \times \text{slant height}]$$

$$\frac{\partial S}{\partial a} + \lambda \frac{\partial V}{\partial a} = 0$$

$$\Rightarrow 8b + 4\sqrt{a^2 + h^2} + \frac{4a^2}{\sqrt{a^2 + h^2}} + \lambda \left[8ab + \frac{8ah}{3} \right] = 0 \quad \dots(1)$$

$$\frac{\partial S}{\partial b} + \lambda \frac{\partial V}{\partial b} = 0 \Rightarrow 8a + 4\lambda a^2 = 0 \quad \dots(2)$$

$$\frac{\partial S}{\partial h} + \lambda \frac{\partial V}{\partial h} = 0 \Rightarrow \frac{4ah}{\sqrt{a^2 + h^2}} + \frac{4}{3}\lambda a^2 = 0 \quad \dots(3)$$

$$\text{From (2)} \quad \lambda a + 2 = 0 \Rightarrow \lambda a = -2 \quad \dots(4)$$

$$\text{From (3)} \quad 12ah + 4\lambda a^2\sqrt{a^2 + h^2} = 0$$

$$\Rightarrow 3h + \lambda a\sqrt{a^2 + h^2} = 0 \quad \dots(5)$$

Substituting the value of λa from (4) in (5), we get

$$3h - 2\sqrt{a^2 + h^2} = 0 \Rightarrow 9h^2 = 4a^2 + 4h^2 \Rightarrow 4a^2 = 5h^2$$

$$a = \frac{\sqrt{5}}{2}h$$

Substituting $\lambda a = -2$ and $a = \frac{\sqrt{5}}{2}h$ in (1) and simplifying, we get

$$8b + 4\sqrt{\frac{5h^2}{4} + h^2} + \frac{5h^2}{\sqrt{\frac{5h^2}{4} + h^2}} - 2\left[8b + \frac{8h}{3}\right] = 0$$

$$\Rightarrow 8b + 6h + \frac{10h}{3} - 16b - \frac{16h}{3} = 0 \quad \Rightarrow -8b + 4h = 0 \quad \Rightarrow b = \frac{h}{2}.$$

Thus, when $a = \frac{\sqrt{5}}{2}h$ and $b = \frac{h}{2}$ we get the stationary value of S . **Ans.**

Example 100. Find the maximum and minimum distances of the point (3, 4, 12) from the sphere $x^2 + y^2 + z^2 = 1$.

Solution. Let the co-ordinates of the given point be (x, y, z), then its distance (D) from (3, 4, 12).

$$D = \sqrt{(x-3)^2 + (y-4)^2 + (z-12)^2}$$

$$\Rightarrow F(x, y, z) = (x-3)^2 + (y-4)^2 + (z-12)^2$$

$$x^2 + y^2 + z^2 = 1$$

$$\phi(x, y, z) = x^2 + y^2 + z^2 - 1$$

$$\frac{\partial F}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 2(x-3) + 2\lambda x = 0 \quad \dots(1)$$

$$\frac{\partial F}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 2(y-4) + 2\lambda y = 0 \quad \dots(2)$$

$$\frac{\partial F}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 2(z-12) + 2\lambda z = 0 \quad \dots(3)$$

Multiplying (1) by x, (2) by y and (3) by z and adding, we get

$$(x^2 + y^2 + z^2) - 3x - 4y - 12z + \lambda(x^2 + y^2 + z^2) = 0$$

$$1 - 3x - 4y - 12z + \lambda = 0 \quad \dots(4)$$

$$\text{From (1)} \quad x = \frac{3}{1+\lambda} \quad \dots(5)$$

$$\text{From (2)} \quad y = \frac{4}{1+\lambda} \quad \dots(6)$$

$$\text{From (3)} \quad z = \frac{12}{1+\lambda} \quad \dots(7)$$

Putting these values of x, y, z in (4), we have

$$1 + \lambda - \frac{9}{1+\lambda} - \frac{16}{1+\lambda} - \frac{144}{1+\lambda} = 0 \Rightarrow (1+\lambda)^2 = 169 \Rightarrow 1 + \lambda = \pm 13$$

Putting the value of $1 + \lambda$ in (5), (6) and (7) we have the points

$$\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right) \text{ and } \left(\frac{-3}{13}, \frac{-4}{13}, \frac{-12}{13}\right).$$

$$\text{The minimum distance} = \sqrt{\left(3 - \frac{3}{13}\right)^2 + \left(4 - \frac{4}{13}\right)^2 + \left(12 - \frac{12}{13}\right)^2} = 12$$

$$\text{The maximum distance} = \sqrt{\left(3 + \frac{3}{13}\right)^2 + \left(4 + \frac{4}{13}\right)^2 + \left(12 + \frac{12}{13}\right)^2} = 14 \quad \text{Ans.}$$

Example 101. If $u = ax^2 + by^2 + cz^2$ where $x^2 + y^2 + z^2 = 1$ and $lx + my + nz = 0$ prove that stationary values of 'u' satisfy the equation

$$\frac{l^2}{a-u} + \frac{m^2}{b-u} + \frac{n^2}{c-u} = 0$$

Solution. We have, $u = ax^2 + by^2 + cz^2$... (1)

Let $\phi = x^2 + y^2 + z^2 - 1$... (2)

$\psi = lx + my + nz$... (3)

$$\frac{\partial u}{\partial x} = 2ax, \quad \frac{\partial u}{\partial y} = 2by, \quad \frac{\partial u}{\partial z} = 2cz$$

$$\frac{\partial \phi}{\partial x} = 2x, \quad \frac{\partial \phi}{\partial y} = 2y, \quad \frac{\partial \phi}{\partial z} = 2z$$

$$\frac{\partial \psi}{\partial x} = l, \quad \frac{\partial \psi}{\partial y} = m, \quad \frac{\partial \psi}{\partial z} = n$$

By Lagrange's method

$$\frac{\partial u}{\partial x} + \lambda_1 \frac{\partial \phi}{\partial x} + \lambda_2 \frac{\partial \psi}{\partial x} = 0, \quad 2ax + 2x\lambda_1 + \lambda_2 l = 0 \quad \dots (4)$$

$$\frac{\partial u}{\partial y} + \lambda_1 \frac{\partial \phi}{\partial y} + \lambda_2 \frac{\partial \psi}{\partial y} = 0, \quad 2by + 2y\lambda_1 + \lambda_2 m = 0 \quad \dots (5)$$

$$\frac{\partial u}{\partial z} + \lambda_1 \frac{\partial \phi}{\partial z} + \lambda_2 \frac{\partial \psi}{\partial z} = 0, \quad 2cz + 2z\lambda_1 + \lambda_2 n = 0 \quad \dots (6)$$

Multiplying (4), (5) and (6) by x , y and z respectively and adding, we get

$$(2ax^2 + 2by^2 + 2cz^2) + (2x^2 + 2y^2 + 2z^2)\lambda_1 + (lx + my + nz)\lambda_2 = 0$$

$$2u + 2\lambda_1 = 0, \quad \lambda_1 = -u$$

Putting the value of λ_1 in (4), (5) and (6), we get

$$2ax - 2xu + \lambda_2 l = 0, \quad x = \frac{-\lambda_2 l}{2(a-u)}$$

$$2by - 2yu + \lambda_2 m = 0, \quad y = \frac{-\lambda_2 m}{2(b-u)}$$

$$2cz - 2zu + \lambda_2 n = 0, \quad z = \frac{-\lambda_2 n}{2(c-u)}$$

Putting the values of x , y , z in (3), we get

$$\frac{-\lambda_2 l^2}{2(a-u)} + \frac{-\lambda_2 m^2}{2(b-u)} + \frac{-\lambda_2 n^2}{2(c-u)} = 0 \Rightarrow \frac{l^2}{a-u} + \frac{m^2}{b-u} + \frac{n^2}{c-u} = 0 \quad \text{Proved.}$$

EXERCISE 1.19

1. Show that the greatest value of $x^m y^n$ where x and y are positive and $x + y = a$ is $\frac{m^m \cdot n^n \cdot a^{m+n}}{(m+n)^{m+n}}$, where a is constant.
2. Using Lagrange's method (of multipliers), find the critical (stationary values) of the function $f(x, y, z) = x^2 + y^2 + z^2$, given that $z^2 = xy + 1$. **Ans.** $(0, 0, -1), (0, 0, 1)$.

3. Decompose a positive number 'a' into three parts so that their product is maximum.

Ans. $\frac{a}{3}, \frac{a}{3}, \frac{a}{3}$

4. The sum, of three numbers is constant. Prove that their product is a maximum when they are equal.

5. Using the method of Lagrange's multipliers, find the largest product of the numbers x, y and z when

$$x + y + z^2 = 16.$$

Ans. $\frac{4096}{25\sqrt{5}}$

6. Using the method of Lagrange's multipliers, find the largest product of the numbers x, y and z when $x^2 + y^2 + z^2 = 9$.

Ans. $3\sqrt{3}$

7. Find a point in the plane $x + 2y + 3z = 13$ nearest to the point $(1, 1, 1)$ using the method of Lagrange's multipliers.

Ans. $\left(\frac{3}{2}, 2, \frac{5}{2}\right)$

8. Using the Lagrange's method (of multipliers), find the shortest distance from the point $(1, 2, 2)$ to the sphere $x^2 + y^2 = 36$.

Ans. 3

9. Find the shortest and the longest distances from the point $(1, 2, -1)$ to the

$$x^2 + y^2 + z^2 = 24.$$

Ans. $\sqrt{6}, 3\sqrt{6}$

10. The sum of the surfaces of a sphere and a cube is given. Show that when the sum of the volumes is least, the diameter of the sphere is equal to the edge of the cube.

11. The electric time constant of a cylindrical coil of wire can be expressed approximately by

$$K = \frac{mxyz}{ax + by + cz}$$

where z is the axial length of the coil, y is the difference between the external and internal radii and x is the mean radius; a, b, m and c represent positive constants. If the volume of the coil is fixed, find the values of x and y which make the time constant K as large as possible.

12. If $u = \frac{a^3}{x^2} + \frac{b^3}{y^2} + \frac{c^3}{z^2}$, where $x + y + z = 1$, prove that the stationary value of u is given by

$$x = \frac{a}{a+b+c}, y = \frac{b}{a+b+c}, z = \frac{c}{a+b+c}$$

13. Find maximum value of the expression $\sum_{i=1}^n a_i x_i$ with $\sum_{i=1}^n x_i^2 = 1$,

where $a_1, a_2, a_3, \dots, a_n$ are positive constants.

Ans. $(a_1^2 + a_2^2 + \dots + a_n^2)^{\frac{1}{2}}$

14. If r is the distance of a point on conic $ax^2 + by^2 + cz^2 = 1$, $lx + my + nz = 0$ from origin, then the stationary values of r are given by the equation.

$$\frac{l^2}{1-ar^2} + \frac{m^2}{1-br^2} + \frac{n^2}{1-cr^2} = 0$$

(A.M.I.E.T.E., Winter 2002)

15. If x and y satisfy the relation $ax^2 + by^2 = ab$, prove that the extreme values of function $u = x^2 + xy + y^2$ are given by the roots of the equation $4(u-a)(u-b) = ab$

(A.M.I.E.T.E., Winter 2000)

16. Use the Lagrange's method of undetermined multipliers to find the minimum value of $x^2 + y^2 + z^2$ subject to the conditions $x + y + z = 1$, $xyz + 1 = 0$.

17. Test the function $f(x, y) = (x^2 + y^2)e^{-(x^2 + y^2)}$ for maxima and minima for points not on the circle $x^2 + y^2 = 1$.

18. Find the absolute maximum and minimum values of the function

$$f(x, y) = cx^2 + y^2 - x \text{ over the region } 2x^2 + y^2 \leq 1$$

(AMIETE, Dec. 2008)